# Chapter 1

# HMC Tuning Interfaces

HMC tuning determines which frozen kernels are suitable for further sampling. For each declared trajectory length $L$, BayesFilter measures proposed step sizes $\epsilon$, independently verifies the surviving pairs, and retains every pair that passes. A failed pair can motivate another measured proposal. A passing pair does not end the search or establish posterior convergence.

## 1.1 One candidate-set procedure

The public functions `tune_hmc_kernel` and `tune_fixed_transport_hmc_kernel` share `HMCTuningCandidateSetController`. Both return `HMCTypedCandidateSetRun`, whose `result` contains the full `HMCTuningCandidateSetResult`. Their preparation differs with the target and coordinates; their candidate scheduling and retention do not.

Ordinary `HMCKernelTuningConfig` prepares a windowed mass matrix or an explicit identity metric. `FixedTransportHMCKernelTuningConfig` prepares identity mass in the coordinates of a reconstructable frozen transport. A caller with already prepared geometry supplies `HMCControllerConfig` and the numerical binding's `typed_adapter`. The conditional `TensorFlowHMCKernelTuningConfig` proposal-field route also uses the shared scheduler, but retains its weaker mechanics authority.

The capability registry `HMC_TUNING_INTERFACE_CAPABILITIES` identifies the supported target and coordinate combinations. Chain runners, stage helpers, discovery routines, and historical selectors cannot independently issue new candidate-set tuning evidence. Unsupported legacy runner customizations fail before execution. Historical single-kernel results remain readable, with their original identities; they are not the result of the current procedure.

Listing 1.1: Inspecting supported target and coordinate routes.

```
"""Select an HMC tuning route from BayesFilter's capability registry."""

from __future__ import annotations

import sys
from pathlib import Path

REPOSITORY_ROOT = Path(__file__).resolve().parents[2]
sys.path.insert(0, str(REPOSITORY_ROOT))

from bayesfilter.inference import (
```

```
    active_hmc_tuning_routes,
    hmc_tuning_interface_capability,
)


def require_artifact_authority(interface_name: str) -> str:
    """Return the qualified public tuner name or reject a lower-level route."""

    capability = hmc_tuning_interface_capability(interface_name)
    if capability.interface_kind != "public_tuner" or not capability.
    artifact_authority:
        replacement = capability.replacement or "no supported replacement"
        raise ValueError(
            f"{capability.qualified_name} is {capability.interface_kind}, not a "
            f"public artifact-authority tuner; use {replacement}"
        )
    return capability.qualified_name


ordinary = require_artifact_authority("tune_hmc_kernel")
fixed_transport = require_artifact_authority("tune_fixed_transport_hmc_kernel")
assert {record.interface_name for record in active_hmc_tuning_routes()} == {
    "tune_hmc_kernel",
    "tune_fixed_transport_hmc_kernel",
}

try:
    require_artifact_authority("run_full_chain_neural_force_hmc")
except ValueError as error:
    expected_rejection = str(error)
else:  # pragma: no cover - the registry contract requires rejection
    raise AssertionError("a chain runner was incorrectly accepted as a tuner")

try:
    require_artifact_authority("tune_hmc_kernel_robust_broad_grid")
except ValueError as error:
    expected_diagnostic_rejection = str(error)
else:  # pragma: no cover - the registry contract requires rejection
    raise AssertionError("a diagnostic helper was incorrectly accepted as a tuner")

print(ordinary)
print(fixed_transport)
print(expected_rejection)
print(expected_diagnostic_rejection)
```

## 1.2 Preparation and immutable geometry

Geometry is a shared responsibility rather than an unspecified tuner default. The caller supplies the initial center and may supply a local geometry hypothesis; the tuner validates that hypothesis and owns the subsequent affine construction and adaptation. Under windowed adaptation, it tries a supplied negative log-posterior Hessian, an initial covariance, and positive diagonal parameter scales, in that order, before falling back to identity. Thus a negative Hessian means

$$H(\theta_0) = -\nabla^2_\theta \ell(\theta_0), \qquad \Sigma_0 \approx H(\theta_0)^{-1}, \tag{1.1}$$

in the same unconstrained coordinates as the supplied center. With geometry fallback enabled, a rejected higher-priority hypothesis is recorded before the next is tried; with fallback disabled, it is a hard error. The explicit fixed- identity policy ignores all hints. Identity is a useful mechanics baseline, not evidence that an anisotropic posterior has identity geometry.

When no hint is supplied, the initializer uses unit curvature frequencies; it does not estimate the target Hessian. A finite density or a near-zero score at the initial point therefore does not establish a stable initial step. Strongly scaled targets can fail the bootstrap before windowed adaptation or candidate search begins. Inspect that preparation failure and supply checked geometry in the active coordinates when available. Increasing the subsequent candidate budget does not repair a bootstrap that never reaches its handoff.

Bootstrap epsilon decisions use the mean Metropolis probability over all recorded proposals, including rejected proposals and excluding discarded burnin:

$$\bar{\alpha} = \frac{1}{n} \sum_{t=1}^{n} \exp\{\min(\log r_t, 0)\}.$$

The realized accepted fraction includes the additional uniform accept/reject draw and remains a separate movement diagnostic. For example, a recorded mean probability of 0.6846 lies within the configured $[0.65, 0.75]$ band even when $13/16 = 0.8125$ proposals were accepted. That binary fraction must not cause an upward epsilon repair. Missing, nonfinite or incomplete probability traces stop the screen. The ordinary fixed-kernel band and the optional startup floor have different preparation roles; neither establishes final candidate or posterior validity.

A target-domain exception permits a bounded smaller bootstrap trial only if the adapter specifically classifies it and the repository failure record locates it in a proposal target callback after a finite pre-transition state. Initial-state, retained-state, unattributed and unrelated execution failures remain terminal. The failed round retains its original assertion and first failure; it receives no invented acceptance probability. A smaller finite parent and the failed epsilon can nominate a logarithmic midpoint; otherwise the configured multiplier reduces epsilon. The next trial recomputes $L$, records any clamp and consumes the existing repair budget. A failed reusable runner is replaced, and only a subsequent completed screen can supply the repaired preparation pair. Target assertions and Metropolis correction remain part of every trial.

An optional finite startup probe is available through `HMCKernelTuningConfig(bootstrap_initialization_rounds=20)`. Zero is the default; a positive integer bounds the number of rounds. Each round tests four independent momentum proposals and shrinks the step when proposals are invalid or acceptance is below the configured lower repair floor. The probe preserves the target, affine mass, failed proposals and separate seed stream. Its selected pair must still pass fresh bootstrap and operational adaptation. In this mode, bootstrap uses its `warmup_startup_only` role: high acceptance does not trigger upward refinement before adaptation, and the starting step is not declared a stability ceiling.

Probe wall times remain in the geometry's full artifact hash for audit, but are excluded from the separate numerical geometry hash used by new candidate scopes. Otherwise, two identical preparations could receive different candidate seeds merely because a probe took longer. Numerical probe outcomes, seed provenance, mass and starts remain bound to the numerical identity. Existing checkpoints retain their saved scopes and streams.

The probe follows the declared target-status policy. With `none`, it checks finite retained and proposed states, values and scores without manufacturing status data. With `per_chain_step`, complete valid telemetry is also required. Invalid initial or retained states stop preparation. A finite startup nomination neither qualifies a final candidate nor establishes convergence.

Preparation success need not mean that the mass changed. The `allow_valid_incumbent` policy permits zero empirical updates; inspect `operational_metric_update_count` when the initial geometry is poorly scaled. A regression diagnosis retained identity geometry throughout preparation and subsequently failed posterior equilibration despite verified tuning candidates. Ordinary operational preparation runs one chain; it constructs the four-chain bank for candidate verification afterwards. Its metric assessment preserves temporal order. For explicitly supplied multiple chains, temporal information is estimated separately within each chain. Split R-hat is reported when available and never vetoes or delays a metric proposal.

The proposed position covariance is the finite-window centered second moment, divided by $N - 1$, with correlations shrunk toward zero while preserving marginal variances. It is a geometry hypothesis during adaptation, not an unbiased posterior covariance estimate. State-count and temporal-information floors, numerical rank, condition and regularization-discrepancy checks screen this hypothesis. Positive-definiteness, coordinate reconstruction, target value/score parity and a fresh reasonable-step check are required before using its transform.

Ordinary preparation records each completed window's metric decision and failed checks in `preparation_progress.json`. It also reports whether the scheduled windows can meet the state-count floors. The `standard` preset is a local diagnostic allocation: with the default metric policy, its 150 transitions in six dimensions contain slow windows of 30, 60 and 30, each shorter than the dense minimum of 64. The explicit `serious` preset funds a larger preparation. Inspect its actual decisions as well; neither a larger allocation nor an accepted metric update establishes posterior equilibration. Rejections preserve the incumbent and identify whether evidence, transform construction, affine parity or step qualification failed.

At a metric change, the fresh step search starts from the previous coordinate system's bounded dual-averaging average. Clipping in log space before exponentiation prevents a large unconstrained adaptation proposal from exhausting the search before any usable step is tested. This old bound limits only the initial hypothesis: the search may expand in the new coordinates and must independently qualify its step. Rejected probes preserve their starting value and attempts. A nonfinite warmup trace identifies its failed fields and first indices; the failure still vetoes preparation.

The optional `metric_evidence_policy="finite_window"` keeps the same covariance formula and numerical checks while treating temporal ESS as explanatory. The default `temporal_information` retains the ESS floor. With the optional policy, automatic preparation allocates slow windows large enough to reach the existing dense state-count floor when the declared budget permits, or the diagonal floor when only that is affordable. It preserves the total transitions and initial/final buffers, merging an undersized final slow remainder. Insufficient total capacity remains visible. This schedule change removes avoidable count failures; it does not establish adequate covariance information or posterior equilibration. Directly supplied schedules and the default policy retain their declared window lengths. This option permits an early geometry proposal before the chain has explored well enough to estimate posterior covariance precisely. For any fixed nonsingular

affine factor $A$, $\theta = c + Az$ has transformed score $A^\mathsf{T}\nabla_\theta \ell$; exact-score Metropolis-adjusted sampling in the transformed coordinates does not require $AA^\mathsf{T}$ to equal the posterior covariance. Its usefulness still depends on exploration and numerical health. Adaptation must finish before candidate verification and retained draws.

The option remains experimental. Its first matched regression preparation applied two updates and later failed a nonfinite warmup trace. The existing reasonable-step probe checks a few momenta at one state and cannot guarantee stability everywhere visited later. Preparation failures remain failures, and neither policy turns metric qualification into posterior convergence.

The optional `metric_probe_num_results` extends each of the four independent metric-boundary probes to a short chain at fixed epsilon and fixed leapfrog count. Its default is one transition. A larger allocation, for example sixteen, tests evolving positions and checks every proposed and retained endpoint, score, momentum and energy correction before the existing acceptance bracket can qualify the step. The evidence records the probe length. Retained-state inconsistency or unclassified execution failure stops preparation; a failed proposal rejects that probe. This remains an experimental preparation option: short chains cannot guarantee global stability. Later warmup windows, candidate measurement and independent verification still require their own numerical checks. It neither adds R-hat admission nor retries an invalid warmup window.

The separate optional `preparation_max_restarts` bounds recovery from rejected nonfinite proposals; its default is zero. Recovery requires finite retained states, values and scores, consistent acceptance decisions and valid declared telemetry. The entire failed preparation attempt is archived and its adaptation statistics discarded. A full new schedule starts from the failed window's independently validated starting state and qualified transform, with fresh streams and reset covariance and dual-averaging statistics. A new probe may only shrink from half the consumed failing step. High acceptance can nominate this conservative preparation step; it does not relax final candidate acceptance or independent verification. A changed metric still requires fresh qualification in its new coordinates. All discarded work and probe work counts toward finite attempt and execution budgets. Retained or shared invalidity and unclassified execution errors remain fatal. This ordinary-preparation option is experimental: three restarts is a development allocation, not a universal robustness guarantee.

Listing 1.2: Binding a local covariance estimate in the target coordinates.

```
"""Bind an accepted local covariance to the ordinary public HMC tuner.

The executable ‘‘main‘‘ uses stubs, so this listing can be checked without
estimating a covariance or launching HMC.
"""

from __future__ import annotations

import sys
from collections.abc import Callable, Mapping, Sequence
from pathlib import Path
from types import SimpleNamespace
from typing import Any

REPOSITORY_ROOT = Path(__file__).resolve().parents[2]
sys.path.insert(0, str(REPOSITORY_ROOT))
```

```
from bayesfilter.inference import (
    HMCKernelTuningConfig,
    estimate_sequential_map_covariance,
    tune_hmc_kernel,
)


def tune_from_fresh_local_covariance(
    *,
    adapter: Any,
    value_and_score_fn: Callable[[Any], tuple[Any, Any]],
    initial_positions: Sequence[Any],
    config: HMCKernelTuningConfig,
    covariance_kwargs: Mapping[str, Any] | None = None,
    covariance_estimator: Callable[..., Any] = estimate_sequential_map_covariance,
    tuner: Callable[..., Any] = tune_hmc_kernel,
) -> Any:
    """Estimate, accept, and bind one center/covariance pair in one coordinate system.
    """

    covariance_result = covariance_estimator(
        value_and_score_fn,
        initial_positions,
        **dict(covariance_kwargs or {}),
    )
    if not covariance_result.accepted:
        raise RuntimeError("local covariance was not accepted")
    if covariance_result.map_candidate is None or covariance_result.covariance is None
    :
        raise RuntimeError("accepted covariance result is incomplete")
    return tuner(
        adapter=adapter,
        initial_position=covariance_result.map_candidate,
        initial_covariance=covariance_result.covariance,
        config=config,
    )


def main() -> Mapping[str, Any]:
    """Exercise only the covariance-to-tuner argument binding."""

    accepted = SimpleNamespace(
        accepted=True,
        map_candidate=(0.25, -0.25),
        covariance=((2.0, 0.1), (0.1, 0.75)),
    )
    observed: dict[str, Any] = {}

    def covariance_stub(*_args: Any, **_kwargs: Any) -> Any:
        return accepted

    def tuner_stub(**kwargs: Any) -> Mapping[str, Any]:
        observed.update(kwargs)
        return {"status": "arguments_bound_without_hmc"}

    result = tune_from_fresh_local_covariance(
        adapter=object(),
        value_and_score_fn=lambda position: (position, position),
```

```
        initial_positions=((0.0, 0.0),),
        config=HMCKernelTuningConfig.smoke(
            target_scope="docs_covariance_first_binding"
        ),
        covariance_estimator=covariance_stub,
        tuner=tuner_stub,
    )
    assert observed["initial_position"] == accepted.map_candidate
    assert observed["initial_covariance"] == accepted.covariance
    assert result["status"] == "arguments_bound_without_hmc"
    return result


if __name__ == "__main__":
    print(main()["status"])
```

`prepare_operational_windowed_mass_handoff` freezes the operational metric and actual post-warmup start bank. The numerical factory `bind_hmc_candidate_set_execution_from_preparation` preserves both its bootstrap and final affine layers. The explicit `bind_hmc_candidate_set_execution` factory instead accepts a frozen mass or supported frozen transport payload. A bound scope records target, data/prior lineage, source dependencies, coordinates, metric, starts, warmup, dtype, backend, and XLA settings. Changing them requires a new scope.

For the proposal-field preparation, four raw-coordinate starts $q_i$ and positive parameter scales $s$ give the initial factor $A = \mathrm{diag}(s)$:

$$c = \frac{1}{4}\sum_{i=1}^{4} q_i, \qquad z_i = (q_i - c)A^{-\mathsf{T}}, \qquad q_i = c + z_i A^{\mathsf{T}}. \tag{1.2}$$

Here starts are represented as row vectors. Caller order is preserved. A diagnostic may explicitly replicate one start; candidate preparation requires its declared four-chain bank. Affine metric windows use $L = 1$ and the configured adaptation counts. Their proposals receive fresh fixed-kernel measurements.

The ordinary preparation config and numerical binding default to GPU/XLA execution. TensorFlow memory growth is configured before device initialization. Small CPU/non-XLA checks are explicit reference or debugging exceptions. Execution policy does not qualify an arbitrary target: the binding still checks its full-chain capability and the consumer must establish value/score fidelity.

## 1.3 Broad search, repairs, and further evidence

The ordinary convenience grid is $L \in \{3, 5, 9, 13, 18, 25\}$, and its legacy preparation config fixes the maximum at 25. The proposal-field config declares its own maximum. Supplied initial, refinement and expansion grids must obey the applicable maximum. An explicit `epsilon_by_l` may supply several proposals at each $L$; `initial_epsilon` is a warm start when no grid is supplied. Every pair is measured independently. A shared numeric proposal at two values of $L$ transfers no acceptance evidence between them. The final-metric epsilon from operational preparation supplies the starting step and the inherited search ceiling. The short probe does not establish a stability limit or guarantee a suitable pair below that ceiling for every $L$. An explicit `candidate_search_bound_expansion_steps=1` widens the exploration cap by the existing repair factor, preserving the final

metric and coordinates. Zero retains the inherited ceiling. Every explored pair still requires its own health checks, measurement and fresh verification. This optional setting is a search hypothesis whose calibration remains separate from its implementation.

Direct preparation bindings expose `preparation_bound_expansion_steps`. They intersect the requested domain with the finite expanded cap, save the original bound and final metric/coordinate identities, and issue a new search identity. Existing scopes and verification receipts are immutable. Preparation failures preserve their stage and available veto causes in `preparation_progress.json` before raising an exception. Nonfinite diagnostic values are marked explicitly in the JSON record, preserving the original failure even when its numerical summaries cannot be finite.

A multiplicative repair can jump from high acceptance directly to nonfinite trajectories. That nonfinite result supplies no valid acceptance direction, so it cannot form the opposing-evidence interval used by refinement. The optional `explore_failed_intervals=True` controller setting allows funded refinement rounds to explore the unvisited geometric interior between a valid directional parent and its nearest numerically rejected same-$L$ child. Each proposal is a new same-family child of the valid parent; both source receipts are recorded, and the failed endpoint stays rejected. This interval is an exploration hypothesis, not a stability bound or an assertion of monotone acceptance. Missing telemetry, unknown invalidity, invalid retained states and shared corruption cannot supply such endpoints. Family, candidate and work limits still apply, and every proposed pair needs measurement and independent verification. The option is disabled by default. An explicit intermediate grid in a fresh scope remains available. Neither procedure exhausts the continuous epsilon domain.

The automatic preparation translation enables one pilot per $L$ and one bounded refinement round. A direct controller config can disable these optional stages for an explicitly supplied grid. A pilot proposes a frozen pair; measurement and fresh verification qualify it. Refinement applies declared epsilon factors and additional $L$ hypotheses to every surviving family. Exact pairs are deduplicated, and all proposals remain subject to candidate and work limits. The reciprocal convenience factors $(0.8, 1.25)$ are hypotheses about useful nearby proposals, not universal numerical optima.

Work proceeds in closed cohorts. Every funded admitted peer receives its reserved stage before a repair child enters the next cohort. A peer whose full required work cannot fit the remaining budget is explicitly deferred, permitting other affordable reserved work to finish. Its evidence allocation is preserved and the search remains incomplete. A first pass does not remove later candidates. Directional evidence creates an immutable child with the same $L$, mass, target, coordinates, and initial bank. The child has a new epsilon and must receive its own measurement and verification. A parent may fail movement because an oversized step rejects nearly everything; that promotion veto does not suppress a supported smaller-epsilon repair. A trajectory alert can propose only declared additional $L$ values, as separate candidates. It never silently changes an existing candidate's $L$.

For the replicated v7 policy, controller policy version 4 closes the initial primary pilot/measurement cohort before dispatching optional measurement extensions or repair descendants. After a candidate is screened, its pending independent verification ladder has priority over those optional descendants. This prevents a long repair branch from starving fresh verification under a wall-time budget; it does not skip a primary peer, grant membership before verification, alter evidence allocation, or rank candidates. Resumed version-3 checkpoints retain their recorded ordering. Versions 1 and 2 remain readable

historical results but cannot resume under version 3 or 4.

The distinction between proposal and evidence matters for fixed-length HMC. For a transition from $x$, the acceptance probability is

$$a(\epsilon, L \mid x) = \mathbb{E}[\min\{1, \exp(-\Delta H)\} \mid x]\,. \tag{1.3}$$

The reported probability mean averages the computed Metropolis probabilities over visited states and sampled momenta. The realized acceptance rate also contains the uniform accept/reject decisions: for each proposal with probability $a$, that indicator has expectation $a$. Both summaries describe acceptance in the explored region; neither measures convergence. For the unit harmonic oscillator, one leapfrog matrix has determinant one and trace $2 - \epsilon^2$. In the stable range its conjugate eigenvalues therefore give the phase

$$\vartheta(\epsilon) = \arccos(1 - \epsilon^2/2), \qquad 0 < \epsilon < 2. \tag{1.4}$$

The trajectory phase is $L\vartheta(\epsilon)$. Near a return phase, acceptance can be high with little net movement. A multiplicative epsilon repair is thus another hypothesis to measure. The controller remembers the latest valid directional evidence for each exact pair and prioritizes unvisited geometric interior points toward nearby opposing observations. It retains this evidence across restart. Finite refinement also investigates unresolved directional intervals when no original family survives. It does not assume monotone acceptance or infer a pass at an unmeasured point. When a proposal crosses the declared domain, its unvisited boundary can be measured before that direction is exhausted.

Inconclusive measurement or verification receives the predeclared `evidence_rungs`; the convenience multipliers are $(1, 2, 4)$. Each rung is an independently seeded run of the unchanged candidate. Unfinished work stays resumable; exhaustion gives `inconclusive_at_cap`. The reported compatibility intervals support an operational screen, not a claim of nominal confidence coverage across repeated looks. A cap does not turn an inconclusive candidate into a verified member.

## 1.4 Acceptance, numerical health, and posterior diagnostics

Acceptance probabilities fluctuate even under a fixed kernel. A temporal block below the practical band and another above it therefore do not by themselves show a change in the chain's acceptance distribution. The default raw crossing screen is a heuristic, and its separate four-chain compatibility interval does not calibrate the crossing decision. Independent random seeds also do not make four differently initialized chain means identically distributed.

For a stationary acceptance sequence $a_{it}$ with an applicable central limit theorem, uncertainty in its mean depends on the long-run variance $\sigma_i^2 = \gamma_i(0) + 2\sum_{k\geq 1}\gamma_i(k)$, where $\gamma_i(k)$ is its lag-$k$ autocovariance. Independence between $m$ chains of length $n$ then gives the approximation

$$\widehat{\mathrm{Var}}(\bar{a}) = \frac{1}{m^2 n}\sum_{i=1}^{m}\widehat{\sigma}_i^2.$$

This follows by adding the variances of the independent chain means and dividing by $m^2$. Batch means and lugsail estimate these long-run variances; the formulas and assumptions

are discussed in the diagnostics chapter. A positive estimate cannot establish adequate batch length, stationarity or sufficient burn-in.

An experimental acceptance diagnostic now preserves the original tuning decision while computing these uncertainty estimates separately. It compares all temporal windows within each chain, so opposing drifts cannot cancel by averaging across chains. It subtracts simultaneous marginal interval endpoints: if both means are covered, their difference is covered regardless of their covariance. A difference interval wholly outside a declared tolerance supports a material discrepancy; an interval wholly inside supports compatibility for that contrast; overlap with the tolerance boundary remains unresolved. The approximate marginal intervals still require calibration, including across the declared evidence looks and candidate search.

The October 1 validation did not support a new default. Under strongly persistent stationary simulations, the tested batch rules underestimated uncertainty and reported false conflicts. Increasing the batch length reduced some conflicts without establishing adequate coverage. The new diagnostic therefore has no tuning-artifact authority. Its finite-start model checks and BGS replays are separate from posterior convergence, and R-hat remains reporting-only during tuning.

### 1.4.1 Experimental fixed-horizon repeated trials

The explicitly selected v7 policy, `HMCReplicatedAcceptancePolicy`, uses the same two public tuners and candidate-set controller. It has not replaced the v5 default or the optional v6 screen. Both the search configuration and the numerical execution configuration must declare the same policy.

For a frozen kernel and labeled start $i$, repetition $r$ discards a fixed prefix of $W$ transitions and measures the following $T$ acceptance probabilities. Define

$$Y_{ri} = \frac{1}{T} \sum_{t=W+1}^{W+T} \min\{1, \exp(\ell_{rit})\}, \qquad \mu_i = \mathbb{E}[Y_{ri} \mid q_{i0}, \text{frozen kernel}],$$

where $\ell_{rit}$ is the log Metropolis acceptance ratio. Every repetition restarts the complete frozen start bank with a fresh recorded random stream. Thus the independent observations are whole start-bank repetitions; the four starts need not have equal $\mu_i$, and their scores within a repetition need not be independent. The bounded scores estimate finite-horizon acceptance from the declared starts. They do not establish a stationary acceptance rate or adequate burn-in.

In this policy, evidence-rung multipliers increase the number of repetitions while keeping $W$ and $T$ fixed. New trials append to existing evidence; chunks within a trial continue its state. All $T$ measurements count, including the remainder when temporal windows have unequal lengths. This differs from the legacy draw-length extension described above.

Qualification requires simultaneous containment of every start's expectation interval within the declared broad qualification region. The narrower preferred region supplies supported directions for epsilon repair from the weighted mean. A pooled mean cannot conceal a start outside qualification. A supported start violation requests preparation investigation; insufficient information remains inconclusive. Independent verification uses fresh streams and a separately declared family error allocation. The search and verification levels are separate statements, not one combined error level.

Numerical health still determines whether a candidate can be retained. Temporal contrasts investigate preparation and R-hat remains reporting-only; neither is an acceptance-

admission requirement. Every verified member is retained. Reconstruction checks raw trials, endpoints, seeds and the cost of each attempted native chunk. A lost native call can be retried under the same stream and charged again, but it supplies at most one completed trial score.

The explicit release-validation profile sets `min_normalized_return_displacement=0.0`, so that first-to-last distance is reporting-only. Two close endpoints can occur by chance along a moving path; they alone do not establish immobility. Adjacent movement, repeated states, persistent short-cycle returns, divergence and finite state/target checks remain active, including discarded-prefix health. Regression checks preserve the original failed Gaussian trial and frozen-path and cycle controls. The explicit profile has a distinct policy identity and needs fresh tuning. Legacy defaults and historical outcomes retain their original meaning, and these health checks do not certify mixing or posterior convergence.

Detailed trial health remains in the numerical records; controller observations summarize its counts without discarding those records. Intermediate checkpoint writes reuse unchanged persisted evidence. Single-trial work boundaries, all dispatch exits, explicit saves, export and reload retain full validation. Internal batched dispatch may reuse unchanged history between work items; failed and paused exits still validate it fully. Attempted work is recorded before the native call, including a call whose result is lost.

The float64 interval implementation uses a conservative arithmetic guard. Independent high-precision references check selected endpoint and threshold-adjacent cases at 64, 1,024 and 16,384 repetitions on CPU/XLA and GPU/XLA. These finite checks do not constitute an error theorem for all inputs or hardware, or establish full-search delivery and affordability.

The framework-free preflight computes required work before evaluating the target. Actual state-space pricing must then establish whether the fixed-trial allocation is affordable and useful. Known-law calibration, numerical recovery, model delivery, GPU parity and independent posterior assessment answer different questions. Passing the first two cannot by itself establish the remaining three or justify a new default.

### 1.4.2 Experimental randomized HMC and microcanonical transitions

An already prepared exact-score binding may now specify `HMCCandidateExecutionConfig.transition_profile` using `HMCTransitionProfile`. The supported experimental families are ordinary HMC with leapfrog integration and fully refreshed MAMS with isokinetic velocity-Verlet integration. Both enter the same public tuners and shared candidate-set controller. A position-field binding is insufficient: these profiles require the score of the declared scalar target. MAMS requires dimension greater than one and identity geometry in the prepared coordinates. Automatic preparation configurations reject these profiles. The caller first obtains a frozen geometry, declares integrator-specific epsilon hypotheses and uses `HMCControllerConfig` with its numerical binding. An HMC warm-up bound cannot silently become a bound for the MAMS integrator.

For the `uniform_duration` option, a candidate records a positive step size $\epsilon$ and a maximum count $L_{\max}$. Each chain and transition draws an independent $U \sim \mathrm{Uniform}[0, 1)$ and uses

$$\tau_{\max} = \epsilon L_{\max}, \qquad L = \max\{1, \lceil U L_{\max} \rceil\}.$$

Thus the legacy candidate field `leapfrog_steps` is a maximum count for this profile. The complete `FrozenTransitionSpec` identifies the law, integrator, correction and full refreshment. A repaired $\epsilon$ changes $\tau_{\max}$ as well and therefore defines a new candidate. The initial implementation does not independently parameterize an arbitrary continuous maximum duration. The `fixed` option retains a deterministic count for matched comparisons.

For MAMS, each outer transition independently refreshes a uniform sphere direction. Its symmetric force-half-step, position-step, force-half-step composition uses the sphere-Jacobian correction derived in Chapter 2; an ordinary kinetic-energy correction would be wrong for this transition. Full refreshment makes positions sufficient at outer-transition boundaries. Momentum or direction, duration and the Metropolis decision have separate stateless streams. The numerical runner supports same-seed continuation with a recorded transition offset; retained archives use the existing fresh-chunk stream and predecessor checks. Persistent and within-trajectory stochastic MAMS, LAPS and MALT are not implemented by this profile.

The optional `propose_trajectory_candidates` helper runs a discarded ChEES-rate or SNAPER pilot with explicit `TrajectoryPilotConfig` settings. The TFP 0.25 adaptation fixes metric variance to one in the prepared coordinates and disables Halton jitter. It retains the source's shared IID pilot jitter; the frozen candidate instead has independent chain jitter. The returned `epsilon_by_l` supplies hypotheses to the same binding's shared search. Neither the pilot objective nor its draws qualify a candidate. Fresh measurement and independent verification remain necessary.

For full-refresh MAMS, the optional helper is `propose_mams_trajectory_candidates`. Its explicit `MAMSPilotConfig` separates the physical travel request from integrator resolution. For the existing integer-uniform count $N \in \{1, \dots, K\}$,

$$E[\epsilon N] = \epsilon \frac{K+1}{2}, \qquad E[(\epsilon N)^2] = \epsilon^2 \frac{(K+1)(2K+1)}{6}.$$

Given a requested mean $\ell \geq \epsilon$, `compile_mean_duration` rounds $2\ell/\epsilon - 1$ to the nearest integer, with ties upward. Without an active count bound, the mean differs from the request by at most $\epsilon/4$. A request below one step or beyond the count support fails explicitly. The helper records the realized moments and quantization; it does not alter the frozen transition law.

The proposer uses bounded geometric epsilon bracketing, followed by settling and separately diagnosed continuous windows at a frozen policy. Only these windows supply autocorrelations. Changing-policy draws and reset-start verification trials are never concatenated for that calculation. The paper's harmonic-coordinate length rule is a proposal hypothesis, supplemented by a declared duration neighborhood and a separately labeled slow-observable extension. Squared coordinates, radial quantities and frozen projections guard against judging exploration from means alone. Constant, nonpositive or lag-limited estimates remain unavailable. These diagnostics have no posterior convergence or tuning-membership authority.

For typed uniform-duration profiles, `repair_trajectory_policy="preserve_mean_duration"` explicitly requests count recompilation when epsilon changes. The first repair adopts its primary candidate's realized mean, and every descendant preserves that one request rather than successively rounding the rounded child mean. Configuration v3 and repair-action v2 record the request and parent/child counts; earlier configurations retain their fixed-count semantics. Every child receives fresh measurement and verification. This option cannot be combined with legacy same-count pilot/refinement stages.

Pilot progress and completed numerical windows can be recorded in a fresh directory and replayed into another fresh directory under the identical source, configuration, binding and backend. Interrupted attempts remain charged. The pilot checks wall allocation at window boundaries; serious runs therefore need measured window prices and a workload-aware supervisor. Engineering regressions establish these mechanics, not global exploration or efficiency.

These profiles require batched chain execution, one independent trial per native call, and the explicit v8 replicated policy `qualification_mode="lower_bound"`. For a declared acceptance floor $a_{\min}$, its qualification region is $[a_{\min}, 1]$. Every start's simultaneous lower bound must reach $a_{\min}$. Qualification precedes preferred-region repair, so supported high acceptance does not by itself reject a candidate or force its step size upward. Numerical health, movement and recurrence checks still apply. R-hat, posterior ESS and MCSE cannot gate, rank, repair or delay tuning. Fixed HMC may use this same policy when comparing kernels under matched evidence requirements.

The new v8 policy and typed transition identities preserve the meaning of legacy artifacts. Work preflight and reservations include maximum value/score and separate status evaluations, while execution reports actual calls and realized counts, including inactive rows evaluated by batched kernels. Checked mechanics archives can export and replay every verified member. The research consumer `modern_tuning_consumer.run_frozen_trajectory_search` records each primary policy, each repair descendant and every primary policy left uncreated by a resource stop. It uses the binding's declared epsilon domain; an upper bound of one is not imposed on MAMS. Candidate ancestry determines a repair child's identity even when another family contains the same numerical pair. Every verified identity is exported and checked on reload. Before expensive target preparation, the full-model consumer also checks whether the scheduled independent trials can possibly qualify a candidate. It evaluates the actual confidence procedure at the most favorable bounded trial scores, separately for measurement and verification, with the full candidate error allocation. An impossible design stops before sampling. Passing this check does not establish statistical power; for a two-sided rule, the separate endpoint bounds are only necessary conditions for joint qualification. The public tuners retain their original qualification rules.

For the full-data predator–prey, SIR and structural-filter experiments, `scripts/run_modern_hmc_frozen_full_model.py` reads an explicit design containing the transition, search, geometry, repair allowance and optional posterior settings. It dispatches to the same public tuner. Posterior assessment visits every verified member and reports finite-filter reference agreement separately, using the declared precision estimator. A member that fails a posterior check remains tuning-qualified. An interrupted run retains completed outcomes and records unfinished members; it supplies no missing verification or convergence evidence. These consumers do not establish that the proposed search covers all useful durations or that an adaptation method improves posterior delivery.

Stateless seeds do not by themselves guarantee identical trajectories across TensorFlow compilation modes. In the current diagnostic environment, CPU/XLA and GPU/XLA share the tested random streams, while non-XLA execution produces different draws at the same seed. Numerical parity comparisons therefore need matched random inputs or matching compiled modes. Backend and compilation mode remain part of the recorded execution identity.

The claim-bearing retained builder rejects these experimental profiles pending statistical, posterior and native GPU qualification. Short nonlinear runs and CPU regressions

establish integration properties, not posterior accuracy, efficiency or release readiness.

### 1.4.3 Legacy compatibility and shared numerical health

The default `HMCAcceptancePolicy` uses four independent chain means, temporal blocks, and a 90% compatibility interval. Its first allocation requires at least four blocks of sixteen decisions per chain. After health, chain-conflict, temporal-conflict and trajectory checks, acceptance passes when the compatibility interval overlaps the practical band, lies inside the repair band, and every chain mean lies inside the repair band. For example, mean 0.76 and interval $[0.73, 0.79]$ can pass with practical band $[0.65, 0.75]$ and repair band $[0.55, 0.85]$. The rule tests compatibility; it does not rank candidates by closeness to 0.70. Acceptance, movement and numerical health have distinct roles. Finite accepted/proposed states and targets, endpoint score finiteness, momentum, Metropolis state consistency and declared target status determine whether the numerical evidence is usable. Candidate-local failure rejects that pair; shared execution corruption stops the scope and disables all its replay. A declared target-domain exception preserves the attempted seeds and failure without inventing an acceptance value, so other candidates can continue. Shared invalidity preserves the history of earlier verification while revoking current replay. Released reservations cause affordable deferred work to be reconsidered within the same invocation. Available native divergence vetoes promotion of the current pair. When the acceptance evidence is valid and supports a directional repair, that veto does not prevent a newly measured child. Acceptance compatibility and promotion eligibility are separate typed fields, so a rejected pair's in-band acceptance receipt remains readable alongside its verified peers. Both routes check numerical health over discarded warmup as well. The proposal-field route checks finite states, log acceptance and energy errors before excluding warmup from acceptance statistics.

$\widehat{R}$ is reporting-only during tuning. High, missing, nonfinite or failed-to-compute values cannot reject, rank, repair or delay a candidate. ESS and short-chain timing are descriptive diagnostics. They do not silently replace the acceptance/health screen or discard a viable member. Posterior reports identify the sample window actually assessed, its declared requirements, each check and each failed check. This distinguishes incompatible chains, inadequate bulk or tail information, missing binary outcomes, unmet precision and an additional consumer veto. Window counts are separate from cumulative burn-in counts. The sequential controller enforces consecutive warmup passes; a passing window alone does not satisfy a multiple-check policy. These explanatory fields do not change any admission or stopping threshold.

Posterior warmup and cumulative model-coordinate $\widehat{R}$/ESS assessment remain separate requirements for inference.

## 1.5 A frozen nonlinear transport

Let the base log density in model coordinates be $\ell_\theta(\theta)$, and let one frozen differentiable bijection map latent coordinates into those model coordinates,

$$\theta = T(z), \qquad J_T(z) = \frac{\partial T(z)}{\partial z}.$$

The fixed-transport HMC target is

$$\ell_z(z) = \ell_\theta(T(z)) + \log \left|\det J_T(z)\right|, \tag{1.5}$$

and its matching score is

$$\nabla_z \ell_z(z) = J_T(z)^\mathsf{T} \nabla_\theta \ell_\theta(T(z)) + \nabla_z \log \left|\det J_T(z)\right|. \tag{1.6}$$

Dropping either Jacobian term changes the target or its derivative. Such a calculation is wrong relative to the transformed-target claim.

The fixed-transport preparation reconstructs a supported diagonal-affine, legacy dense-IAF or configured IAF/conditional-DSF map from `frozen_transport_payload`, checks its live identity, and uses identity mass in latent coordinates. Starts are supplied in those coordinates. Every proposed $(L, \epsilon)$ then enters the same measurement, repair and fresh-verification procedure. Arbitrary transport callbacks without a supported reconstruction are rejected. Changing a map changes the scope; receipts or chain states from different maps are never pooled.

Positive funnel tuning tests assume a supplied frozen whitening or partially whitening map. Learning a useful map is a separate preparation problem. For the model $v \sim \mathcal{N}(0, \sigma^2)$ and $x_i \mid v \sim \mathcal{N}(0, e^v)$ with two children, the exact map $v = \sigma z_0$, $x_i = e^{v/2} z_i$ has log determinant $\log \sigma + v$. Substitution in the transformed density cancels both conditional scale terms and gives three independent standard normals. This provides an algebraic control without assuming that a learner discovers the map. A partial map $x_i = e^{s(v)} z_i$ instead leaves conditional standard deviation $e^{v/2 - s(v)}$ in the latent coordinates; it tests tuning with known residual curvature.

Bounding $s(v)$ alone does not bound the residual stiffness: if $s$ remains bounded as $v \to -\infty$, the latent conditional precision $e^{2s(v)-v}$ diverges. A useful diagnostic partial-whitening family instead sets $s(v) = v/2 + \delta(v)$ with $\delta(v) = a \tanh(v/(2\sigma))$. For $S = z_1^2 + z_2^2$, its log determinant is $\log \sigma + v + 2\delta(v)$, and substitution in the normalized funnel density yields

$$U(z) = \tfrac{1}{2} z_0^2 + \tfrac{1}{2} e^{2\delta(\sigma z_0)} S - 2\delta(\sigma z_0) + \tfrac{3}{2} \log(2\pi).$$

The child-coordinate curvature lies in $[e^{-2a}, e^{2a}]$ for $a \geq 0$; the zero-amplitude limit recovers the exact control. The entire Hessian is still unbounded because derivatives with respect to $z_0$ contain $S$. The validation fixture uses $\sigma = 3$ and $a = 1/2$ as an explicit development hypothesis, not a learned-map or sampling default. Even the exact latent Gaussian control can require many draws for precise means of the model's heavy-tailed children.

The untransformed centered funnel remains a stress test for bounded search and honest numerical rejection. An empty candidate set there does not by itself establish a tuning defect. For a supplied map, tuning searches the declared step sizes and trajectory lengths in the frozen coordinates and independently verifies every survivor. A finite search need not succeed for every imperfect map. Comparisons at common model starts must apply the inverse chart and inverse map before passing latent starts to the tuner, then check the forward roundtrip. Posterior precision and exploration in the original model remain separate from candidate qualification.

Listing 1.3: Fixed-transport candidate-set tuning.

```
"""Small CPU/reference candidate-set example; no posterior or ranking claim."""
from __future__ import annotations
import os
import sys
from pathlib import Path
os.environ.setdefault("CUDA_VISIBLE_DEVICES", "-1")
os.environ.setdefault("TF_FORCE_GPU_ALLOW_GROWTH", "true")
```

```
sys.path.insert(0, str(Path(__file__).resolve().parents[2]))
import tensorflow as tf
from bayesfilter.inference import (
    HMCControllerConfig, HMCCandidateExecutionConfig, HMCAcceptancePolicy,
    HMCKernelTuningConfig, FixedTransportHMCKernelTuningConfig,
    ValueScoreCapability, tune_hmc_kernel, tune_fixed_transport_hmc_kernel,
)

class AnalyticGaussianAdapter:
    parameter_dim = 2
    def adapter_signature(self):
        return "docs-candidate-set-gaussian-v1"
    def value_score_capability(self):
        return ValueScoreCapability(value_score_authority="graph_native",
    xla_hmc_ready=False,
            runtime_backend="tensorflow", target_scope="docs_candidate_set",
            evidence_path=__file__, nonclaims=("small CPU reference fixture",))
    def log_prob_and_grad(self, position):
        position = tf.convert_to_tensor(position, tf.float64)
        return -.5 * tf.reduce_sum(position**2, axis=-1), -position

adapter = AnalyticGaussianAdapter()
starts = tf.constant([[-1.,-.5],[-.3,.2],[.4,-.2],[1.,.5]], tf.float64)
# This reduced grid bounds the documentation smoke; the ordinary convenience
# policy uses L=(3,5,9,13,18,25) and optional pilots/refinement.
search = HMCControllerConfig(primary_l_grid=(3,5),
    epsilon_by_l=((3,(1.1,1.3)),(5,(1.1,1.3))), evidence_rungs=(1,))
execution = HMCCandidateExecutionConfig(measurement_num_results=64,
    verification_num_results=64, num_warmup_steps=8, seed=(20260914,71),
    acceptance_policy=HMCAcceptancePolicy(), target_status_trace_policy="none",
    use_xla=False, non_xla_reason="small documentation CPU reference")

from bayesfilter.inference.neutra_artifacts import load_frozen_neutra_artifact
frozen = {"schema":"bayesfilter.neutra.frozen_affine_diag.v1",
    "transport_id":"docs-identity", "dimension":2,
    "target_signature":adapter.adapter_signature(), "log_jacobian_available":True,
    "shift":[0.,0.], "raw_scale":[0.,0.]}
loaded = load_frozen_neutra_artifact(frozen, expected_target_signature=adapter.
    adapter_signature())
run = tune_fixed_transport_hmc_kernel(base_adapter=adapter, fixed_transport=loaded.
    transport,
    frozen_transport_payload=frozen, initial_position=starts,
    config=FixedTransportHMCKernelTuningConfig(initial_step_size=1.3,
        leapfrog_grid=(3,5), use_xla=False, target_scope="docs_candidate_set"),
    search_config=search, execution_config=execution,
    target_lineage={"model":"standard Gaussian", "data":"none", "prior":"standard
    normal"},
    source_paths=[__file__])
assert run.result.payload()["nominee_id"] is None
assert len(run.result.candidates) >= 4
print(run.result.completion_status, run.result.verified_candidate_ids)
```

## 1.6 Typed deterministic proposal-field mechanics

`bind_neural_force_hmc_tuning_runner` binds a declared position-only field to its coordinate-consistent endpoint potential. The public facade accepts that binding with `TensorFlowHMCKernelTunin`

Affine preparation remains specific to this transition, while the shared controller schedules its candidate measurements, repairs and verification. The result is conditional mechanics evidence and cannot become an exact-score numerical retained member. The endpoint function still needs target-specific validation.

The typed objects use potential, not log-density, signs. If $\ell(\theta) = \log \pi(\theta)$, the exact endpoint object is

$$U(\theta) = -\ell(\theta). \tag{1.7}$$

For the scalar-potential-gradient semantic, the proposal force is $F(\theta) = \nabla U(\theta) = -\nabla \ell(\theta)$. The kick uses

$$p_{n+1/2} = p_n - \frac{\epsilon}{2} F(q_n), \qquad q_{n+1} = q_n + \epsilon M^{-1} p_{n+1/2}, \tag{1.8}$$

followed by the symmetric terminal kick and momentum flip.

The v2 binding also permits an honestly labeled deterministic position-only proposal field $B(q)$ that need not equal $\nabla U(q)$. Position-only kicks remain volume-preserving shears, and the symmetric schedule remains reversible. The exact endpoint Metropolis rule is

$$\alpha = \min\left\{1, \exp\left[-U(q') - K(p') + U(q) + K(p)\right]\right\}. \tag{1.9}$$

Under these conditions, the endpoint potential determines the invariant target; a poor or sign-reversed proposal field can destroy efficiency without by itself changing that target. A wrong endpoint potential, missing coordinate change, momentum-dependent field, asymmetric schedule, or direct state update invalidates that conclusion.

For an affine mass coordinate $\theta = c + Fz$, the repository wrapper applies the chain rule to the proposal field and endpoint potential. The affine log-Jacobian is constant and is deliberately omitted from energy differences under the declared `constant_omitted` convention. Both raw objects and that convention are bound into runner schema `bayesfilter.hmc_tuning_runner_binding.v2`.

Listing 1.4: Typed proposal field and exact endpoint-potential binding.

```
"""Construct the typed deterministic-field tuning inputs without launching HMC."""

from __future__ import annotations

import sys
from collections.abc import Mapping
from pathlib import Path
from typing import Any

REPOSITORY_ROOT = Path(__file__).resolve().parents[2]
sys.path.insert(0, str(REPOSITORY_ROOT))

import tensorflow as tf

from bayesfilter.inference import (
    DETERMINISTIC_POSITION_ONLY_PROPOSAL_FIELD_SEMANTICS,
    FourChainMeanBandAcceptancePolicy,
    FrozenPositionOnlyForce,
    FrozenTargetPotential,
    TensorFlowHMCKernelTuningConfig,
    bind_neural_force_hmc_tuning_runner,
    tune_hmc_kernel,
)
```

```
def build_binding() -> Any:
    """Bind a deterministic non-gradient proposal field to an exact potential."""

    def proposal_field(position: tf.Tensor) -> tf.Tensor:
        position = tf.convert_to_tensor(position, tf.float64)
        return tf.stack(
            (
                position[..., 0] + 0.2 * position[..., 1],
                0.4 * position[..., 1],
            ),
            axis=-1,
        )

    def exact_endpoint_potential(position: tf.Tensor) -> tf.Tensor:
        position = tf.convert_to_tensor(position, tf.float64)
        return 0.5 * tf.reduce_sum(tf.square(position), axis=-1)

    return bind_neural_force_hmc_tuning_runner(
        force=FrozenPositionOnlyForce(
            function=proposal_field,
            identity="docs-deterministic-proposal-field-v1",
            semantics=DETERMINISTIC_POSITION_ONLY_PROPOSAL_FIELD_SEMANTICS,
            coordinate_system="raw",
        ),
        target=FrozenTargetPotential(
            function=exact_endpoint_potential,
            identity="docs-exact-endpoint-potential-v1",
            coordinate_system="raw",
        ),
        target_scope="docs_neural_force_binding",
    )


def build_mechanics_config() -> TensorFlowHMCKernelTuningConfig:
    """Declare an intentionally tiny, non-authoritative documentation budget."""

    return TensorFlowHMCKernelTuningConfig(
        parameter_dimension=2,
        evidence_role="diagnostic_only",
        mass_window_results=(1,),
        step_adaptation_results=64,
        verification_results=64,
        max_leapfrog_steps=5,
        initial_step_size=0.1,
        budget_provenance="bounded documentation mechanics diagnostic",
        initial_step_size_provenance="documentation convenience hypothesis",
        geometry_provenance="unit-scale documentation diagnostic",
        target_scope="docs_neural_force_binding",
        acceptance_policy=FourChainMeanBandAcceptancePolicy(
            overall_band=(0.65, 0.75),
            per_chain_band=(0.55, 0.85),
        ),
        target_accept_prob=0.70,
        verification_repair_rounds=0,
        step_repair_factor=2.0,
        mass_shrinkage=0.10,
        covariance_jitter=1.0e-9,
```

```
        eigenvalue_floor=1.0e-9,
        max_condition_number=1.0e8,
        seed=(20260905, 1),
    )


def tune_deterministic_field(
    *,
    adapter: Any,
    initial_position: Any,
    parameter_scales: Any,
    output_dir: str | Path,
) -> Any:
    """Show the only valid public dispatch for this mechanics-only branch."""

    return tune_hmc_kernel(
        adapter=adapter,
        initial_position=initial_position,
        parameter_scales=parameter_scales,
        config=build_mechanics_config(),
        runner_binding=build_binding(),
        output_dir=output_dir,
    )


def main() -> Mapping[str, Any]:
    """Return both typed payloads; no tuner or chain runner is called."""

    binding_payload = build_binding().payload()
    config_payload = build_mechanics_config().payload()
    assert binding_payload["artifact_authority"] is False
    assert binding_payload["force_semantics"] == (
        DETERMINISTIC_POSITION_ONLY_PROPOSAL_FIELD_SEMANTICS
    )
    assert binding_payload["endpoint_target_coordinate_system"] == "raw"
    assert config_payload["artifact_authority"] is False
    assert config_payload["trajectory_candidate_policy"] == (
        "powers_of_two_then_explicit_cap"
    )
    return {"binding": binding_payload, "config": config_payload}


if __name__ == "__main__":
    print(main()["binding"]["binding_hash"])
```

## 1.7 Persistence, resource limits, and replay

`verified_candidate_ids` retains every verified member with no implicit nominee. `viable_candidate_ids` also includes validating and terminally inconclusive candidates; their individual states must be inspected. A verified member remains usable when another candidate exhausts its budget, unless the scope itself is invalid. Descriptive differences among members do not establish a statistically supported ranking.

A fresh output directory preserves `execution_spec.json`, immutable numerical evidence and chunk files, and the atomic `tuning_checkpoint.json`. All observations, in-

cluding failed measurements and zero-verified searches, are retained. Complete results use `candidate_set_result.json` and are never overwritten. Resource failures pause with their reason; unexpected exceptions preserve state and propagate. Optional diagnostic errors cannot erase completed numerical evidence. Malformed provider data is recorded as an execution error before it enters the observation history; retry retains the attempted cost. Valid shared-failure observations remain recorded. If the same live binding later records shared invalidity, even a member verified earlier becomes unavailable for replay.

`resume_hmc_candidate_set_tuning` reconstructs the same geometry, policy, evidence, candidates and work queue using the original target. It checks source and target drift, tensor checksums, numerical versions, and scope identity. Position-field mechanics separately preserve preparation and observations for `resume_position_field_candidate_tuning`, which also requires the original runner binding. Their `controller_checkpoint.json` also references immutable completed chunks, permitting restart within a longer evidence stage. Controller-only reload cannot create missing numerical evidence.

Before frozen execution exists, `preparation_progress.json` records preparation phases, elapsed time and failures. Ordinary preparation checks time at available progress boundaries; proposal-field affine preparation is one compiled graph checked before and after its call. Failed preparation must be retried in a fresh directory. Numerical resume requires a frozen scope with the same source and controller policy; historical results remain readable when those identities change, but cannot be silently resumed under new rules.

Call budgets count attempted work. Candidate reservations protect mandatory stages; extensions and retries can use remaining free budget. A separate transition-times-$(L+1)$ estimate accounts conservatively for gradient work. The scheduler quotes unfinished work, and each chunk is charged and checkpointed before its native call. Resume does not recharge saved completed chunks; an interrupted native call remains conservatively charged even when it returned no usable evidence. Elapsed time includes recorded preparation. Numerical chunks are bounded by `chunk_max_results`, whose convenience default is 256. Time limits are checked between chunks; Python cannot preempt a native operation or compilation. A hard process limit must be applied externally. When preparation and search time limits are both supplied, the smaller limit applies. The first-call timing includes any tracing/compilation rather than pretending that it measures steady-state kernel cost.

For all-member handoff, `export_hmc_candidate_retained_runners` exports every verified ID and `load_hmc_candidate_retained_runners` reloads those paths. The existing compact member files share one evidence bundle. Common evidence is checked once per call; every member's identity and verified endpoint is checked separately. Individual-member loading remains compatible. Later runs and exports revalidate the binding, so grouped loading creates no persistent trust cache. Partial delivery cannot price a complete search. Native finite-value and Metropolis-state checks use a compiled reduction and transfer their Boolean results together. Diagnostic vectors are materialized once at the reporting boundary. The health rules and serialized values are preserved; saved-trace parity does not establish full-search affordability.

For the v7 policy, `HMCCandidateExecutionConfig.replicated_trial_batch_size` optionally groups independent repetitions in one exact-score batched call. The default is one and the implementation bounds the group at 32. Each complete discarded-prefix-plus-measured path must fit within one chunk. The target evaluates the combined rows, while every repetition preserves its original TFP stateless seed schedule, momentum, Metropolis randomness and health/score record. Horizons, repetition counts and candidate identities

do not merge. All attempted repetitions are charged before execution; a lost call consumes budget and retries the same streams. Charges are saved as a group before the call and returned rows together afterwards; a serialization failure saves its completed prefix before propagating. Batch size is bound into the execution identity, so old checkpoints cannot resume under changed settings. Serial, threaded and position-field routes reject this option. Native numerical parity, durable recovery and complete-search affordability require separate checks; batching does not by itself establish a supported v7 release.

Select a member by explicit ID and supply its result and numerical binding to `build_retained_bound_hmc_archive_runner_from_candidate_set_result`. The two frozen-kernel variants share these checks, with the claim-eligible variant additionally requiring actual GPU/XLA evidence. A child's own passing verification is required; its failed ancestor need not pass. Export with `runner.export(path)` and reconstruct with `load_hmc_candidate_retained_runner(path,adapter=original_target)`. Fresh retained blocks continue from the verified or preceding endpoint with frozen settings and fresh seeds. Freshness checks include every tuning stage and chunk, attempted partial chunks and predecessor retained blocks. Attempted seeds are reconstructed from the durable work/accounting records after member export and reload, including failed native calls that returned no trace. Tuning draws remain excluded. `runner.run_sequential` connects the explicitly chosen member to the shared cumulative posterior controller, preserving the member's transition, chain topology and target-status requirements. The generic entry point is `run_hmc_posterior(member=runner,config=config)`. Diagnostics use model coordinates and any declared scientific functionals. Posterior rejection does not change tuning membership. Chapter **??** distinguishes adaptation, discarded equilibration and retained precision.

The status `complete` means the declared search reached terminal dispositions; its verified set may be empty. It does not prove the absence of a useful kernel. The recorded limiting reason distinguishes the candidate cap, the family repair count, the epsilon domain and exhaustion of representable unvisited proposals. At a refinement cap, the remaining count includes only eligible pairs not yet admitted in that round. Reporting collections resolve explicit scope/search/member identities and require every included search to complete, independently of their ordering.

Public option conflicts are rejected before numerical preparation. An issued binding rejects redundant search, execution, provenance or transport overrides. Automatic routes may accept the shared search and execution configs. For the proposal-field route, legacy `step_adaptation_results` supplies the fixed-pair pilot count, while `verification_results` supplies measurement and verification counts. The shared execution configuration can replace these allocations and supply an explicit `pilot_num_results`. Pilots never qualify a pair. The legacy adaptive selector remains historical. Config payloads label their legacy candidate-policy fields as historical helper metadata; the active result's shared configs determine candidate stages. Proposal-field execution defaults to XLA; a diagnostic exception supplies `non_xla_reason`.

Exact-score TFP bindings optionally share a compiled runner across leapfrog counts at each chunk size with `execution_config.reuse_leapfrog_graphs=True`; the default is false. Each cache belongs to one frozen target and geometry with fixed chain topology, trace policy and backend. Independent scalar chains keep their original seed folding. The complete execution artifact records this option and restart preserves it. Within the same source snapshot it does not change numerical seeds; a source change still changes scope identity and can change seeds. The proposal-field route rejects the option. Reuse can

reduce first-call compilation while increasing warmed execution cost, so assess its cost on complete fits. Candidate qualification and posterior diagnostics are unchanged.

Retained-member exports share one checksummed evidence bundle beside their compact member files. Copy the bundle when moving the members, or use `runner.export(path, portable=True)` for a standalone file. Numerical analyses are cached by current evidence content; mutations and changed source identities still invalidate reuse. Predecessor history is checked iteratively. These mechanics tests do not qualify target-scale memory, restart latency, posterior convergence, or scientific validity. Consumer evidence must also establish complete target/data/prior identity and any intermediate integrator telemetry required by that target.

Listing 1.5: Ordinary candidate-set invocation.

```
"""Small CPU/reference candidate-set example; no posterior or ranking claim."""
from __future__ import annotations
import os
import sys
from pathlib import Path
os.environ.setdefault("CUDA_VISIBLE_DEVICES", "-1")
os.environ.setdefault("TF_FORCE_GPU_ALLOW_GROWTH", "true")
sys.path.insert(0, str(Path(__file__).resolve().parents[2]))
import tensorflow as tf
from bayesfilter.inference import (
    HMCControllerConfig, HMCCandidateExecutionConfig, HMCAcceptancePolicy,
    HMCKernelTuningConfig, FixedTransportHMCKernelTuningConfig,
    ValueScoreCapability, tune_hmc_kernel, tune_fixed_transport_hmc_kernel,
)

class AnalyticGaussianAdapter:
    parameter_dim = 2
    def adapter_signature(self):
        return "docs-candidate-set-gaussian-v1"
    def value_score_capability(self):
        return ValueScoreCapability(value_score_authority="graph_native",
    xla_hmc_ready=False,
            runtime_backend="tensorflow", target_scope="docs_candidate_set",
            evidence_path=__file__, nonclaims=("small CPU reference fixture",))
    def log_prob_and_grad(self, position):
        position = tf.convert_to_tensor(position, tf.float64)
        return -.5 * tf.reduce_sum(position**2, axis=-1), -position

adapter = AnalyticGaussianAdapter()
starts = tf.constant([[-1.,-.5],[-.3,.2],[.4,-.2],[1.,.5]], tf.float64)
# This reduced grid bounds the documentation smoke; the ordinary convenience
# policy uses L=(3,5,9,13,18,25) and optional pilots/refinement.
search = HMCControllerConfig(primary_l_grid=(3,5),
    epsilon_by_l=((3,(1.1,1.3)),(5,(1.1,1.3))), evidence_rungs=(1,))
execution = HMCCandidateExecutionConfig(measurement_num_results=64,
    verification_num_results=64, num_warmup_steps=8, seed=(20260914,71),
    acceptance_policy=HMCAcceptancePolicy(), target_status_trace_policy="none",
    use_xla=False, non_xla_reason="small documentation CPU reference")

run = tune_hmc_kernel(adapter=adapter, initial_position=starts,
    config=HMCKernelTuningConfig.smoke(mass_policy="fixed_identity",
        target_scope="docs_candidate_set"),
    search_config=search, execution_config=execution,
    target_lineage={"model":"standard Gaussian", "data":"none", "prior":"standard
```

```
    normal"},
    source_paths=[__file__])
assert run.result.payload()["nominee_id"] is None
assert len(run.result.candidates) >= 4
print(run.result.completion_status, run.result.verified_candidate_ids)
```

Listing 1.6: Numerical candidate evidence and explicit retained-member replay.

```
"""Numerical candidate-set/reload smoke; engineering evidence, not a posterior.

GPU/XLA is the default. --cpu-reference is a deliberate CPU/non-XLA exception.
The small Gaussian grid, widened acceptance band, and draw counts are fixture
choices that bound test cost; consumers must specify their own tuning policy.
"""
from __future__ import annotations

import argparse
import json
import os
from pathlib import Path
import subprocess
import sys
import time

REPO_ROOT = Path(__file__).resolve().parents[2]
sys.path.insert(0, str(REPO_ROOT))


class GaussianTarget:
    parameter_dim = 2

    def adapter_signature(self):
        return "docs-candidate-retained-gaussian-v1"

    def value_score_capability(self):
        from bayesfilter.inference import ValueScoreCapability
        return ValueScoreCapability(value_score_authority="graph_native",
            xla_hmc_ready=True, full_chain_xla_diagnostic_ready=True,
            target_scope="docs_candidate_retained", runtime_backend="tensorflow",
            evidence_path=__file__, nonclaims=("known Gaussian mechanics fixture",))

    def log_prob_and_grad(self, position):
        import tensorflow as tf
        position = tf.convert_to_tensor(position, tf.float64)
        return -0.5 * tf.reduce_sum(position * position, axis=-1), -position

    def target_status_telemetry(self, position):
        import tensorflow as tf
        shape = tf.shape(position)[:-1]
        return {"status_code": tf.zeros(shape, tf.int32),
            "valid_pre_regularized_score": tf.ones(shape, tf.bool),
            "floor_count_value": tf.zeros(shape, tf.int32)}


def run_example(output_dir: Path, *, cpu_reference: bool = False):
    os.environ["TF_FORCE_GPU_ALLOW_GROWTH"] = "true"
    if cpu_reference:
        os.environ["CUDA_VISIBLE_DEVICES"] = "-1"
```

```
import tensorflow as tf
from bayesfilter.runtime.gpu_memory_policy import
configure_tensorflow_gpu_memory_growth
memory = configure_tensorflow_gpu_memory_growth(tf, require_gpu=not cpu_reference)
from bayesfilter.inference import (
    HMCAcceptancePolicy, HMCCandidateExecutionConfig, HMCControllerConfig,
    PrecomputedMassArtifact, bind_hmc_candidate_set_execution, tune_hmc_kernel,
    build_retained_bound_hmc_archive_runner_from_candidate_set_result,

build_claim_bearing_retained_frozen_kernel_hmc_adapter_from_candidate_set_result,
    load_hmc_candidate_retained_runner,
)
from bayesfilter.inference.hmc import FullChainHMCConfig,
build_independent_chain_tfp_hmc_runner

target = GaussianTarget()
mass = PrecomputedMassArtifact(position=[0.,0.], covariance=tf.eye(2, dtype=tf.
float64),
    factor=tf.eye(2, dtype=tf.float64), adapter_signature=target.adapter_signature
(),
    position_role="reference_center", covariance_source="known Gaussian fixture")
binding = bind_hmc_candidate_set_execution(adapter=target,
    initial_position=tf.constant([[-1.,-.5],[-.3,.2],[.4,-.2],[1.,.5]], tf.float64
),
    mass_artifact=mass, target_scope="docs_candidate_retained",
    target_lineage={"model": "standard_normal_2d", "data": "none", "prior": "
target_itself"},
    source_paths=[__file__], scope_id="docs-bridge", search_id="example-1",
    epsilon_domain=(.01,1.95), repair_factor=1.1, max_repairs_per_family=0,
    config=HMCCandidateExecutionConfig(measurement_num_results=128,
        verification_num_results=128, num_warmup_steps=8, seed=(20260914,11),
        acceptance_policy=HMCAcceptancePolicy(practical_region=(.55,.85),
repair_region=(.5,.9)),
        target_status_trace_policy="per_chain_step", use_xla=not cpu_reference,
        non_xla_reason="CPU reference smoke" if cpu_reference else None))
run = tune_hmc_kernel(adapter=target, initial_position=binding.
initial_active_state,
    candidate_set_adapter=binding.typed_adapter, output_dir=output_dir / "tuning",
    config=HMCControllerConfig(primary_l_grid=(2,3),
        epsilon_by_l=((2,(1.1,1.3,1.5)),(3,(1.1,1.3,1.5))),
        total_budget_units=40, repair_reserve_units=3))
if not run.result.verified_candidate_ids:
    raise RuntimeError("no verified member in the bounded engineering fixture")
candidate_id = run.result.verified_candidate_ids[0]  # Representative, no ranking
claim.
builder = (build_retained_bound_hmc_archive_runner_from_candidate_set_result if
cpu_reference
           else
build_claim_bearing_retained_frozen_kernel_hmc_adapter_from_candidate_set_result)
runner = builder(candidate_set_result=run.result, candidate_id=candidate_id,
retained_binding=binding)
member_path = runner.export(output_dir / "member.json")
first = runner.run(num_results=12, seed=(71,1), output_dir=output_dir / "
retained_1")
restored = load_hmc_candidate_retained_runner(member_path, adapter=GaussianTarget
(),
                                              claim_eligible=not cpu_reference)
second = restored.run(num_results=12, seed=(71,2), output_dir=output_dir / "
```

```
retained_2",
                        previous_archive=first["archive_path"])
# Match both compilation mode and device: stateless seeds do not promise
# identical random draws between XLA and non-XLA implementations.
with tf.device("/CPU:0" if cpu_reference else "/GPU:0"):
    reference = build_independent_chain_tfp_hmc_runner(binding._active_adapter,
        binding.initial_active_state, FullChainHMCConfig(num_results=12,
            num_burnin_steps=0, step_size=runner.candidate.epsilon,
            num_leapfrog_steps=runner.candidate.leapfrog_steps, seed=(71,2),
            use_xla=not cpu_reference, target_scope="docs_candidate_retained"))
    reference_result = reference.run(current_state=first["final_active_state"],
mode="serial")
max_error = float(tf.reduce_max(tf.abs(second["samples"] - reference_result.
samples)))
if max_error > 1e-10:  # Existing float64 reference-comparison tolerance.
    raise AssertionError(f"retained transition parity failed: {max_error}")
from bayesfilter.inference.hmc_candidate_set_execution import _trace_from_payload,
 _tensor_from_payload
archive = json.loads(Path(second["archive_path"]).read_text())
trace = _trace_from_payload(archive["trace"])
samples = _tensor_from_payload(archive["active_samples"])
initial = _tensor_from_payload(archive["initial_active_state"])
# Independent deterministic leapfrog reference, using the actual captured
# momentum. This checks CPU versus GPU arithmetic without comparing RNGs.
with tf.device("/CPU:0"):
    q0 = tf.concat([initial[None], samples[:-1]], axis=0)
    p0 = tf.identity(trace["initial_momentum"])
    q, p = q0, p0
    eps = tf.constant(runner.candidate.epsilon, tf.float64)
    for _ in range(runner.candidate.leapfrog_steps):
        p = p - .5 * eps * q
        q = q + eps * p
        p = p - .5 * eps * q
    expected_log_accept = .5 * tf.reduce_sum(q0*q0 + p0*p0 - q*q - p*p, axis=-1)
    position_error = float(tf.reduce_max(tf.abs(q - trace["proposed_state"])))
    momentum_error = float(tf.reduce_max(tf.abs(p - trace["final_momentum"])))
    acceptance_error = float(tf.reduce_max(tf.abs(expected_log_accept - trace["
log_accept_ratio"])))
if max(position_error, momentum_error, acceptance_error) > 1e-10:
    raise AssertionError("deterministic CPU leapfrog/energy reference failed")
compiled_counts = [r._runner.experimental_get_tracing_count()
    for ensemble in binding._runners.values() for r in ensemble._runners]
if not all(count == 1 for count in compiled_counts):
    raise AssertionError("unbounded/repeated runner tracing")
return {"status": "passed", "verified_candidate_ids": run.result.
verified_candidate_ids,
    "candidate_count": len(run.result.candidates), "selected_member": candidate_id
,
    "member_hash": runner.member_hash, "member_path": str(member_path),
    "retained_archives": [first["archive_path"], second["archive_path"]],
    "maximum_same_backend_transition_error": max_error,
    "cpu_reference_errors": {"position": position_error, "momentum":
momentum_error, "log_accept": acceptance_error},
    "compiled_runner_trace_counts": compiled_counts,
    "memory_policy": memory, "runtime_policy": binding._runtime,
    "source_closure": binding._spec["source_closure"],
    "execution_config": binding.config.payload(),
    "gpu_allocator": tf.config.experimental.get_memory_info("GPU:0") if not
```

```
    cpu_reference else None,
        "inference_status": "engineering parity only; no statistical ranking or
    posterior admission"}


def main():
    parser = argparse.ArgumentParser(description=__doc__)
    parser.add_argument("--output-dir", type=Path, required=True)
    parser.add_argument("--cpu-reference", action="store_true")
    args = parser.parse_args()
    args.output_dir.mkdir(parents=True, exist_ok=False)
    started = time.monotonic()
    manifest = {"command": sys.argv, "git_commit": subprocess.check_output(
        ["git","rev-parse","HEAD"], cwd=REPO_ROOT, text=True).strip(),
        "python": sys.executable, "cpu_reference": args.cpu_reference,
        "gpu_intentionally_hidden": args.cpu_reference, "data_version": "
    known_Gaussian_fixture_v1",
        "plan": "docs/plans/bayesfilter-typed-candidate-retained-bridge-plan
    -2026-09-14.md"}
    try:
        manifest["result"] = run_example(args.output_dir, cpu_reference=args.
    cpu_reference)
    except Exception as exc:
        manifest["result"] = {"status": "failed", "error": type(exc).__name__ + ": " +
     str(exc)}
        raise
    finally:
        manifest["wall_seconds"] = time.monotonic() - started
        (args.output_dir / "validation_manifest.json").write_text(json.dumps(manifest,
     indent=2) + "\n")
    print(json.dumps({"status": manifest["result"]["status"], "wall_seconds": manifest
    ["wall_seconds"]}))


if __name__ == "__main__":
    main()
```

## 1.8 What the validation experiments establish

The tuning procedure is tested through several distinct experiments. Numerical oracles compare target values, scores, coordinate Jacobians and leapfrog updates with independent calculations. Frozen-kernel experiments test invariance using independent reference draws and a reversible random-position construction. The latter uses Algorithm 2 and Proposition 2.3 of Gandy and Scott [2021]; it applies to frozen reversible kernels. An experiment may replace its kernel $K$ by $K^s$, composing $s$ complete Metropolis transitions with independent randomness at each substep. Repeated composition of this same reversible kernel preserves reversibility; section 2.2 of that paper proposes this extension to improve sensitivity to subtle errors. Sensitivity still needs measurement for the chosen alternative. The option `kernel_power` applies only to the frozen invariance experiment. It does not make posterior draws independent. Finite states and log acceptance ratios are checked at every substep, including earlier substeps when the final one is finite.

A separate optional wrapper implements Algorithm 3 of Gandy and Scott [2021]. Each look uses an independently seeded complete experiment, with the sample count

increased once after the first look. For overall level $\alpha$, at most $k$ looks and a fixed family of $d$ component tests, initialize $\beta = \alpha/k$ and $\gamma = \beta^{1/k}$. At each look, set $q = d \min_j p_j$. Reject when $q \leq \beta$, stop without rejection when $q > \gamma + \beta$, and otherwise continue with $\beta \leftarrow \beta/\gamma$. Theorem 3.1 bounds the probability of rejection under the null by $\alpha$ when the look vectors are independent and each component p-value is superuniform. Fresh experiments are essential: repeatedly testing cumulative posterior draws would not satisfy that independence condition. Numerical failures invalidate the affected experiment; they do not count as defect detection. This wrapper is a validation experiment and does not alter the tuning acceptance screen, warmup checks or posterior precision stopping.

The Gaussian mechanics experiment also reconstructs the Metropolis log ratio from an independent analytic density and the actual TFP endpoint momenta. It checks that the returned state agrees with the recorded accept/reject decision. This arithmetic check can expose a reversed energy correction even when a short distributional experiment has little power to detect its effect. For an identity-metric Gaussian, an additional fixed-look test uses the sum of squared standardized endpoint coordinates. Starting from independent exact Gaussian draws makes the endpoints independent under the correct frozen kernel, so this sum has a chi-squared distribution with degrees of freedom equal to the number of draws times the dimension. The experiment adjusts for all declared radius, rank and energy tests. Adjacent states from one chain cannot replace independent endpoints in this calculation. Null controls and complete repeated experiments measure size and power for the declared defect; one detected error does not establish broad sensitivity. Controller tests check that every qualified pair is retained, each repair keeps its own L, and fresh verification and restart preserve the numerical record. Full-procedure experiments then call the public tuner and run the separate posterior controller. Small integration cases reconstruct every verified member. Larger replicated experiments may predeclare one member by tuning identity before sampling; all other verified members remain recorded as unassessed for that experiment.

Simulation-based calibration generates a parameter from a proper prior and a dataset conditional on that parameter. If the fitted outputs are independent exact posterior draws, the generating parameter's randomized rank among them is uniform. This supplies the null distribution for the test. The implemented design uses independent complete fits within each dataset and one fixed output per fit. It includes data-dependent likelihood quantities because parameter ranks alone can miss a procedure that ignores the observations. Failed fits remain in the denominator; conditional ranks from successful fits cannot establish calibration for a procedure that sometimes fails. Candidate siblings share a dataset and are not independent calibration replications. The rank argument follows Talts et al. [2020]; the likelihood quantities follow the sensitivity analysis of Modrák et al. [2023].

The later M15 experiment obtained all 246 planned fitted outputs over 82 fresh datasets, using three independently fitted outputs per dataset. Its rank tests did not reject, but three rank draws give a coarse test. Independent analytic controls in M16 showed that the 32-dataset normal design detected a shift of one quarter of a posterior standard deviation in only 27 of 256 experiments, and a half-standard-deviation shift in 64 of 256. The corresponding nonrejection therefore supplies little evidence against those errors. These controls measure the rank statistic's sensitivity; they do not measure defect power of the complete HMC procedure. The full denominators, uncertainty and source versions are recorded in the M15 and M16 result notes under `docs/plans/`.

An analytic posterior permits an additional direct check of fitted outputs. For a zero-

mean normal prior with variance $\tau^2$ and independent observations with known variance $\sigma^2$, the conditional posterior has variance $s_y^2 = (\tau^{-2}+n\sigma^{-2})^{-1}$ and mean $\mu_y = s_y^2 \sum_i y_i/\sigma^2$. If a predetermined output $X_j$ from each independent complete fit has that conditional posterior law, then $Z_j = (X_j - \mu_y)/s_y$ are independent standard normals. Consequently, $\sum_j Z_j/\sqrt{J}$ is standard normal and $\sum_j Z_j^2$ has a $\chi^2_J$ distribution. Two two-sided tests, each at half the declared size, check location and second moment. The union bound controls their combined size under this correct-output null. The second check can expose errors that leave the mean unchanged.

The diagnostic fixes the member, chain and terminal-draw rule before seeing posterior outcomes. Missing, capped or unqualified fits make its planned experiment incomplete; they are never counted as successful defect detections. Shared-stream controls cannot count as independent fits. The normal laws above describe the hypothesis under test. Adaptive stopping does not prove that a finite HMC endpoint follows them, and these tests do not establish coverage of intervals for stopped posterior means. Arithmetic adversaries and activated whole-fit defects answer different questions and are reported separately.

An invariant kernel can remain motionless or revisit only a few states. Identity and two-cycle controls therefore accompany the distributional tests. Similarly, an MCSE estimator can perform adequately on fixed-length stationary arrays while intervals based on it have poor coverage at a data-dependent stopping time. Separate experiments compare the actual controller's stopped means and medians with available exact references, retaining cap and unavailable outcomes. A fixed-length arm reuses the same verified kernel with independent randomness and predeclared discarded and retained counts. Mixture mode probabilities check global exploration that a local mean or small reported MCSE may miss. Missing arms remain in the comparison denominator; pointwise uncertainty intervals from a small number of replications do not establish a ranking or nominal sequential coverage. No R-hat, ESS or MCSE assessment changes tuning membership. Coverage is reported both over all planned fits and conditional on an available interval. Repeating a fit to one fixed dataset measures conditional behavior; it is distinct from prior-predictive simulation-based calibration. A longer fixed-count arm can also have inaccurate uncertainty estimates, so disagreement with nominal coverage does not by itself identify optional stopping as the cause.

Qualified posterior delivery, interval coverage, and their joint event answer different questions. A covered interval reported at a precision cap contributes to coverage, but the fit has not delivered the requested precision. The validation report records all three events for every declared quantity and keeps missing fits in the planned denominator. A stopping design may declare a minimum success probability and compare it with a pointwise exact-binomial lower confidence bound. Conditional coverage among available intervals remains descriptive; these pointwise screens give no simultaneous or sequential coverage guarantee. Multiple selected kernels are reported in separate member slots, with one independent observation per complete fit in each slot.

The September 2026 campaign observed this distinction directly. A Gaussian mixture run passed its local posterior checks after 6,500 discarded and 2,000 retained transitions per chain, but all retained draws occupied the right mode. Its estimated mean was 4.956, against the exact mean 2, with reported MCSE 0.038. Thus a small within-mode uncertainty estimate did not measure error relative to the full posterior. Target-specific posterior assessment should include relevant mode or other global quantities and examine sensitivity to starting locations. The saved result and its limited replication evidence are docu-

mented in `docs/plans/bayesfilter-inference-validation-24h-result-2026-09-16.md`. Validation designs may now declare the mixture probability $P(x < 0)$ as a monitored posterior mean. It receives the same diagnostic and precision checks as the other declared quantities; a constant event indicator supplies no usable precision estimate. A separate experiment starts chains in both known modes. These checks address the saved failure mechanism without establishing discovery of unknown modes. Synthetic acceptance calibration separately measures the candidate screen near its practical and repair boundaries, including repeated evidence allocations. It does not establish posterior validity. The separate numerical acceptance experiment calls the public tuner on a declared identity-metric Gaussian pair. Every repeated search measures that exact pair from the fixed tuning starts, with no repair children. Each measurement survivor receives its own fresh verification. Independent exact Gaussian anchors supply a stationary acceptance reference using actual Metropolis transitions and a separate endpoint-energy check. That reference neither supplies tuning starts nor determines qualification. Stationary acceptance and acceptance over a finite window from fixed starts are different quantities; their difference is part of the experiment. Observed qualification rates therefore describe the operational compatibility screen, not nominal sequential coverage or automatic-preparation reliability. The M16 experiment completed 384 CPU-reference and 192 GPU/XLA searches around stationary acceptance values from .64 to .76. Pairs on both sides of the nominal [.65,.75] band could qualify. This is consistent with the implemented uncertainty-aware compatibility rule, whose finite-start evidence measures a different quantity from the independent stationary reference. Separately, 128 complete sequential validation experiments per device detected the declared reversed-Metropolis-ratio defect every time. Three of the four baseline/no-op null intervals were still too wide for the declared size-precision screen. That result supports detection of this defect while leaving null-size precision and sensitivity to other defects unresolved. Stopped-interval reporting compares the monitored mode probability with its independent mixture CDF. That quantity remains in the planned denominator when warmup ends without retained draws.

Automatic initialization and supplied epsilon proposals are distinct subjects of validation. A supplied proposal can exceed the final prepared metric's bound; native initialization uses preparation's own starting epsilon. Either search can return no verified member under its finite domain and budget. A complete calibration experiment must preserve those missing fits, even when all observed ranks among successful fits look uniform. Parallel dataset groups are combined only when declared together with identical settings and frozen source; pilot and repaired experiments remain separate.

A complementary diagnostic uses all retained draws from one qualified fit to test its posterior mean. With $\theta \sim N(0, \tau^2)$ and conditionally independent observations $y_i \mid \theta \sim N(\theta, \sigma^2)$, completing the square in the prior times likelihood gives the independent reference

$$s_*^2 = (\tau^{-2} + n\sigma^{-2})^{-1}, \qquad m_* = s_*^2 \sigma^{-2} \sum_{i=1}^{n} y_i.$$

The validation engine `reference_mean` compares $|\widehat{m} - m_*|/\widehat{\mathrm{MCSE}}$ with the declared normal critical value. It recomputes lugsail MCSE independently from the archived model draws; the reference never enters tuning. The normal approximation defines the alarm, but its achieved false-alarm and detection rates require independent complete fits with fresh data. A capped, missing or unqualified fit is a possible alarm in the conservative

null rate and a nondetection in the defect rate. This answers a location-error question distinct from endpoint invariance and SBC; arithmetic and execution tests alone do not establish its power.

Matched consumer comparisons use the same public ordinary tuner and posterior controller. The explicit noncentered eight-schools and Bayesian regression adapters reproduce the pinned posteriordb Stan laws, observations and coordinate Jacobians. A member is selected by tuning identity before posterior sampling; every other verified member remains recorded. The ten-chain Stan reference is used only for the subsequent comparison, with lugsail uncertainty from both finite samples. Agreement within a declared practical margin is a target-specific accuracy screen. It neither makes the reference exact nor establishes calibration for another dataset or consumer. Exact inputs, commands and results are recorded in `docs/plans/bayesfilter-hmc-repair-m8-result-2026-09-18.md`.

The later M17 matrix completed 21 model/route fits and four matched-reference fits. Eleven matrix members and three reference members passed their full declared posterior checks. Centered funnels produced no verified member; mixtures reached the warmup cap; rotated Gaussian, noncentered funnel and CPU eight-schools fits reached retained precision caps. Their verified kernels remain in the tuning set. All four reference fits met the declared mean agreement margins, illustrating why reference mean agreement and requested precision must be reported separately. Six conditional position-field cases also exercised fresh verification and durable restart. Fixed nonlinear maps in this matrix were not trained. Exact model/device outcomes and source versions are recorded in `docs/plans/bayesfilter-hmc-repair-m17-result-2026-09-21.md`. The original MacroFinance target and matched posterior reference remain a separate missing comparison.

Cost pilots can save a host profile of automatic preparation, candidate search, export/reload and posterior assessment. The profile includes failed attempts; a framework call may include both compilation and execution time. Such a pilot identifies costs to investigate before funding more replications. It does not establish statistical calibration or rank the viable kernels by efficiency. Use the validation CLI's `run` command with `--profile-execution` for paired comparisons: it preserves the numerical design and random streams. With isolated fits, the profiler runs in the numerical child after framework imports, with setup time recorded separately. It records Python call boundaries without native builtins tracing, avoiding an observed Python 3.13/TensorFlow interaction that loses outer calls. This is host cost attribution, not device kernel timing. The run index records profile availability. A completed fit is never repeated to replace a missing profile, and profiling failures do not replace numerical outcomes. Profiling also consumes wall time and can change which work fits within a finite execution budget. Hold its setting fixed in cost comparisons and price the setting intended for confirmation.

Repeated validation fits can use a fresh process for each complete fit, with an explicit per-fit timeout inside the total experiment budget. This bounds the lifetime of process-local framework caches. Each child preserves its full candidate set, posterior chunks and independent assessment before normal exit; the coordinator records process failures separately from saved numerical results. Missing or abnormally terminated fits remain in the replication denominator. Numerical search, accuracy, stopping and reference-mean designs enable this optional mode with `options.isolate_fits` and `options.fit_process_timeout_seconds`. The setting does not change the HMC transition, tuning criteria or posterior thresholds.

For exact-score validation pipelines, `--reuse-leapfrog-graphs` carries the optional

compiled-runner policy through the cell and isolated child. It defaults to off; the reference-mean engine already uses reuse. The run index and child manifest record the choice, and resume rejects a changed execution policy. A development wrapper's reuse setting must be supplied explicitly when moving to the isolated executor.

Sequential suites may use `--share-unused-budget` to carry unused time from settled cells to later cells. The sum of the original cell allocations remains the ceiling; unstarted cells retain their own allocations, and a resumed cell cannot reclaim time already spent elsewhere. Parallel sharing is rejected. This avoids stranding a completed cell's spare time while a later fit is cut short by its local forecast.

Machine workload affects how long a numerical chunk takes. An isolated design may therefore declare a finite contention allowance in `options.timeout_policy.max_extension_seconds`, which defaults to zero. The supervisor grants that allowance only when recent committed numerical work and fresh workload evidence are both present. For GPU work it matches the selected device UUID to compute-process telemetry and excludes the worker's own process group. GPU utilization alone is insufficient. Missing telemetry or an ambiguous numeric device selection cannot justify an extension. Normalized CPU load is a saturation heuristic, not proof that another process caused the slowdown. These observations describe execution conditions, not sampler quality.

The parent publishes an atomic execution allowance. Preparation, candidate measurement and verification, and posterior sampling check it between numerical chunks. Its deadline is separate from the frozen tuning configuration, so a workload extension preserves candidate identities and random streams. An active TensorFlow call cannot be stopped at such a boundary until it returns; the hard process cap remains necessary. Startup, telemetry, retries and cleanup all count against the allocation, and the inherited outer deadline reserves time for the coordinator's report. An optional no-progress stall interval must allow for measured compilation and chunk times; it is disabled by default.

The v7 confirmation worker accepts an explicit `--allowance` from its bounded supervisor. This replaces its internal wall clocks, including the controller and between-chunk clock, while preserving configured numerical work limits. Without that option, the configured wall limit still applies. A monitor that merely writes heartbeats does not activate the allowance. Progress includes v1 and v2 tuning checkpoints and completed export, evidence reconstruction and member-reload operations; duplicate completions and copied old checkpoints do not earn additional time.

With `extension_mode="observed_intervals"`, trusted contention intervals can earn bounded extra time while useful work continues, including during closeout. This scheduling allowance is not an estimate of lost GPU compute. The extension cap and enclosing campaign deadline remain binding. Checkpoint continuation uses its original source; a change of source creates a different scope and cannot be presented as preservation of the old random streams.

A shared GPU need not become empty before useful work can proceed. The validation policy `gpu_admission_mode="shared"` admits work with trusted device telemetry and available memory even when another compute process is present. The compatibility mode `gpu_admission_mode="idle"` waits for those processes to leave. Both modes require memory growth and bounded admission waiting; neither guarantees a fixed share of device throughput.

An explicit `max_contention_retries` permits bounded automatic recovery after an incomplete cooperative budget stop or supervised timeout with observed contention. The retry resumes the same source, design and committed numerical checkpoints. It spends

only the unspent portion of the original fit allowance, including its contention reserve and all startup costs, and cannot renew that reserve. The attempt limit also applies across coordinator restarts. Completed posterior assessments, numerical errors and invalid artifacts do not trigger this recovery. Complete-fit pricing includes every recovery attempt.

When the original fit cap is exhausted, a campaign may explicitly reassign some of its remaining time. The separate campaign continuation coordinator runs the original frozen numerical worker with the same design, seeds and checkpoint paths. The saved execution also binds the exact `CUDA_VISIBLE_DEVICES` selection; the same GPU model on another device is not interchangeable for checkpoint continuation. The coordinator checks this binding before launch. A device-binding startup failure can be repaired only after unchanged evidence and absence of numerical progress are verified, the original GPU selection is restored, and failed attempts remain charged. It records an enlarged cumulative allocation, counts all earlier attempts, checks that completed evidence remains unchanged, and enforces the enclosing deadline and additional-attempt limit. Only an incomplete resource stop with durable progress is eligible. Completed assessments are reused, including unfavorable results; numerical failures require separate diagnosis. The complete fit cost includes every attempt. This reallocation spends the existing campaign budget and never restarts its clock.

The common-pool dispatcher connects this continuation to the campaign queue. It schedules individual fits across models and returns a resource-stopped fit with durable progress to the queue after its peers. Each allocation is bounded by the absolute deadline. A finite attempt limit is optional; the state-space continuation instead allows further turns while durable evidence and elapsed time advance. These time quanta govern scheduling; they do not change numerical stopping rules. Unfinished pilots and main fits share one queue, and the coordinator refreshes complete workload costs and eligibility after each turn. A completed matching pilot can therefore admit waiting fits without a separate stage boundary. Each child receives only its next bounded allocation beyond recorded prior cost. A price must match the declared numerical workload, including its reference resolution. Posterior success is not a pricing criterion, and completed unfavorable assessments remain final. Unstarted and incomplete fits remain in the original inventory.

Stopped-interval assessment distinguishes analytic functionals from numerical quadrature references. For a checked quadrature reference, the diagnostic report compares the nominal MCSE interval with the reference value plus or minus its recorded resolution and domain sensitivity. It reports containment, partial overlap or disjointness. The sensitivity is neither a certified integration error bound nor a confidence interval, so these comparisons cannot contribute to exact-reference coverage counts or establish sequential coverage. An absent joint posterior reference remains unavailable. The report can use saved final checks without rerunning a chain or overwriting its original assessment.

Supervision records preserve the progress observations, workload samples, extension decision and termination reason. Budget exhaustion after any declared recovery makes the fit unavailable and allows the next independent fit to run. Exhausted fit allocations remain in the original denominator and are not restarted by ordinary resume; additional time requires the separate campaign allocation above. Receipts distinguish a shorter cell allowance from an exhausted fit allowance. A cell-limited partial fit can resume on the same source/design within its unspent cumulative fit time. When only a running preparation progress file exists, the coordinator preserves that file before replaying the same preparation seed; it does not treat the progress log as a checkpoint. Completed

results are preserved. A stall, corrupt checkpoint, missing assessment or unexpected process failure stops the cell for diagnosis. Cooperative posterior stops retain committed chunks and a pause record, allowing a later explicitly funded invocation to replay the same streams without mistaking an incomplete member for a finished result. Changing the execution policy creates a new validation design; historical confirmation results cannot be silently relabelled as results of that new policy.

A bounded sibling study can instead predeclare several members using `member_rule="shortest_verified_l"`, `posterior_members="selected"` and an explicit `posterior_member_count`. The rule orders distinct verified leapfrog counts increasingly and takes the first candidate identifier within each. It records the list before posterior sampling and preserves any shortage. Each ordinal member is assessed separately across complete fits; siblings do not increase the number of independent replications. A failed member does not suppress the next member or remove a tuning candidate. This is a bounded cost hypothesis, not an established mixing ranking, and posterior outcomes cannot choose a winner from the same assessment draws.

The executable suite definitions are documented in `docs/validation/README.md`. The state-space K0–K7 inventory calls the actual QR and SVD Kalman filters and a nonlinear sigma-point filter. It covers unknown location, persistence and noise scales, near-unit persistence, small observation noise, longer series and a multivariate model. Independent dense Gaussian, Kalman-recursion and checked low-dimensional integration references test the declared targets. The nonlinear reference describes the same sigma-point approximation; a separate short-horizon integration diagnoses approximation error. The multivariate parameter posterior has no established independent oracle. CPU mechanics and public-procedure regressions do not establish GPU execution, posterior calibration or a new tuning default.

The following table is generated from the saved suite reports. C, H and P denote current-source executions, historical-source executions and planned designs, including retries. These are counts of designs, not models declared correct or independent replications. CPU reference work and GPU/XLA work remain separate. A completed controller call with no retained draws does not establish posterior coverage; missing stopping intervals remain in the denominator. Affine and dense nonlinear transports occupy separate rows. Each experiment declares its evidence scope. Incomplete confirmation cannot establish calibration; statistical uncertainty and unresolved coverage are reported with the underlying observations.

The table covers suite reports. Separately executed phase diagnostics, including the matched posteriordb comparisons above, are recorded in their phase results. An unavailable external eight-schools row refers to that original suite's missing reference bundle; it does not negate the later explicitly bound eight-schools comparison. Neither supplies the absent MacroFinance reference.

Automatically inferred suite requirements identify each planned design, so a completed step-size or kernel-power setting cannot cover an unrun setting in the same category. Explicit category requirements may request any matching assessed design. Older saved plans without design identifiers retain that category-level meaning; their individual rows determine whole-suite completion.

| Target | Route/device | Question | C / H / P | Findings |
|---|---|---|---|---|
| Banana | Dense IAF / CPU | Accuracy | 0 / 1 / 1 | pipeline assessed |

| Target | Route/device | Question | C / H / P | Findings |
|---|---|---|---|---|
| Banana | Dense IAF / GPU | Accuracy | 0 / 1 / 1 | pipeline assessed |
| Banana | Affine transport / CPU | Accuracy | 0 / 1 / 1 | pipeline assessed |
| Banana | Affine transport / GPU | Accuracy | 0 / 1 / 1 | pipeline assessed |
| Banana | Frozen kernel / CPU | Invariance | 0 / 2 / 2 | no discrepancy detected |
| Banana | Frozen kernel / CPU | Mechanics | 0 / 3 / 3 | mechanics passed |
| Banana | Frozen kernel / GPU | Mechanics | 0 / 1 / 2 | mechanics passed, not run |
| Banana | Automatic / GPU | Accuracy | 0 / 2 / 2 | pipeline assessed |
| Beta | Frozen kernel / CPU | Invariance | 0 / 2 / 2 | no discrepancy detected |
| Beta | Frozen kernel / CPU | Mechanics | 0 / 3 / 3 | mechanics passed |
| Beta | Frozen kernel / GPU | Mechanics | 0 / 1 / 2 | mechanics passed, not run |
| Beta | Prepared / CPU | Accuracy | 0 / 1 / 1 | pipeline assessed |
| Beta-binomial | Frozen kernel / CPU | Mechanics | 0 / 3 / 3 | mechanics passed |
| Beta-binomial | Frozen kernel / GPU | Mechanics | 0 / 1 / 2 | mechanics passed, not run |
| Beta-binomial | Automatic / CPU | SBC | 0 / 65 / 66 | calibration incomplete, no discrepancy detected, timed out |
| Beta-binomial | Automatic / CPU | Stopping | 0 / 49 / 49 | pipeline assessed |
| Beta-binomial | Automatic / GPU | Accuracy | 0 / 2 / 3 | pipeline assessed, timed out |
| Beta-binomial | Automatic / GPU | SBC | 0 / 7 / 8 | no discrepancy detected, not run |
| Beta-binomial | Automatic / GPU | Stopping | 0 / 34 / 34 | pipeline assessed |
| Beta-binomial | Reference / CPU | SBC | 0 / 2 / 2 | no discrepancy detected |
| Cauchy | Frozen kernel / CPU | Invariance | 0 / 2 / 2 | no discrepancy detected |
| Cauchy | Frozen kernel / CPU | Mechanics | 0 / 3 / 3 | mechanics passed |
| Cauchy | Frozen kernel / GPU | Mechanics | 0 / 1 / 2 | mechanics passed, not run |
| Cauchy | Automatic / CPU | Accuracy | 0 / 2 / 2 | pipeline assessed |
| Cauchy | Automatic / GPU | Accuracy | 0 / 1 / 1 | pipeline assessed |
| Dirichlet | Dense IAF / CPU | Accuracy | 0 / 1 / 1 | pipeline assessed |

| Target | Route/device | Question | C / H / P | Findings |
|---|---|---|---|---|
| Dirichlet | Dense IAF / GPU | Accuracy | 0 / 1 / 1 | pipeline assessed |
| Dirichlet | Frozen kernel / CPU | Invariance | 0 / 2 / 2 | no discrepancy detected |
| Dirichlet | Frozen kernel / CPU | Mechanics | 0 / 3 / 3 | mechanics passed |
| Dirichlet | Frozen kernel / GPU | Mechanics | 0 / 1 / 2 | mechanics passed, not run |
| Dirichlet | Prepared / CPU | Accuracy | 0 / 1 / 1 | pipeline assessed |
| Dirichlet | Prepared / GPU | Accuracy | 0 / 1 / 3 | not run, pipeline assessed, timed out |
| Eight schools | External / CPU | Accuracy | 0 / 0 / 2 | unavailable |
| Funnel | Frozen kernel / CPU | Invariance | 0 / 2 / 2 | no discrepancy detected |
| Funnel | Frozen kernel / CPU | Mechanics | 0 / 3 / 3 | mechanics passed |
| Funnel | Frozen kernel / GPU | Mechanics | 0 / 2 / 3 | mechanics passed, not run |
| Funnel | Automatic / CPU | Accuracy | 0 / 4 / 5 | failed, no verified members |
| Funnel | Automatic / GPU | Accuracy | 0 / 6 / 8 | failed, no verified members, not run |
| Funnel noncentered | Automatic / CPU | Accuracy | 0 / 5 / 5 | pipeline assessed, posterior incomplete |
| Funnel noncentered | Automatic / GPU | Accuracy | 0 / 6 / 7 | pipeline assessed, posterior incomplete, timed out |
| Gamma | Frozen kernel / CPU | Invariance | 0 / 2 / 2 | no discrepancy detected |
| Gamma | Frozen kernel / CPU | Mechanics | 0 / 3 / 3 | mechanics passed |
| Gamma | Frozen kernel / GPU | Mechanics | 0 / 2 / 3 | mechanics passed, not run |
| Gamma | Prepared / CPU | Accuracy | 0 / 1 / 1 | pipeline assessed |
| Gaussian | Controller / CPU | Acceptance | 0 / 22 / 22 | acceptance rates estimated |
| Gaussian | Controller / CPU | Search | 0 / 2 / 2 | inventory passed |
| Gaussian | Dense IAF / GPU | Accuracy | 0 / 3 / 3 | no verified members, pipeline assessed |
| Gaussian | Affine transport / CPU | Accuracy | 0 / 2 / 2 | pipeline assessed |
| Gaussian | Affine transport / GPU | Accuracy | 0 / 2 / 2 | pipeline assessed |
| Gaussian | Frozen kernel / CPU | Acceptance | 0 / 10 / 10 | actual hmc acceptance measured |
| Gaussian | Frozen kernel / CPU | Invariance | 0 / 2 / 2 | no discrepancy detected |
| Gaussian | Frozen kernel / CPU | Mechanics | 0 / 3 / 3 | mechanics passed |

| Target | Route/device | Question | C / H / P | Findings |
|---|---|---|---|---|
| Gaussian | Frozen kernel / CPU | Power | 0 / 9 / 9 | power estimated |
| Gaussian | Frozen kernel / GPU | Invariance | 0 / 1 / 1 | no discrepancy detected |
| Gaussian | Frozen kernel / GPU | Mechanics | 0 / 5 / 6 | mechanics passed, not run |
| Gaussian | Frozen kernel / GPU | Power | 0 / 18 / 21 | not run, power estimated |
| Gaussian | Automatic / CPU | Accuracy | 0 / 1 / 2 | pipeline assessed, timed out |
| Gaussian | Automatic / GPU | Accuracy | 0 / 2 / 4 | pipeline assessed, timed out |
| Gaussian | Prepared / CPU | Acceptance | 0 / 6 / 6 | acceptance rates estimated |
| Gaussian | Prepared / CPU | Accuracy | 0 / 1 / 2 | pipeline assessed, timed out |
| Gaussian | Prepared / CPU | Stopping | 0 / 4 / 5 | pipeline assessed, timed out |
| Gaussian | Prepared / GPU | Acceptance | 0 / 22 / 22 | acceptance rates estimated |
| Gaussian | Prepared / GPU | Stopping | 0 / 3 / 5 | not run, pipeline assessed |
| Gaussian | Reference / CPU | Stopping | 0 / 6 / 6 | diagnostics assessed |
| LGSSM location | Frozen kernel / CPU | Mechanics | 0 / 3 / 3 | mechanics passed |
| LGSSM location | Frozen kernel / GPU | Mechanics | 0 / 1 / 2 | mechanics passed, not run |
| LGSSM location | Automatic / CPU | SBC | 0 / 18 / 18 | no discrepancy detected |
| LGSSM location | Automatic / CPU | Stopping | 0 / 16 / 16 | pipeline assessed |
| LGSSM location | Automatic / GPU | Accuracy | 0 / 2 / 3 | pipeline assessed, timed out |
| LGSSM location | Automatic / GPU | SBC | 0 / 1 / 2 | calibration incomplete, not run |
| LGSSM location | Automatic / GPU | Stopping | 0 / 1 / 1 | pipeline assessed |
| LGSSM location | Reference / CPU | SBC | 0 / 2 / 2 | no discrepancy detected |
| Macrofinance | External / CPU | Accuracy | 0 / 0 / 2 | unavailable |
| Mixture | Frozen kernel / CPU | Invariance | 0 / 2 / 2 | no discrepancy detected |
| Mixture | Frozen kernel / CPU | Mechanics | 0 / 3 / 3 | mechanics passed |
| Mixture | Frozen kernel / GPU | Mechanics | 0 / 1 / 2 | mechanics passed, not run |
| Mixture | Prepared / CPU | Accuracy | 0 / 2 / 2 | posterior incomplete |
| Mixture | Prepared / CPU | Stopping | 0 / 3 / 4 | incomplete, pipeline discrepancy, timed out |
| Mixture | Prepared / GPU | Accuracy | 0 / 2 / 2 | posterior incomplete |

| Target | Route/device | Question | C / H / P | Findings |
|---|---|---|---|---|
| Mixture | Prepared / GPU | Stopping | 0 / 4 / 6 | not run, pipeline discrepancy, posterior incomplete |
| Mixture | Reference / CPU | Stopping | 0 / 2 / 2 | diagnostics assessed |
| Normal-normal | Frozen kernel / CPU | Mechanics | 0 / 3 / 3 | mechanics passed |
| Normal-normal | Frozen kernel / GPU | Mechanics | 0 / 1 / 2 | mechanics passed, not run |
| Normal-normal | Automatic / CPU | SBC | 0 / 91 / 91 | calibration incomplete, no discrepancy detected |
| Normal-normal | Automatic / CPU | Stopping | 0 / 49 / 49 | pipeline assessed |
| Normal-normal | Automatic / GPU | SBC | 0 / 11 / 22 | calibration incomplete, cancelled, no discrepancy detected, not run, timed out |
| Normal-normal | Automatic / GPU | Stopping | 0 / 34 / 34 | pipeline assessed |
| Normal-normal | Prepared / CPU | SBC | 0 / 17 / 17 | no discrepancy detected |
| Normal-normal | Reference / CPU | Power | 0 / 17 / 17 | power estimated |
| Normal-normal | Reference / CPU | SBC | 0 / 2 / 2 | no discrepancy detected |
| Regression | External / CPU | Accuracy | 0 / 0 / 2 | unavailable |
| Rotated gaussian | Affine transport / CPU | Accuracy | 0 / 1 / 1 | posterior incomplete |
| Rotated gaussian | Affine transport / GPU | Accuracy | 0 / 1 / 2 | not run, posterior incomplete |
| Rotated gaussian | Frozen kernel / CPU | Invariance | 0 / 2 / 2 | no discrepancy detected |
| Rotated gaussian | Frozen kernel / CPU | Mechanics | 0 / 3 / 3 | mechanics passed |
| Rotated gaussian | Frozen kernel / GPU | Mechanics | 0 / 1 / 2 | mechanics passed, not run |
| Rotated gaussian | Automatic / CPU | Accuracy | 0 / 1 / 1 | pipeline assessed |
| Rotated gaussian | Automatic / GPU | Accuracy | 0 / 3 / 3 | pipeline assessed |
| Student t | Frozen kernel / CPU | Invariance | 0 / 2 / 2 | no discrepancy detected |
| Student t | Frozen kernel / CPU | Mechanics | 0 / 3 / 3 | mechanics passed |
| Student t | Frozen kernel / GPU | Mechanics | 0 / 1 / 2 | mechanics passed, not run |
| Student t | Automatic / CPU | Accuracy | 0 / 1 / 1 | pipeline assessed |

## 1.9 Historical readers and complete interface inventory

Initial geometry and bootstrap screening are implemented in `hmc_geometry.py` and `hmc_bootstrap.py`, respectively. Windowed preparation, its timeout policy and the frozen-mass/start-bank handoff are implemented in `hmc_mass_adaptation.py`. Public imports and historical `hmc_kernel_tuning` aliases resolve to the same definitions. Automatic preparation calls these implementations directly; public presets, configuration translation and geometry-scaled budgets live in `hmc_configuration.py`. The old module retains historical orchestration and readers, with aliases for the extracted definitions. These helpers do not issue candidate-set tuning authority. Numerical bindings include their source files and shared preparation helpers. Seed identifiers remain stable, while live seed records name the current implementation files. Historical results retain their original source provenance.

Earlier single-kernel ordinary, fixed-transport and typed TensorFlow results retain their original identities and readers. Their compatibility replay roles and older `claim_bearing_blockers` remain historical evidence; a caller-edited authority flag or blocker list cannot grant authority. The active procedure uses the candidate-set result and the member's own numerical evidence. Discovery, refinement, `select_fixed_transport_candidate_set`, and stage helpers remain diagnostics. The generated registry below identifies their replacement public route and their limits.

| Interface | Status | Target, tuning, verification, and use |
|---|---|---|
| `bayesfilter.inference.hmc_tuning_dispatch.tune_hmc_kernel` | public tuner; tested supported; authority | Target: artifact authority requires an ordinary exact value/score target; the typed position-field branch additionally requires a repository-issued binding and exact endpoint target but remains mechanics-only. Coordinates: ordinary adapter coordinates with repository-owned affine mass coordinates. Mass: windowed adaptation by default or explicit fixed identity from config. Epsilon: shared per-L pilot, exact-pair measurement, bounded directional repair, and independently seeded evidence rungs. L: shared primary L grid (3, 5, 9, 13, 18, 25), explicit optional all-survivor refinement and expansion; retain every verified pair; typed full-refresh MAMS accepts discarded duration/epsilon proposals and explicit preserve_mean_duration repair without posterior gates. Verification: fresh fixed-kernel verification; default TFP runner requires typed acceptance, health, and minimum draws; rank-normalized split/folded R-hat is retained as an explanatory diagnostic with threshold metadata (1.01), not a tuning handoff gate; explicit experimental v7 uses independent fixed-horizon start-bank repetitions with separate search/verification error allocations. ESS: disabled for ordinary tuning admission; retained posterior ESS is separate. Forbidden: arbitrary bare runner callback; engineering_probe_covariance_multiplier as a public ordinary mode; transferring epsilon qualification across different L candidates; modern profiles as statistically qualified or claim-eligible release evidence; fixed nonlinear transport without its transformed target contract; retained posterior convergence claim. |
| `bayesfilter.inference.fixed_transport_hmc_tuning_tf.tune_fixed_transport_hmc_kernel` | public tuner; tested supported; authority | Target: identity-bound frozen transport with exact transformed value including Jacobian and matching score. Coordinates: fixed transport latent z coordinates. Mass: fixed identity mass in z; no ordinary windowed mass adaptation. Epsilon: shared per-L pilot or explicit epsilon proposals, exact-pair measurement and bounded independently verified repairs. L: all declared (epsilon, L) pairs measured; all-survivor refinement and retention; typed full-refresh MAMS supports discarded duration/epsilon proposals and explicit preserve_mean_duration repair. Verification: fresh fixed-kernel verification for every survivor; declared evidence rungs; R-hat reporting-only; explicit experimental v7 appends independent repetitions at fixed prefix and horizon. ESS: disabled for kernel tuning admission; cumulative posterior ESS is separate. Forbidden: arbitrary position-only force without a frozen transport; ordinary mass adaptation claim; reuse after transport identity changes; modern profiles as statistically qualified or claim-eligible release evidence; legacy directional diagnostic policy as an artifact-authority handoff. |

| Interface | Status | Target, tuning, verification, and use |
|---|---|---|
| `bayesfilter.inference.hmc_candidate_set_tuning.tune_hmc_candidate_set` | diagnostic helper; diagnostic only; no authority | Target: not authoritative; use the active replacement target contract. Coordinates: not authoritative; use the active replacement coordinates. Mass: not authoritative for canonical handoff. Epsilon: not authoritative for canonical handoff. L: not authoritative for canonical handoff. Verification: cannot issue canonical fresh-verification admission. ESS: not authoritative for canonical handoff. Forbidden: canonical tuning artifact; retained-kernel handoff; public tuning recommendation. Replacement: tune_hmc_kernel. |
| `bayesfilter.inference.fixed_transport_candidate_selection.select_fixed_transport_candidate_set` | diagnostic helper; diagnostic only; no authority | Target: not authoritative; use the active replacement target contract. Coordinates: not authoritative; use the active replacement coordinates. Mass: not authoritative for canonical handoff. Epsilon: not authoritative for canonical handoff. L: not authoritative for canonical handoff. Verification: cannot issue canonical fresh-verification admission. ESS: not authoritative for canonical handoff. Forbidden: canonical tuning artifact; retained-kernel handoff; public tuning recommendation. Replacement: tune_fixed_transport_hmc_kernel. |
| `bayesfilter.inference.hmc_robust_broad_grid.tune_hmc_kernel_robust_broad_grid` | diagnostic helper; diagnostic only; no authority | Target: not authoritative; use the active replacement target contract. Coordinates: not authoritative; use the active replacement coordinates. Mass: not authoritative for canonical handoff. Epsilon: not authoritative for canonical handoff. L: not authoritative for canonical handoff. Verification: cannot issue canonical fresh-verification admission. ESS: not authoritative for canonical handoff. Forbidden: canonical tuning artifact; retained-kernel handoff; public tuning recommendation. Replacement: tune_hmc_kernel. |
| `bayesfilter.inference.hmc_fixed_metric_grid_search.run_fixed_metric_grid_search` | diagnostic helper; diagnostic only; no authority | Target: not authoritative; use the active replacement target contract. Coordinates: not authoritative; use the active replacement coordinates. Mass: not authoritative for canonical handoff. Epsilon: not authoritative for canonical handoff. L: not authoritative for canonical handoff. Verification: cannot issue canonical fresh-verification admission. ESS: not authoritative for canonical handoff. Forbidden: canonical tuning artifact; retained-kernel handoff; public tuning recommendation. Replacement: tune_hmc_kernel. |

| Interface | Status | Target, tuning, verification, and use |
|---|---|---|
| `bayesfilter.inference.hmc_operational_broad_grid.run_operational_broad_grid` | diagnostic helper; diagnostic only; no authority | Target: not authoritative; use the active replacement target contract. Coordinates: not authoritative; use the active replacement coordinates. Mass: not authoritative for canonical handoff. Epsilon: not authoritative for canonical handoff. L: not authoritative for canonical handoff. Verification: cannot issue canonical fresh-verification admission. ESS: not authoritative for canonical handoff. Forbidden: canonical tuning artifact; retained-kernel handoff; public tuning recommendation. Replacement: tune_hmc_kernel. |
| `bayesfilter.inference.hmc_operational_broad_grid.run_operational_broad_grid_process_parallel` | diagnostic helper; diagnostic only; no authority | Target: not authoritative; use the active replacement target contract. Coordinates: not authoritative; use the active replacement coordinates. Mass: not authoritative for canonical handoff. Epsilon: not authoritative for canonical handoff. L: not authoritative for canonical handoff. Verification: cannot issue canonical fresh-verification admission. ESS: not authoritative for canonical handoff. Forbidden: canonical tuning artifact; retained-kernel handoff; public tuning recommendation. Replacement: tune_hmc_kernel. |
| `bayesfilter.inference.hmc_budget_ladder.run_fixed_mass_hmc_tuning_budget_ladder` | diagnostic helper; diagnostic only; no authority | Target: not authoritative; use the active replacement target contract. Coordinates: not authoritative; use the active replacement coordinates. Mass: not authoritative for canonical handoff. Epsilon: not authoritative for canonical handoff. L: not authoritative for canonical handoff. Verification: cannot issue canonical fresh-verification admission. ESS: not authoritative for canonical handoff. Forbidden: canonical tuning artifact; retained-kernel handoff; public tuning recommendation. Replacement: tune_hmc_kernel. |
| `bayesfilter.inference.hmc_tuning.run_fixed_mass_step_tuning_diagnostic` | diagnostic helper; diagnostic only; no authority | Target: not authoritative; use the active replacement target contract. Coordinates: not authoritative; use the active replacement coordinates. Mass: not authoritative for canonical handoff. Epsilon: not authoritative for canonical handoff. L: not authoritative for canonical handoff. Verification: cannot issue canonical fresh-verification admission. ESS: not authoritative for canonical handoff. Forbidden: canonical tuning artifact; retained-kernel handoff; public tuning recommendation. Replacement: tune_hmc_kernel. |

| Interface | Status | Target, tuning, verification, and use |
|---|---|---|
| `bayesfilter.inference.hmc_tuning.run_windowed_mass_adaptation_diagnostic` | diagnostic helper; diagnostic only; no authority | Target: not authoritative; use the active replacement target contract. Coordinates: not authoritative; use the active replacement coordinates. Mass: not authoritative for canonical handoff. Epsilon: not authoritative for canonical handoff. L: not authoritative for canonical handoff. Verification: cannot issue canonical fresh-verification admission. ESS: not authoritative for canonical handoff. Forbidden: canonical tuning artifact; retained-kernel handoff; public tuning recommendation. Replacement: tune_hmc_kernel. |
| `bayesfilter.inference.hmc_tuning.run_fixed_trajectory_tuning_diagnostic` | diagnostic helper; diagnostic only; no authority | Target: not authoritative; use the active replacement target contract. Coordinates: not authoritative; use the active replacement coordinates. Mass: not authoritative for canonical handoff. Epsilon: not authoritative for canonical handoff. L: not authoritative for canonical handoff. Verification: cannot issue canonical fresh-verification admission. ESS: not authoritative for canonical handoff. Forbidden: canonical tuning artifact; retained-kernel handoff; public tuning recommendation. Replacement: tune_hmc_kernel. |
| `bayesfilter.inference.hmc_tuning.run_gaussian_dual_averaging_diagnostic` | diagnostic helper; diagnostic only; no authority | Target: not authoritative; use the active replacement target contract. Coordinates: not authoritative; use the active replacement coordinates. Mass: not authoritative for canonical handoff. Epsilon: not authoritative for canonical handoff. L: not authoritative for canonical handoff. Verification: cannot issue canonical fresh-verification admission. ESS: not authoritative for canonical handoff. Forbidden: canonical tuning artifact; retained-kernel handoff; public tuning recommendation. Replacement: tune_hmc_kernel. |
| `bayesfilter.inference.hmc_kernel_tuning.run_hmc_start_bank_diagnostic` | diagnostic helper; diagnostic only; no authority | Target: not authoritative; use the active replacement target contract. Coordinates: not authoritative; use the active replacement coordinates. Mass: not authoritative for canonical handoff. Epsilon: not authoritative for canonical handoff. L: not authoritative for canonical handoff. Verification: cannot issue canonical fresh-verification admission. ESS: not authoritative for canonical handoff. Forbidden: canonical tuning artifact; retained-kernel handoff; public tuning recommendation. Replacement: tune_hmc_kernel. |

| Interface | Status | Target, tuning, verification, and use |
|---|---|---|
| `bayesfilter.inference.fixed_transport_hmc_candidate_discovery_tf.discover_fixed_transport_hmc_candidates` | diagnostic helper; diagnostic only; no authority | Target: not authoritative; use the active replacement target contract. Coordinates: not authoritative; use the active replacement coordinates. Mass: not authoritative for canonical handoff. Epsilon: not authoritative for canonical handoff. L: not authoritative for canonical handoff. Verification: cannot issue canonical fresh-verification admission. ESS: not authoritative for canonical handoff. Forbidden: canonical tuning artifact; retained-kernel handoff; public tuning recommendation. Replacement: tune_fixed_transport_hmc_kernel. |
| `bayesfilter.inference.fixed_transport_hmc_candidate_discovery_tf.run_fixed_transport_hmc_candidate_campaign` | diagnostic helper; diagnostic only; no authority | Target: not authoritative; use the active replacement target contract. Coordinates: not authoritative; use the active replacement coordinates. Mass: not authoritative for canonical handoff. Epsilon: not authoritative for canonical handoff. L: not authoritative for canonical handoff. Verification: cannot issue canonical fresh-verification admission. ESS: not authoritative for canonical handoff. Forbidden: canonical tuning artifact; retained-kernel handoff; public tuning recommendation. Replacement: tune_fixed_transport_hmc_kernel. |
| `bayesfilter.inference.fixed_transport_hmc_candidate_discovery_tf.refine_fixed_transport_hmc_candidates` | diagnostic helper; diagnostic only; no authority | Target: not authoritative; use the active replacement target contract. Coordinates: not authoritative; use the active replacement coordinates. Mass: not authoritative for canonical handoff. Epsilon: not authoritative for canonical handoff. L: not authoritative for canonical handoff. Verification: cannot issue canonical fresh-verification admission. ESS: not authoritative for canonical handoff. Forbidden: canonical tuning artifact; retained-kernel handoff; public tuning recommendation. Replacement: tune_fixed_transport_hmc_kernel. |
| `bayesfilter.inference.generic_hmc_tuning.run_generic_hmc_tuning_orchestration` | historical helper; historical only; no authority | Target: not authoritative; use the active replacement target contract. Coordinates: not authoritative; use the active replacement coordinates. Mass: not authoritative for canonical handoff. Epsilon: not authoritative for canonical handoff. L: not authoritative for canonical handoff. Verification: cannot issue canonical fresh-verification admission. ESS: not authoritative for canonical handoff. Forbidden: canonical tuning artifact; retained-kernel handoff; public tuning recommendation. Replacement: tune_hmc_kernel. |

| Interface | Status | Target, tuning, verification, and use |
|---|---|---|
| `bayesfilter.inference.generic_hmc_tuning.orchestrate_generic_hmc_tuning` | historical helper; historical only; no authority | Target: not authoritative; use the active replacement target contract. Coordinates: not authoritative; use the active replacement coordinates. Mass: not authoritative for canonical handoff. Epsilon: not authoritative for canonical handoff. L: not authoritative for canonical handoff. Verification: cannot issue canonical fresh-verification admission. ESS: not authoritative for canonical handoff. Forbidden: canonical tuning artifact; retained-kernel handoff; public tuning recommendation. Replacement: tune_hmc_kernel. |
| `bayesfilter.inference.fixed_trajectory_hmc_tuning_v2.run_tiny_gaussian_fixed_trajectory_hmc_tuning_v2` | historical helper; historical only; no authority | Target: not authoritative; use the active replacement target contract. Coordinates: not authoritative; use the active replacement coordinates. Mass: not authoritative for canonical handoff. Epsilon: not authoritative for canonical handoff. L: not authoritative for canonical handoff. Verification: cannot issue canonical fresh-verification admission. ESS: not authoritative for canonical handoff. Forbidden: canonical tuning artifact; retained-kernel handoff; public tuning recommendation. Replacement: tune_hmc_kernel. |
| `bayesfilter.inference.hmc_kernel_tuning.run_hmc_tune_verify_repair_loop` | stage helper; internal only; no authority | Target: prevalidated ordinary adapter, geometry, and bootstrap artifacts. Coordinates: same adapter and mass-coordinate lineage as public tuner. Mass: internal windowed adaptation and repair. Epsilon: internal fixed-mass tuning and repair. L: internal leapfrog-count selection and repair. Verification: internal fresh fixed-kernel verification. ESS: disabled for ordinary tuning admission. Forbidden: final one-call API; canonical artifact authority. Replacement: tune_hmc_kernel. |
| `bayesfilter.inference.hmc.run_full_chain_tfp_hmc` | chain runner; internal only; no authority | Target: adapter exact value and matching score. Coordinates: caller-supplied fixed mass coordinates. Mass: not owned; consumes supplied coordinates. Epsilon: fixed or config-requested dual averaging only. L: fixed caller-supplied leapfrog count. Verification: none by itself. ESS: none by itself. Forbidden: standalone full-tuning claim; canonical artifact authority. Replacement: tune_hmc_kernel. |
| `bayesfilter.inference.fixed_transport_hmc_mechanics_tf.run_fixed_transport_full_chain_tfp_hmc` | chain runner; internal only; no authority | Target: preconstructed exact transformed value/score adapter. Coordinates: frozen-transport latent z coordinates. Mass: not owned; consumes the supplied fixed coordinate system. Epsilon: fixed or config-requested dual averaging only. L: fixed caller-supplied leapfrog count. Verification: none by itself. ESS: none by itself. Forbidden: standalone full-tuning claim; candidate-selection claim; canonical artifact authority. Replacement: tune_fixed_transport_hmc_kernel. |

| Interface | Status | Target, tuning, verification, and use |
|---|---|---|
| `bayesfilter.inference.neural_force_hmc.run_full_chain_neural_force_hmc` | chain runner; diagnostic only; no authority | Target: frozen position-only force plus exact deterministic endpoint potential. Coordinates: native affine mass coordinates when bound; direct identity fallback is diagnostic only. Mass: not owned; identity in supplied mass coordinates. Epsilon: fixed or config-requested dual averaging only. L: fixed caller-supplied leapfrog count. Verification: none by itself. ESS: none by itself. Forbidden: standalone full-tuning claim; mass-tuning claim; leapfrog-count-tuning claim; canonical artifact authority. Replacement: tune_hmc_kernel. |
| `bayesfilter.inference.neural_force_hmc.bind_neural_force_hmc_tuning_runner` | runner binding factory; tested supported; no authority | Target: coordinate-consistent frozen force and exact raw-coordinate endpoint potential. Coordinates: TensorFlow tuning branch native affine mass wrapper; identity fallback for a direct low-level call is diagnostic only. Mass: owned by the typed TensorFlow mechanics branch. Epsilon: owned by the typed TensorFlow mechanics branch. L: shared broad L grid (3, 5, 9, 13, 18, 25) within the declared maximum; explicit all-survivor and unresolved-interval refinement; retain every verified pair. Verification: fresh injected-runner fixed-kernel health and acceptance verification. ESS: disabled for ordinary tuning admission. Forbidden: standalone tuning authority; bare callable injection; transformed target masquerading as ordinary raw coordinates. Replacement: tune_hmc_kernel. |

# Chapter 2

# Modern Hamiltonian Monte Carlo for Structural State-Space Inference

A posterior distribution may take a long time to explore because its geometry is difficult, because each likelihood evaluation is expensive, or because the sampler uses its evaluations poorly. These explanations lead to different repairs. A nonlinear change of coordinates can remove a funnel without making a Kalman recursion cheaper. A parallel filter can reduce evaluation latency without making a poorly chosen trajectory explore the posterior. A new trajectory rule can reduce gradient work without correcting an inaccurate likelihood. For structural state-space inference, the useful question is therefore which combination of geometry, dynamics, adaptation and filtering produces accurate estimates of economically meaningful quantities at a reasonable total cost.

Recent work offers several answers. ChEES and SNAPER learn trajectory lengths that suit parallel computation; MALT and MEADS use persistent or partially refreshed dynamics; MAMS changes the dynamics to a microcanonical construction; and LAPS uses an unadjusted ensemble transient before switching to adjusted sampling. Neural transport and Pathfinder address geometry and initialization. Parallel Gaussian message passing and gradient-informed particle methods exploit the state-space structure itself [Hoffman et al., 2021, Sountsov and Hoffman, 2022, Riou-Durand and Vogrinc, 2023, Hoffman and Sountsov, 2022, Robnik et al., 2025, Robnik and Seljak, 2026, Hoffman et al., 2019, Zhang et al., 2022, Särkkä and García-Fernández, 2020, Corenflos and Finke, 2024].

This chapter develops the mathematical constructions needed to assess these methods. The central arguments are given in full: the probability measure being sampled, the integration and acceptance corrections, the quantities adapted, and the diagnostic assumptions. General ergodicity and optimal-scaling theorems are cited with their conditions rather than asserted for arbitrary economic models. The discussion covers work published or revised during approximately 2016–2026, with older HMC and particle-MCMC foundations included where they are needed to make the account self-contained. Author benchmarks are evidence about their stated experiments. They are not measurements of BayesFilter, MacroFinance or `dsge_hmc`.

## 2.1 The posterior, its coordinates and its computational cost

### 2.1.1 Marginal parameters and latent trajectories are different targets

Let $y_{1:T}$ be observations, $s_{0:T}$ latent states and $\vartheta$ structural parameters. An initial density $m_0$, transition density $m_t$ and observation density $g_t$ define

$$p(\vartheta, s_{0:T} \mid y_{1:T}) \propto p(\vartheta) m_0(s_0 \mid \vartheta) \prod_{t=1}^{T} m_t(s_t \mid s_{t-1}, \vartheta) g_t(y_t \mid s_t, \vartheta), \tag{2.1}$$

$$L(\vartheta) = \int m_0(s_0 \mid \vartheta) \prod_{t=1}^{T} m_t(s_t \mid s_{t-1}, \vartheta) g_t(y_t \mid s_t, \vartheta)\, ds_{0:T}, \tag{2.2}$$

$$p(\vartheta \mid y_{1:T}) \propto p(\vartheta) L(\vartheta). \tag{2.3}$$

Sampling (2.1) introduces an unknown at every state and time. Sampling (2.3) can be low-dimensional even when the state vector and observation horizon are large. For a linear Gaussian model, Kalman filtering evaluates the integral exactly in real arithmetic. A nonlinear filter instead defines an approximation $\widehat{L}(\vartheta)$. An exact MCMC transition for $p(\vartheta)\widehat{L}(\vartheta)$ does not turn that approximation into $p(\vartheta)L(\vartheta)$. Sampler error and likelihood approximation error must be analyzed separately.

The prediction-error decomposition makes the cost visible:

$$\ell(\vartheta) = \log L(\vartheta) = \sum_{t=1}^{T} \log p(y_t \mid y_{1:t-1}, \vartheta). \tag{2.4}$$

An integrator may evaluate $\nabla \ell$ hundreds of times for one retained transition. Reducing the time per evaluation and reducing the number of evaluations are distinct opportunities. A useful accounting is

$$C_{\text{total}} = C_{\text{compile}} + C_{\text{initialize/map}} + C_{\text{adapt/verify}} + C_{\text{retained}}. \tag{2.5}$$

Each term includes likelihood, derivative, integration, diagnostic and host work. When $B$ chains are batched, write $c_\nabla(B)$ for the measured latency of one batched gradient call. Neither $c_\nabla(B) = c_\nabla(1)$ nor $c_\nabla(B) = Bc_\nabla(1)$ should be assumed. Accelerator advantages depend on the tensor shapes, arithmetic intensity, memory and control flow of the actual target [Sountsov et al., 2024].

### 2.1.2 Constraints and a frozen transport

Suppose the model chart $\vartheta = C(q)$ maps $q \in \mathbb{R}^d$ bijectively to the admissible parameter space. A further frozen transport $q = F(z)$ may improve geometry. Assume the maps are differentiable with nonsingular Jacobians on the domain of use. Both maps act between $d$-dimensional coordinate domains, so $J_C, J_F \in \mathbb{R}^{d \times d}$. Assume twice differentiability when differentiating their log determinants. With $J_C$ and $J_F$ denoting those Jacobians, change of variables gives

$$\log \widetilde{\pi}_q(q) = \log p(C(q)) + \ell(C(q)) + \log |\det J_C(q)|, \tag{2.6}$$

$$\log \widetilde{\pi}_z(z) = \log \widetilde{\pi}_q(F(z)) + \log |\det J_F(z)|. \tag{2.7}$$

The tilde indicates that a normalization constant is unnecessary. The score must differentiate the same expression. The chain rule and Jacobi's determinant formula imply

$$\nabla_z \log \widetilde{\pi}_z(z) = J_F(z)^\top \nabla_q \log \widetilde{\pi}_q(F(z)) + \nabla_z \log |\det J_F(z)|, \tag{2.8}$$

$$\partial_{z_j} \log |\det J_F(z)| = \operatorname{tr}\{J_F(z)^{-1} \partial_{z_j} J_F(z)\}. \tag{2.9}$$

To obtain the second line, use $d \det J = \det J \operatorname{tr}(J^{-1} dJ)$ and divide by $\det J$; its sign is locally constant when $J$ is nonsingular. Freezing the learned coefficients of $F$ removes adaptation during sampling. It does not remove the derivative through $F(z)$ or the log determinant.

For example, the positive chart $\vartheta = e^q$ adds $q$ to the log density. Its score is $e^q \partial_\vartheta [\log p(\vartheta) + \ell(\vartheta)] + 1$. Omitting the 1 computes the score of a different density. A numerical acceptance statistic cannot establish that the intended target was used.

## 2.2 Corrected HMC and the scope of its guarantee

### 2.2.1 Hamiltonian invariance

From this point let $x \in \mathbb{R}^d$ denote the coordinates actually sampled, let $\pi(x) \propto e^{-U(x)}$ be normalizable, and suppose $U$ is twice continuously differentiable wherever the smooth-flow argument is used. Assume the exact flow exists over the time interval under consideration. Introduce $p \in \mathbb{R}^d$ with $p \sim \mathcal{N}(0, M)$ and fixed symmetric positive-definite mass $M \in \mathbb{R}^{d \times d}$. The joint density is proportional to $e^{-H(x,p)}$, where

$$H(x, p) = U(x) + K(p), \qquad K(p) = \tfrac{1}{2} p^\top M^{-1} p. \tag{2.10}$$

Hamilton's equations are

$$\dot{x} = M^{-1} p, \qquad \dot{p} = -\nabla U(x). \tag{2.11}$$

Along a solution,

$$\frac{dH}{dt} = \nabla U(x)^\top M^{-1} p - (M^{-1} p)^\top \nabla U(x) = 0. \tag{2.12}$$

The phase-space velocity has divergence zero: $M^{-1} p$ has no $x$ dependence and $-\nabla U(x)$ has no $p$ dependence. Liouville's formula then gives a unit Jacobian for the flow. Energy and volume preservation together preserve $e^{-H} dx\, dp$. Momentum reversal $R(x, p) = (x, -p)$ reverses time because $K$ is even. These are the foundations of HMC [Neal, 2011, Sections 2 and 3].

The leapfrog approximation consists of three shears:

$$p_{1/2} = p - \tfrac{\epsilon}{2} \nabla U(x), \tag{2.13}$$

$$x' = x + \epsilon M^{-1} p_{1/2}, \tag{2.14}$$

$$p' = p_{1/2} - \tfrac{\epsilon}{2} \nabla U(x'). \tag{2.15}$$

The kick Jacobian is $\left(\begin{smallmatrix} I & 0 \\ -\epsilon \nabla^2 U/2 & I \end{smallmatrix}\right)$ and the drift Jacobian is $\left(\begin{smallmatrix} I & \epsilon M^{-1} \\ 0 & I \end{smallmatrix}\right)$. Both determinants equal one. The palindromic ordering also gives $R \Phi_\epsilon R = \Phi_\epsilon^{-1}$. Consequently $\Psi = R \Phi_\epsilon^L$ is an involution. Numerical energy error remains; the next argument explains how to correct it.

### 2.2.2 Metropolis correction for a general involution

**Proposition 2.1** (Deterministic Metropolis correction)**.** *Let $\rho(w)$ be a density relative to a specified volume measure, and let $\Psi$ be a differentiable bijection with $\Psi^2 = I$. Write $J(w) = |\det D\Psi(w)|$ in local coordinates for that measure. The proposal $w' = \Psi(w)$, accepted with*

$$a(w) = \min\left\{1, \frac{\rho(\Psi(w))J(w)}{\rho(w)}\right\}, \tag{2.16}$$

*and otherwise left at $w$, satisfies detailed balance.*

*Proof.* Differentiating $\Psi(\Psi(w)) = w$ gives $J(\Psi(w))J(w) = 1$. Hence

$$\begin{aligned}\rho(w)a(w) &= \min\{\rho(w), \rho(\Psi(w))J(w)\} \\ &= \rho(\Psi(w))a(\Psi(w))J(w).\end{aligned}$$

Multiplication by $dw$, followed by the change of variables $w' = \Psi(w)$, equates accepted probability flow in both directions. Rejected probability stays at the same state and is automatically symmetric. □

For leapfrog, $J = 1$ and $\rho \propto e^{-H}$, so

$$a = \min\{1, e^{-\Delta H}\}, \qquad \Delta H = H(x', p') - H(x, p). \tag{2.17}$$

The proof shows why a different integrator with non-unit Jacobian needs a different correction. It also shows the limit of the guarantee: invariance means that a stationary input remains stationary. It does not state how quickly arbitrary initial states approach stationarity or whether a finite run visits every relevant mode.

### 2.2.3 A deterministic force need not be a gradient

Replace $-\nabla U(x)$ in a kick by a deterministic position-only field $b(x)$. The map $(x, p) \mapsto (x, p + hb(x))$ still has triangular Jacobian with determinant one, inverse given by $-h$, and the required momentum-reversal symmetry. Its symmetric composition with a drift can therefore be corrected against the actual endpoint Hamiltonian using (2.17). It need not be a symplectic Hamiltonian integrator, and it need not be efficient, but the volume-and-involution proof does not require $b$ to be a gradient.

This observation does not authorize an arbitrary force in every HMC extension. Energy conservation in (2.12), the microcanonical stationary-density calculation below, and score-based equipartition identities use the actual gradient of the declared potential. History-dependent, noisy or statefully changing forces also need an enlarged state and a new reversibility argument. The inspected BGS documentation in `dsge_hmc` declares an endpoint UKF target and a position-only field not asserted to be its exact score. Its mechanics must be assessed under that contract rather than silently reclassified as an exact-gradient method.

## 2.3 Trajectory length, resonance and learned adaptation

### 2.3.1 A Gaussian calculation that separates acceptance from exploration

Take $U(x) = x^2/(2\sigma^2)$ and unit mass. Solving the harmonic oscillator in (2.11) gives

$$x_\tau = x_0 \cos(\tau/\sigma) + \sigma p_0 \sin(\tau/\sigma), \tag{2.18}$$

$$p_\tau = p_0 \cos(\tau/\sigma) - \frac{x_0}{\sigma} \sin(\tau/\sigma). \tag{2.19}$$

With fresh $p_0 \sim \mathcal{N}(0,1)$ at each transition, the stationary position chain is an AR(1) process with coefficient $r = \cos(\tau/\sigma)$. For $|r| < 1$,

$$\mathrm{Corr}(X_0, X_k) = r^k, \qquad \mathrm{Corr}(X_0^2, X_k^2) = r^{2k}. \tag{2.20}$$

To obtain the second identity, a centered bivariate Gaussian pair with variance $\sigma^2$ and correlation $r^k$ satisfies $\mathbb{E}[X_0^2 X_k^2] = \sigma^4(1+2r^{2k})$; subtract $\sigma^4$ and divide by $\mathrm{Var}(X^2) = 2\sigma^4$. Summing the geometric series gives integrated autocorrelation times

$$\tau_{\mathrm{int}}(X) = \frac{1+r}{1-r}, \qquad \tau_{\mathrm{int}}(X^2) = \frac{1+r^2}{1-r^2}. \tag{2.21}$$

Near a half-period, $r \approx -1$: the mean can look unusually efficient while the squared quantity mixes very slowly. At an exact half-period $X' = -X$, and at a full period $X' = X$. These limiting chains are not covered by the summable-correlation formulas. Both have acceptance one under exact integration, yet neither explores the distribution of $|X|$.

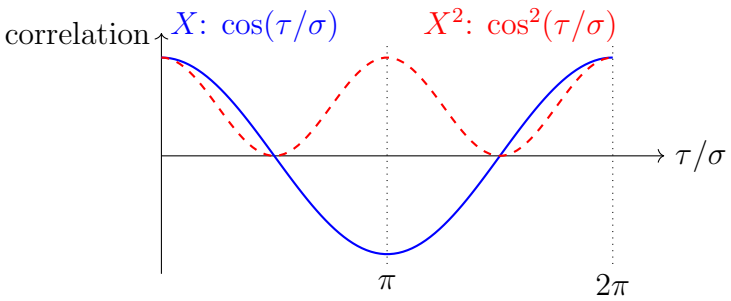



Figure 2.1: Exact Gaussian dynamics. The curves are analytic lag-one correlations, not benchmark results. Half-period trajectories change the sign of a draw but preserve its square; full-period trajectories preserve both.

### 2.3.2 Random lengths and NUTS

If $P_\tau$ preserves $\pi$ for each fixed $\tau$, and $G$ is a state-independent distribution over lengths, then

$$\int \pi(dx) \int P_\tau(x, A) G(d\tau) = \int \pi(A) G(d\tau) = \pi(A). \tag{2.22}$$

Thus randomizing length preserves invariance without a new correction, provided each component kernel is valid and the distribution is fixed during retained sampling. It spreads the risk of resonance. If $\tau \sim \mathrm{Unif}(0, 2\bar{\tau})$ independently at every Gaussian transition, direct integration yields

$$r_1 = \mathbb{E}[\cos(\tau/\sigma)] = \frac{\sin(2a)}{2a}, \tag{2.23}$$

$$r_2 = \mathbb{E}[\cos^2(\tau/\sigma)] = \frac{1}{2} + \frac{\sin(4a)}{8a}, \qquad a = \bar{\tau}/\sigma. \tag{2.24}$$

Here $X$ and $X^2 - \sigma^2$ are eigenfunctions of the mixture with eigenvalues $r_1$ and $r_2$. In particular, choosing $2\bar{\tau} = 2\pi\sigma$ gives $r_1 = 0$ but $r_2 = 1/2$: the mean is decorrelated in one step while variance estimation still has integrated autocorrelation time three.

NUTS uses the observed trajectory to decide how far to simulate [Hoffman and Gelman, 2014, Section 3.1 and Algorithm 3]. In Euclidean coordinates the familiar turning conditions involve $(x_+ - x_-)^\top M^{-1} p_-$ and $(x_+ - x_-)^\top M^{-1} p_+$. Simply stopping when one becomes negative and accepting the endpoint is not the NUTS algorithm: a state-dependent stopping rule is not covered by (2.22). Tree building, candidate selection and the exclusion of asymmetric subtrees provide the required balance. That machinery removes a difficult manual length choice but produces irregular work across chains. The extent of the resulting accelerator cost depends on the implementation and target; it is not determined by the name NUTS alone.

### 2.3.3 ChEES and the stationary squared-jump identity

For a square-integrable function $f$, define

$$D_f = \tfrac{1}{2}\mathbb{E}[(f(X') - f(X))^2]. \tag{2.25}$$

At stationarity, $f(X)$ and $f(X')$ have equal means and variances, so expansion of the square gives

$$D_f = \mathrm{Var}_\pi(f) - \mathrm{Cov}(f(X), f(X')) = \mathrm{Var}_\pi(f)\{1 - \rho_f(1)\}. \tag{2.26}$$

This connects a local movement objective to one-step correlation. It does not determine every higher-lag correlation. In a reversible stationary chain, a spectral representation makes the limitation precise. Normalize the spectral measure $\mu_f$ to have total mass one. If the relevant sums are integrable, then

$$\rho_f(1) = \int \lambda \, \mu_f(d\lambda), \qquad \tau_{\mathrm{int}}(f) = \int \frac{1+\lambda}{1-\lambda} \, \mu_f(d\lambda). \tag{2.27}$$

The function $(1+\lambda)/(1-\lambda)$ is convex on $(-1, 1)$ because its second derivative is $4/(1-\lambda)^3$. Jensen's inequality therefore implies

$$\mathrm{ESS}_f = \frac{n}{\tau_{\mathrm{int}}(f)} \le n \frac{1 - \rho_f(1)}{1 + \rho_f(1)}. \tag{2.28}$$

The right side is an upper bound, not a guarantee that maximizing a jump criterion maximizes actual ESS [Sountsov and Hoffman, 2022, Appendix B].

ChEES uses the centered squared radius

$$f_{\mathrm{ChEES}}(x) = \tfrac{1}{2}\|x-\mu\|^2, \qquad \mu = \mathbb{E}_\pi[X]. \tag{2.29}$$

With the convention in (2.25), the original paper's ChEES criterion is $2D_{f_{\mathrm{ChEES}}}$; this constant factor does not change its maximizer. The center is estimated during adaptation. For independent Gaussian coordinates with variances $\sigma_i^2$, the calculation above gives

$$D_{f_{\mathrm{ChEES}}}(\tau) = \tfrac{1}{2}\sum_i \sigma_i^4 \sin^2(\tau/\sigma_i). \tag{2.30}$$

To derive the coefficient, the variance of $X_i^2/2$ is $\sigma_i^4/2$, its lag-one correlation is $\cos^2(\tau/\sigma_i)$, and cross-coordinate terms vanish. Larger variance directions receive weight $\sigma_i^4$, but sufficiently many moderate directions can still outweigh one slow direction. Also, a large jump obtained using a long trajectory may be poor value per gradient. These motivate SNAPER [Hoffman et al., 2021, Sountsov and Hoffman, 2022].

### 2.3.4 SNAPER, endpoint derivatives and freezing

Let $w$ be a unit principal direction estimated from the current ensemble. SNAPER uses

$$f_w(x) = \{w^\top(x-\mu)\}^2, \qquad C_w(\tau) = \frac{D_{f_w}(\tau)}{\tau}. \tag{2.31}$$

Dividing by $\tau$ accounts for integration cost at fixed epsilon; optimizing along $w$ focuses on a difficult direction rather than the sum of all directions. Both changes remain proxies. The principal covariance direction need not be the most difficult nonlinear or multimodal direction.

The pathwise derivative can be obtained without differentiating every integration step. For an exact trajectory with initial state held fixed, $dx_\tau/d\tau = M^{-1}p_\tau$. Holding $w$ and $\mu$ fixed while differentiating,

$$\nabla f_w(x) = 2w\, w^\top(x-\mu), \tag{2.32}$$

$$\frac{d}{d\tau}\frac{\{f_w(x_\tau)-f_w(x_0)\}^2}{2\tau} = \frac{f_w(x_\tau)-f_w(x_0)}{\tau}\nabla f_w(x_\tau)^\top M^{-1}p_\tau - \frac{\{f_w(x_\tau)-f_w(x_0)\}^2}{2\tau^2}. \tag{2.33}$$

This is the quotient rule plus the endpoint velocity. It explains why a trajectory objective need not require filter Hessians. A numerical algorithm uses an annotated endpoint derivative and treats the rounded step count and accept/reject behavior carefully; the exact derivative above is not an exact derivative of a discontinuous rounded-and-rejected program [Sountsov and Hoffman, 2022, Appendix C]. Such gradients propose adaptation updates. They do not enter the acceptance ratio.

A principal direction can be estimated by stochastic power iteration. If $z = x-\mu$, then $\mathbb{E}[zz^\top w] = \mathrm{Cov}(X)w$. An Oja-type update takes the tangent component

$$\widetilde{w} = w + \eta\{zz^\top w - w(w^\top z)^2\}, \qquad w^+ = \widetilde{w}/\|\widetilde{w}\|. \tag{2.34}$$

Its expectation moves toward covariance eigenvectors, subject to the usual eigenvalue separation and stochastic-approximation conditions. An estimated center, diagonal mass, length parameters and epsilon all change the transition during adaptation. Their final

values must be frozen before the retained phase is interpreted using a fixed-kernel argument.

TensorFlow Probability 0.25.0 provides the kernel `SNAPERHamiltonianMonteCarlo`. It adapts diagonal scaling, the principal direction and trajectory length, but not epsilon; the `sample_snaper_hmc` helper supplies dual averaging. The paper's SGA implementation uses its own synthetic acceptance update, so the helper should not be described as byte-for-byte reproduction of that tuning scheme. The implementation requires at least two chains. Its concentrated-start recommendation is an adaptation choice, not a justification for discarding independent dispersed verification starts. Source availability reduces porting effort; complete target, graph, GPU and checkpoint compatibility still needs testing.

## 2.4 Partial refreshment, MALT and MEADS

### 2.4.1 The Ornstein–Uhlenbeck refreshment

An underdamped Langevin process with mass $M$ and scalar friction $\gamma$ can be written

$$dX_t = M^{-1}P_t\,dt, \qquad dP_t = -\nabla U(X_t)\,dt - \gamma P_t\,dt + \sqrt{2\gamma}M^{1/2}dW_t. \tag{2.35}$$

During a refreshment substep with $X$ fixed, the linear stochastic equation has solution

$$P' = aP + \sqrt{1-a^2}M^{1/2}\xi, \qquad a = e^{-\gamma h}, \quad \xi \sim \mathcal{N}(0, I). \tag{2.36}$$

Indeed, multiply the momentum equation by $e^{\gamma t}$ and integrate. The stochastic integral has covariance $2\gamma M\int_0^h e^{-2\gamma(h-s)}ds = (1-e^{-2\gamma h})M$. If $P \sim \mathcal{N}(0, M)$, then $\mathrm{Var}(P') = a^2M + (1-a^2)M = M$. Moreover $(P, P')$ is jointly Gaussian with equal marginal covariances and symmetric cross-covariance $aM$, so its law is unchanged by exchange. The refreshment kernel $Q$ therefore satisfies

$$e^{-K(p)}Q(p, dp') = e^{-K(p')}Q(p', dp). \tag{2.37}$$

The parameter $a$ trades momentum persistence against randomization. It does not remove the need to choose a numerical step size.

### 2.4.2 Why MALT sums deterministic energy errors

MALT composes refreshment and leapfrog substeps in a symmetric Langevin trajectory [Riou-Durand and Vogrinc, 2023, Sections 2–3]. Consider the complete realized path. Its probability ratio under time reversal contains the endpoint density ratio $e^{-[H_{\mathrm{end}}-H_{\mathrm{start}}]}$, the Jacobians of deterministic maps, and the ratios of refreshment probabilities. The deterministic leapfrog Jacobians are one. From (2.37), each refreshment contributes

$$\frac{Q(p', dp)}{Q(p, dp')} = e^{K(p')-K(p)}. \tag{2.38}$$

Decompose the total energy change into changes during deterministic leapfrog substeps, $\Delta H_j^{\mathrm{D}}$, and refreshment changes $\Delta K_j^{\mathrm{O}}$. The path log ratio becomes

$$-\Delta H_{\mathrm{total}} + \sum_j \Delta K_j^{\mathrm{O}} = -\sum_j \Delta H_j^{\mathrm{D}}. \tag{2.39}$$

The OU terms cancel rather than being counted as numerical energy errors. A reversal of the complete path, with the momentum sign convention and palindromic refreshment schedule, provides the extended-space involution. Accepting with $1 \wedge \exp(-\sum_j \Delta H_j^{\mathrm{D}})$ is therefore the relevant correction. Full momentum refresh at trajectory boundaries allows the accepted position transition to be used without inheriting a persistent rejected momentum.

For unit mass, the deterministic local error can be expressed using positions alone. Let $y = x + hv_{1/2}$, and write $g_x = \nabla U(x)$, $g_y = \nabla U(y)$. The endpoint velocities of the leapfrog substep are

$$v_0 = (y - x)/h + (h/2)g_x, \qquad v_1 = (y - x)/h - (h/2)g_y. \tag{2.40}$$

Expanding the squared norms gives

$$E_h(x, y) = U(y) - U(x) + \tfrac{1}{2}(\|v_1\|^2 - \|v_0\|^2) \tag{2.41}$$

$$= U(y) - U(x) - \tfrac{1}{2}(y - x)^\top (g_y + g_x) + \tfrac{h^2}{8}(\|g_y\|^2 - \|g_x\|^2). \tag{2.42}$$

This reproduces the paper's local error, equation (5). Summing it along the path yields the acceptance exponent. Using only the total endpoint Hamiltonian difference, including refreshment randomness, computes a different ratio.

Adaptive MALT estimates scaling, damping, epsilon and trajectory length [Riou-Durand et al., 2023]. Its empirical results distinguish efficiency per iteration from efficiency per gradient. It is sometimes slightly less gradient-efficient than uniformly randomized HMC using SNAPER adaptation, despite competitive iteration efficiency. The author implementation in TensorFlow Probability's discussion directory uses JAX/FunMC; its location does not make it an installed TensorFlow transition kernel.

### 2.4.3 Generalized HMC and ensemble adaptation

Generalized HMC retains part of its momentum between transitions. One valid construction composes OU refreshment with the involutive leapfrog proposal $R\Phi$, accepts or rejects against $H$, and then applies an unconditional momentum flip $R$. The final flip preserves the joint target. In this equivalent convention, an accepted move ends with the forward momentum, whereas a rejected move flips it. Repeated rejections can therefore undo directional progress; correct momentum handling is part of the algorithm, not a cosmetic implementation detail.

MEADS adds ensemble-based scale adaptation and a persistent slice construction [Hoffman and Sountsov, 2022]. A slice variable has joint unnormalized density

$$\rho(x, p, s) = \mathbf{1}\{0 < s < e^{-H(x,p)}\}. \tag{2.43}$$

Integrating over $s$ gives $e^{-H(x,p)}$. A volume-preserving reversible proposal at fixed $s$ can move only within the slice; a valid update of $s$ must preserve this joint measure. Reusing an arbitrary uniform variate across ordinary MH tests is not sufficient. The paper supplies a particular persistent-slice transition, together with ensemble estimates that avoid direct Hessian evaluation.

The conditional-independence argument behind folded ensemble adaptation can be stated precisely. Partition chains into a block $A$ and its complement. Choose parameters

$\lambda = \Lambda(X_{-A})$ only from the complementary block. Conditional on $X_{-A}$, suppose the transition $P_\lambda$ for $A$ preserves $\pi^{\otimes|A|}$. Then

$$\int \pi^{\otimes|A|}(dx_A)\,\pi^{\otimes|A^c|}(dx_{-A})P_{\Lambda(x_{-A})}(x_A, dx'_A) \tag{2.44}$$

$$= \pi^{\otimes|A|}(dx'_A)\,\pi^{\otimes|A^c|}(dx_{-A}). \tag{2.45}$$

The product target is preserved. Sequentially changing the held-out block preserves it as well. If the parameter estimate uses the very state being updated, the displayed conditional argument no longer applies. MEADS's fold construction must therefore be preserved when claiming its exact ensemble-adaptation property. For a first local implementation, finite adaptation followed by a completely frozen kernel is simpler to validate.

## 2.5 Microcanonical dynamics and the MAMS correction

### 2.5.1 A velocity of fixed norm

MAMS uses a direction $u$ on the unit sphere $S^{d-1}$ instead of a Gaussian momentum [Robnik et al., 2025]. For $d > 1$, define

$$\dot{x} = u, \qquad \dot{u} = -\frac{1}{d-1}(I - uu^\top)g(x), \qquad g(x) = \nabla U(x). \tag{2.46}$$

The projected force changes direction while preserving speed. Indeed,

$$\frac{d}{dt}\|u\|^2 = -\frac{2}{d-1}\{u^\top g - \|u\|^2 u^\top g\} = 0 \quad \text{when } \|u\| = 1. \tag{2.47}$$

The factor $d-1$ is the dimension of the tangent space. It makes this formula undefined at $d = 1$; a one-dimensional specialization requires a different construction, not an arbitrary division guard.

The name microcanonical can obscure an important distinction: these equations are not ordinary canonical Hamiltonian equations in $(x, u)$ with Gaussian kinetic energy. They are a time-rescaled, fixed-speed construction with stationary measure

$$\rho(x, u)\,dx\,d\omega(u) = \pi(x)\,dx\,\frac{d\omega(u)}{|S^{d-1}|}, \tag{2.48}$$

where $d\omega$ is sphere surface measure. The measure, and not merely the dimension of an array storing $u$, determines the Jacobian in the acceptance calculation.

### 2.5.2 Derivation of the stationary measure

For fixed $a \in \mathbb{R}^d$, the tangent field $v(u) = (I - uu^\top)a$ has intrinsic divergence

$$\operatorname{div}_{S^{d-1}} v = -(d-1)u^\top a. \tag{2.49}$$

To see this without a coordinate chart, let $P = I - uu^\top$ and use an orthonormal tangent frame. The ambient derivative is $Dv = -ua^\top - (u^\top a)I$. Taking the tangent trace gives $\operatorname{tr}(PDvP) = -(u^\top a)\operatorname{tr}(P) = -(d-1)u^\top a$, since $Pu = 0$. With $a = -g/(d-1)$, the

velocity field in (2.46) therefore has divergence $u^\top g$. Its position component has zero position divergence. The stationary continuity equation is satisfied because

$$\operatorname{div}(\rho b) = \rho\{\operatorname{div} b + b \cdot \nabla \log \rho\} \tag{2.50}$$

$$= \rho\{u^\top g - u^\top g\} = 0. \tag{2.51}$$

This derivation assumes sufficient smoothness and a nonexplosive flow for the continuity argument. It establishes the invariant density of the exact dynamics. It does not prove ergodicity of the deterministic motion or remove finite-step integration bias [Robnik et al., 2025, Appendix B.4].

The comparison with canonical HMC is useful. In canonical coordinates the flow preserves volume and density separately through energy conservation. Here density changes along the position trajectory and volume changes in the sphere directions compensate. The numerical method must keep track of that compensation.

### 2.5.3 The exact force subflow

Split the dynamics into a drift $A_h : (x, u) \mapsto (x + hu, u)$ and a force subflow $B_h$ holding $x$ fixed. For nonzero $g$, put

$$e = -g/\|g\|, \qquad k = \|g\|/(d-1), \qquad c_0 = e^\top u_0, \qquad \delta = kh. \tag{2.52}$$

Then $\dot{u} = k(e - (e^\top u)u)$. The scalar projection $c = e^\top u$ satisfies $\dot{c} = k(1 - c^2)$. Integrating $dc/(1-c^2) = k\,dt$ gives

$$c_h = \frac{\sinh\delta + c_0\cosh\delta}{\cosh\delta + c_0\sinh\delta}. \tag{2.53}$$

For $|c_0| < 1$ this follows from the addition formula for tanh; the formula also has well-defined limits at $c_0 = \pm 1$. Write $r = u - ce$, with $e^\top r = 0$. The transverse equation is $\dot{r} = -kcr$. Define $D_h = \cosh\delta + c_0\sinh\delta$. Since $d\log D_t/dt = kc_t$, integration yields $r_h = r_0/D_h$. Combining the parallel and perpendicular components gives the closed form

$$B_h(u_0) = \frac{u_0 + e\{\sinh\delta + (\cosh\delta - 1)c_0\}}{D_h}. \tag{2.54}$$

The denominator is positive for finite $\delta$ and $c_0 \in [-1, 1]$, since $\cosh\delta > |\sinh\delta|$. A direct norm check uses

$$1 - c_0^2 + (\sinh\delta + c_0\cosh\delta)^2 = (\cosh\delta + c_0\sinh\delta)^2 = D_h^2. \tag{2.55}$$

Thus the update remains on the sphere. At $g = 0$, the correct limiting update is $B_h(u) = u$; constructing $e$ by dividing by zero is unnecessary.

### 2.5.4 The surface Jacobian and accumulated work

During $B_h$, the sphere divergence is $u^\top g = -(d-1)kc$. Liouville's formula on the sphere and the expression for $D_h$ imply

$$\log J_{B_h}(u_0) = \int_0^h \operatorname{div}_{S^{d-1}} \dot{u}_t\,dt \tag{2.56}$$

$$= -(d-1)\int_0^h kc_t\,dt = -(d-1)\log D_h. \tag{2.57}$$

This is the negative of the paper's quantity $\Delta K = (d-1)\log D_h$ [Robnik et al., 2025, equation (8)]. Here $\Delta K$ denotes a volume correction; it is not a change in $\|u\|^2/2$, which is constant.

A symmetric composition such as $B_{h/2}A_hB_{h/2}$ is reversible under $R(x,u) = (x,-u)$. The drift is volume preserving, while force substeps contribute (2.57). Proposition 2.1 therefore yields

$$\log r = -U(x') + U(x) + \sum_j \log J_{B_j}, \tag{2.58}$$

$$W = U(x') - U(x) + (d-1)\sum_j \log D_j, \tag{2.59}$$

$$a = \min\{1, e^{-W}\}. \tag{2.60}$$

Potential changes telescope across drift substeps. The signs matter: subtracting rather than adding $\Delta K$ in $W$ would produce the wrong accepted measure. The inspected BlackJAX implementation accumulates this integrator work and adds the endpoint log-density difference with precisely this sign.

For numerical stability, the equivalent identity

$$D_h = \tfrac{1}{2}\{(1+c_0)e^{\delta} + (1-c_0)e^{-\delta}\} \tag{2.61}$$

permits a log-sum-exp calculation of $\log D_h$, including limiting zero weights when $c_0 = \pm 1$. The vector update needs a correspondingly stable implementation for large $|\delta|$. A guard that merely clips $\delta$ changes the map and its Jacobian; it cannot silently retain the original correction. Reversibility, surface Jacobian and zero-gradient limits should be tested together.

### 2.5.5 Sphere refreshment and the stochastic variant

A full refresh is $u = \xi/\|\xi\|$ for $\xi \sim \mathcal{N}(0, I)$; rotational invariance gives the uniform sphere law. A partial refresh used in microcanonical implementations has the form [Robnik and Seljak, 2025, Robnik et al., 2025]

$$u' = \frac{u + \eta\xi}{\|u + \eta\xi\|}, \qquad \eta > 0. \tag{2.62}$$

At a fixed state-independent $\eta$, its transition density with respect to surface measure is

$$q(u' \mid u) = (2\pi\eta^2)^{-d/2} \int_0^\infty r^{d-1} \exp\left[-\frac{r^2 + 1 - 2ru^\top u'}{2\eta^2}\right] dr. \tag{2.63}$$

This follows by writing a Gaussian draw in polar coordinates. The expression depends only on $u^\top u'$, so $q(u' \mid u) = q(u \mid u')$ and the uniform sphere law is reversible. The zero-noise limit is the identity transition. In a symmetric stochastic path, these refreshment ratios cancel; the deterministic subflow Jacobians still accumulate. The extended-path argument in MAMS's Appendix A handles the actual noise and rejection conventions. State-dependent refreshment strengths would require rechecking the ratio.

MAMS consequently has several controls: a geometry or preconditioner, epsilon, a trajectory length and, in its Langevin variant, a decoherence scale. The paper uses automatic adaptation, including a preliminary preconditioner estimate—in practice from

NUTS—and length selection using autocorrelation information. Its practical acceptance target is 0.90, with 0.99 used for its funnel experiment; these are author settings, not universal optimal values. The paper calls physical trajectory length $L$. This chapter uses $\tau$ for physical length and $L$ for an integer integration count, with $\tau \approx L\epsilon$, to avoid confusion with BayesFilter's tuning interface.

BayesFilter's experimental full-refresh implementation now separates a requested mean physical duration $\ell$ from the integration step size. The integer count $N$ is uniform from 1 through $K$. The executed duration $\tau = \epsilon N$ therefore has moments

$$\mathbb{E}[\tau] = \epsilon\frac{K+1}{2}, \qquad \mathbb{E}[\tau^2] = \epsilon^2\frac{(K+1)(2K+1)}{6}. \tag{2.64}$$

The compiler rounds $2\ell/\epsilon - 1$ to the nearest admissible integer, with ties rounded upward, and records both the request and its realized mean. Without an active count cap the mean error is at most $\epsilon/4$. An infeasible request is reported explicitly. This uniform integer law differs from the paper's average-preserving count construction; its adaptation coefficients are therefore hypotheses to test locally.

The discarded proposer brackets epsilon at approximately fixed mean travel, then measures separate continuous windows with both controls frozen. The paper's harmonic-coordinate autocorrelation rule can nominate a duration; a separately labeled slow-observable rule and a predeclared duration neighborhood provide additional hypotheses. Coordinate squares, radius squared, frozen projections, proposed displacement and realized displacement including rejection zeros help distinguish small travel from numerical acceptance. Constant, numerically unresolved, or lag-limited correlation estimates remain unavailable. None of these pilot summaries establishes equilibration.

The resulting finite policies enter the same public candidate-set tuners. Each needs fresh measurement and verification; every verified member remains retained. The optional duration-preserving repair changes epsilon and count together and keeps the original family duration request across descendants. The shared frozen transition and Metropolis correction are unchanged. R-hat, posterior ESS and lugsail precision remain posterior assessment criteria, not tuning membership requirements. These changes establish an experimental implementation with mechanics tests; multi-model confirmation is still required before a robustness or efficiency conclusion.

## 2.6 Unadjusted bias and the LAPS warm-up strategy

### 2.6.1 What a finite-step unadjusted chain samples

An exact invariant diffusion or deterministic flow does not imply that its finite-step discretization has the same invariant distribution. A Gaussian calculation exposes the distinction without asymptotic notation. Use the unit-mass leapfrog in (2.15) for $U(x) = x^2/(2\sigma^2)$, refresh momentum independently before each single step, and omit MH. With $h = \epsilon$ and $y = h^2/\sigma^2$, its matrix is

$$\begin{pmatrix} x' \\ p' \end{pmatrix} = \begin{pmatrix} 1 - y/2 & h \\ -h\sigma^{-2}(1 - y/4) & 1 - y/2 \end{pmatrix} \begin{pmatrix} x \\ p \end{pmatrix}. \tag{2.65}$$

For $0 < y < 4$ the position recursion has coefficient $a = 1 - y/2$ with $|a| < 1$. If its stationary variance is $v_h$, then $v_h = a^2 v_h + h^2$. Solving gives

$$v_h = \frac{h^2}{1 - a^2} = \frac{\sigma^2}{1 - y/4}, \qquad b = \frac{v_h}{\sigma^2} - 1 = \frac{y/4}{1 - y/4}. \tag{2.66}$$

Every positive stable step size inflates the variance in this scheme. Increasing the number of iterations only estimates $v_h$ more accurately; it does not make $v_h$ equal to $\sigma^2$. Different splitting conventions can have different bias, so their formulas must not be exchanged silently.

The energy error contains useful information about this bias. Let $A$ be the matrix in (2.65), $D = \operatorname{diag}(\sigma^{-2}, 1)$ and $S = \operatorname{diag}(v_h, 1)$. At the stationary unadjusted input, $z = (x, p) \sim \mathcal{N}(0, S)$, and

$$\Delta H = \tfrac{1}{2} z^\top Q z, \qquad Q = A^\top D A - D. \tag{2.67}$$

For a centered Gaussian vector, expansion using fourth moments gives $\operatorname{Var}(z^\top Q z) = 2\operatorname{tr}(QSQS)$. Hence

$$\operatorname{Var}(\Delta H) = \tfrac{1}{2}\operatorname{tr}(QSQS) = \frac{y^3}{16(1 - y/4)} \tag{2.68}$$

$$= \frac{4b^3}{(1+b)^2} = F(b^2), \qquad F(D) = \frac{4D^{3/2}}{(1 + \sqrt{D})^2}. \tag{2.69}$$

For an explicit check, substitution into $Q$ gives

$$Q_{11} = -\frac{y^2(1 - y/4)}{4\sigma^2}, \qquad Q_{12} = Q_{21} = \frac{hy(1 - y/2)}{4\sigma^2}, \qquad Q_{22} = \frac{y^2}{4}. \tag{2.70}$$

Consequently $Q_{11} v_h = -y^2/4$ and $\frac{1}{2}\operatorname{tr}(QSQS) = y^4/16 + y^3(1 - y/2)^2/[16(1 - y/4)]$. Combining the numerator uses $y(1 - y/4) + (1 - y/2)^2 = 1$ and yields (2.69). This is a relation between the stationary error of this specified integrator and a specified covariance bias, not a relation between arbitrary transient error and arbitrary observables.

For independent Gaussian eigendirections, independent energy errors add. With $b_i = v_{h,i}/\sigma_i^2 - 1$, define

$$D_{\mathrm{cov}} = \frac{1}{d}\sum_i b_i^2, \qquad E = \frac{1}{d}\operatorname{Var}(\Delta H) = \frac{1}{d}\sum_i F(b_i^2). \tag{2.71}$$

In the isotropic case $E = F(D_{\mathrm{cov}})$ exactly. More generally,

$$F'(D) = \frac{2\sqrt{D}(3 + \sqrt{D})}{(1 + \sqrt{D})^3} > 0, \tag{2.72}$$

$$F''(D) = \frac{3 - 4\sqrt{D} - D}{\sqrt{D}(1 + \sqrt{D})^4}. \tag{2.73}$$

The numerator of the second derivative is positive for $0 < D < 11 - 4\sqrt{7}$. If every $b_i^2$ lies in that interval, Jensen's inequality gives $F(D_{\mathrm{cov}}) \le E$, and monotonicity implies $D_{\mathrm{cov}} \le F^{-1}(E)$. This is a sufficient restricted Gaussian bound. The broader statement and its admissible range in Robnik et al. [2026, Theorem 4.2 and Appendix B] require their

stated conditions. Neither argument proves a universal microcanonical or non-Gaussian error bound.

For small $h$, $b = O(h^2)$, $D_{\text{cov}} = O(h^4)$ and $E = O(h^6)$. Thus a sixth-root adjustment $h_{\text{new}} = h(E_{\text{wanted}}/E)^{1/6}$ has a local scaling rationale. Outside that regime, the relation need not be accurate. The cited paper's non-Gaussian experiments include a Brownian-motion case where the proposed bound underpredicts observed bias by about a factor of 1.5. Its June 2026 revision provides useful evidence for adaptation, not a general certificate of approximation error.

### 2.6.2 The ideal bias–contraction schedule

LAPS uses an unadjusted microcanonical transient, followed by adjusted MAMS [Robnik and Seljak, 2026]. Its theoretical motivation can be reconstructed from two assumptions. Let $D$ now denote a genuine distance satisfying the triangle inequality. Assume $K_h$ has invariant distribution $\pi_h$, with

$$D(\pi_h, \pi) \le C_{\text{b}} h^{\kappa}, \tag{2.74}$$

$$D(\rho K_h, \pi_h) \le (1-c) D(\rho, \pi_h), \qquad 0 < c < 1, \tag{2.75}$$

where the constants are uniform over the step sizes being used. Two triangle inequalities imply

$$\begin{aligned} D(\rho K_h, \pi) &\le D(\rho K_h, \pi_h) + D(\pi_h, \pi) && (2.76)\\ &\le (1-c) D(\rho, \pi) + (2-c) C_{\text{b}} h^{\kappa}. && (2.77) \end{aligned}$$

This separates contraction of initialization error from the numerical bias floor. Suppose one transition simulates fixed physical length $\tau$ and therefore costs approximately $\tau/h$ gradient evaluations. If $D_t = D(\rho_t, \pi)$, the lower bound on improvement per gradient is

$$R(h) = \frac{h}{\tau}\{c D_t - (2-c) C_{\text{b}} h^{\kappa}\}. \tag{2.78}$$

Differentiating gives $\tau R'(h) = c D_t - (\kappa+1)(2-c) C_{\text{b}} h^{\kappa}$. The positive maximizer consequently satisfies

$$C_{\text{b}} h_t^{\kappa} = \frac{c D_t}{(2-c)(\kappa+1)}, \qquad D_{t+1} \le \left(1 - \frac{c\kappa}{\kappa+1}\right) D_t. \tag{2.79}$$

This reproduces the argument in LAPS Appendix A. The optimum is of the specified lower bound under uniform contraction and cost assumptions. It is not a universal optimum for arbitrary MCMC. In particular, the moment statistic used in practice below is not a general distance with the triangle inequality, so it cannot simply be substituted into this proof.

### 2.6.3 Equipartition as a measurable but incomplete diagnostic

The target moments needed to measure $D_t$ are generally unknown. LAPS uses an integration-by-parts identity. Let $\rho$ describe the current ensemble, with mean $m_\rho$. Define

$$V_{ij}(\pi, \rho) = \mathbb{E}_\rho[-(X_i - m_{\rho,i}) \partial_j \log \pi(X)], \qquad \widetilde{D} = \frac{1}{d}\|I - V\|_F^2. \tag{2.80}$$

Assume the relevant expectations exist, $\pi$ is differentiable and the boundary term $(x_i - m_{\pi,i})\pi(x)$ vanishes in the integration-by-parts calculation. At the target,

$$V_{ij}(\pi,\pi) = -\int (x_i - m_{\pi,i})\partial_j \pi(x)\,dx \tag{2.81}$$

$$= -\,[\text{boundary term}] + \int \partial_j (x_i - m_{\pi,i})\pi(x)\,dx = \delta_{ij}. \tag{2.82}$$

The identity uses the score of the actual density, including coordinate Jacobians. A generic force $b$ cannot be inserted without a new argument. Bounded domains and heavy tails require particular care with the boundary term and existence of moments.

The implication $\rho = \pi \Rightarrow V = I$ does not have a general converse. For $\pi = \mathcal{N}(0, I)$, $-\nabla \log \pi(x) = x$. If $\rho = \mathcal{N}(m, S)$, then

$$V = \mathbb{E}_\rho[(X - m)X^\top] = S. \tag{2.83}$$

Thus $S = I$ makes the diagnostic zero for every mean $m$, however far it is from the target. The full matrix detects a wrong covariance, but its diagonal reduction

$$\widetilde{D}_{\text{diag}} = \frac{1}{d}\sum_i (1 - V_{ii})^2 \tag{2.84}$$

does not. For instance, $S = \left(\begin{smallmatrix} 1 & r \\ r & 1 \end{smallmatrix}\right)$ with $0 < |r| < 1$ has zero diagonal diagnostic and wrong correlation. Even the full matrix cannot distinguish all non-Gaussian distributions: a centered smooth mixture $\frac{1}{2}\mathcal{N}(-a, 1 - a^2) + \frac{1}{2}\mathcal{N}(a, 1 - a^2)$, $0 < a < 1$, has variance one and therefore $V = 1$, while its fourth moment $3 - 2a^4$ differs from the standard Gaussian value three. These are exact counterexamples, not merely concerns about finite ensemble size.

The LAPS paper explicitly calls $\widetilde{D}$ a proxy and proves separation only on a restricted zero-mean Gaussian family. The examples above explain why one needs additional means, dependence, tail and mode diagnostics when using it for adaptation. Empirical moment fluctuations can also stabilize inside a single missed mode. A stable statistic is informative about that statistic, not all posterior quantities.

### 2.6.4 From the proxy to an operational schedule

In the paper's scheme, select a fraction $0 < C < 1$ and set

$$D_{\text{wanted}} = C\widetilde{D}_t, \qquad E_{\text{wanted}} = F(D_{\text{wanted}}), \qquad h_{t+1} = h_t \left(\frac{E_{\text{wanted}}}{\widehat{E}_t}\right)^{1/6}. \tag{2.85}$$

The first assignment expresses a desired numerical bias below the estimated current error; the second borrows the Gaussian relationship; the third uses the small-step energy-error scaling. Each link has assumptions. In particular, $\widehat{E}_t$ is measured during a transient, whereas (2.69) was derived at an unadjusted stationary law. Finite ensembles, small denominators, heavy tails and a wrong proxy can all mislead the update. A bounded implementation must handle them explicitly and report why adaptation ended.

After the unadjusted stage, LAPS applies a diagonal reparameterization, turns on MH and adapts the final adjusted settings. The paper uses a practical stopping rule based on fluctuations of second moments, not a proof that all initialization bias has

disappeared. For posterior reporting, discard every unadjusted and adaptive draw, freeze all shared settings and assess the final fixed kernel. If chains shared adaptation, freezing stops further coupling but does not instantly make their current states independent or stationary.

The inspected BlackJAX revision contains choices different from the paper. Its desired error is coded as $C\,\mathrm{bias}^{3/8}$, with the paper's $F(C\widetilde{D})$ expression commented out. Its diagonal statistic uses $-x_i\partial_i \log \pi(x)$ without centering, and its fluctuation function uses variance divided by squared mean rather than relative standard deviation. The centered mean-shift counterexample therefore does not apply unchanged to that uncentered code, but neither statistic determines a general distribution. The adjusted epsilon freezes through an acceptance-tolerance flag; the first tolerance-satisfying update can still change epsilon before the next call freezes it. A faithful local port must resolve these definitions and test the setting actually frozen. These observations concern the pinned current source, not an identified source revision of the published benchmark.

### 2.6.5 Gradient work and parallel latency

Table 2.1 reports selected entries from the MAMS paper. The criterion is a normalized squared error in the worst second moment below 0.01, taking median error across at least 128 chains. Tuning work is excluded. A small ratio in one model does not invalidate the method, but it does limit a blanket several-fold claim.

| Author target | NUTS gradients | MAMS gradients | Ratio |
|---|---|---|---|
| Gaussian | 19,652 | 3,249 | 6.05 |
| Banana | 95,519 | 14,078 | 6.78 |
| Bimodal | 210,758 | 139,418 | 1.51 |
| Brownian | 29,816 | 13,528 | 2.20 |
| Stochastic volatility | 843,768 | 430,088 | 1.96 |

Table 2.1: Selected MAMS Table 1 entries [Robnik et al., 2025]. Ratios are descriptive arithmetic on the authors' reported counts, not local timings or uncertainty-qualified rankings for the present consumers.

LAPS's experiments use an NVIDIA A100 with 40 GB. Its Table 1 uses 4,096 chains; Table 2 uses 256. For stochastic volatility, Table 2 reports 29,900 gradients for sequential NUTS, 1,100 per LAPS chain and 281,600 across the LAPS ensemble [Robnik and Seljak, 2026]. The last two figures satisfy $256 \times 1{,}100 = 281{,}600$. Under the idealized model of perfectly synchronized work, its latency ratio to that sequential comparator is

$$\frac{C_{\mathrm{LAPS}}}{C_{\mathrm{NUTS}}} \approx \frac{1{,}100\, c_\nabla(256)}{29{,}900\, c_\nabla(1)}. \tag{2.86}$$

Per-chain work is about 27 times smaller, but total gradient work is about 9.4 times larger. The idealized latency gain requires $c_\nabla(256)/c_\nabla(1) < 29{,}900/1{,}100$ before other overheads. This calculation is not a fair complete local benchmark by itself: a local NUTS baseline may also be batched, compilation and adaptation must be counted, and a requested MCSE may require more than one final draw per chain. The paper provides motivation to measure batch scaling, not permission to assume it for Kalman, UKF or QZ calculations.

## 2.7 Geometry and initialization

### 2.7.1 Affine whitening and nonlinear funnels

For a Gaussian target, let $\Sigma = AA^\top \in \mathbb{R}^{d\times d}$ be its symmetric positive-definite covariance, with $A$ invertible. The map $x = \mu + Az$ gives a standard normal density for $z$. Substitution into (2.7) gives

$$-\tfrac{1}{2}(x-\mu)^\top \Sigma^{-1}(x-\mu) + \log|\det A| = -\tfrac{1}{2}z^\top z + \text{constant}. \tag{2.87}$$

For ordinary Gaussian HMC in the original coordinates, equal oscillation frequencies correspond to $M = \Sigma^{-1}$ in the convention $K(p) = p^\top M^{-1}p/2$. Libraries sometimes expose $M^{-1}$ as their *inverse mass* parameter, so verbal descriptions of mass adaptation are ambiguous unless this convention is stated. Whitening and mass adaptation repair constant linear anisotropy. They cannot generally remove position-dependent conditional scales.

Consider the funnel

$$V \sim \mathcal{N}(0, s^2), \qquad X_i \mid V = v \sim \mathcal{N}(0, e^v), \quad i = 1, \dots, m. \tag{2.88}$$

Its log density, up to a constant, is

$$-\frac{v^2}{2s^2} - \frac{m}{2}v - \frac{1}{2}e^{-v}\sum_i x_i^2. \tag{2.89}$$

The child-coordinate curvature is $e^{-v}$ and becomes arbitrarily large in the neck. Let $v = sz_0$ and $x_i = e^{v/2}z_i$. The Jacobian is block triangular, with determinant $se^{mv/2}$. Adding its logarithm cancels the linear $-mv/2$ term and yields

$$\log \pi_z(z) = -\tfrac{1}{2}z_0^2 - \tfrac{1}{2}\sum_i z_i^2 + \text{constant}. \tag{2.90}$$

Thus exact noncentering solves this analytic geometry problem. It does not show that an affine tuning procedure ought to solve the centered funnel.

A partial-whitening example is equally useful. Use $x_i = e^{v/2-\delta(v)}z_i$ and retain $v$ as a coordinate. Its transformed log density is

$$\log \pi(v, z) = -\frac{v^2}{2s^2} - m\delta(v) - \frac{1}{2}e^{-2\delta(v)}\sum_i z_i^2 + \text{constant}. \tag{2.91}$$

The child Hessian of the negative log density is $e^{-2\delta(v)}I$. If $|\delta(v)| \le a$, its eigenvalues are bounded between $e^{-2a}$ and $e^{2a}$. This bounds that block only: derivatives of $\delta$ create cross terms and curvature in $v$. It is wrong to infer a global full-Hessian bound from this child-coordinate bound. Exact and residual-map fixtures distinguish map correctness from robustness to remaining geometry.

### 2.7.2 What neural transport learns

For base density $\phi(z)$ and invertible $F_\lambda$, the induced density $q_\lambda$ obeys $\log q_\lambda(F_\lambda(z)) = \log \phi(z) - \log|\det J_{F_\lambda}(z)|$. The reverse-KL objective is therefore

$$\mathrm{KL}(q_\lambda \| \pi) = \mathbb{E}_{Z\sim\phi}\big[\log \phi(Z) - \log|\det J_{F_\lambda}(Z)| + U(F_\lambda(Z))\big] + \log Z_\pi. \tag{2.92}$$

The unknown constant $Z_\pi = \int e^{-U(x)}dx$ does not affect optimization. This is a training objective for useful coordinates, not a posterior-error certificate [Hoffman et al., 2019]. Undercoverage of a mode contributes little when that mode is rarely sampled by $q_\lambda$. The final transformed target must still use (2.7), not the learned approximation alone.

Once frozen, the same map can be used by ordinary HMC, SNAPER or MAMS. Consequently a fair kernel comparison holds the map fixed. An end-to-end comparison may let each method construct geometry, but must then count fitting, inverse-map evaluation for matched model starts, forward/log-Jacobian work and transformed gradients. The current BayesFilter architecture policy selects configured Huang DSF/NAF for the October 2026 forward/reverse training study, and the repaired author IAF outside that study. Architecture selection and successful training are separate from map validity, tail coverage and downstream sampling accuracy.

### 2.7.3 Pathfinder from a quadratic expansion

Pathfinder constructs Gaussian approximations along optimization paths [Zhang et al., 2022]. Let $x_k$ be an L-BFGS iterate minimizing $U$, $g_k = \nabla U(x_k)$, and $B_k$ a positive-definite inverse-Hessian approximation. A local quadratic model is

$$U(x) \approx U(x_k) + g_k^\top (x - x_k) + \tfrac{1}{2}(x - x_k)^\top B_k^{-1}(x - x_k). \tag{2.93}$$

Completing the square gives mean and covariance

$$\mu_k = x_k - B_k g_k, \qquad q_k = \mathcal{N}(\mu_k, B_k). \tag{2.94}$$

In particular, the Gaussian is centered at the stationary point of the local quadratic, not necessarily the current iterate. If $s_k = x_{k+1} - x_k$ and $y_k = g_{k+1} - g_k$, the full BFGS inverse update is

$$B_{k+1} = (I - \rho_k s_k y_k^\top) B_k (I - \rho_k y_k s_k^\top) + \rho_k s_k s_k^\top, \qquad \rho_k = (y_k^\top s_k)^{-1}. \tag{2.95}$$

Multiplying by $y_k$ verifies the secant equation $B_{k+1} y_k = s_k$. If $y_k^\top s_k > 0$ and $B_k$ is positive definite, the quadratic form of the update is the sum of a nonnegative transformed $B_k$ form and $\rho_k (s_k^\top v)^2$; their simultaneous vanishing forces $v = 0$. This proves positive definiteness. Limited-memory representations retain selected $(s, y)$ pairs and exploit a diagonal-plus-low-rank form instead of storing a dense $d \times d$ matrix. Negative or negligible curvature must be handled explicitly, not treated as a valid Gaussian covariance.

For each proposed $q_k$, the evidence lower bound satisfies

$$\mathcal{L}(q_k) = \mathbb{E}_{q_k}[\log \widetilde{\pi}(X) - \log q_k(X)] = \log Z_\pi - \mathrm{KL}(q_k \| \pi). \tag{2.96}$$

Maximizing an estimate of $\mathcal{L}$ along the path chooses a promising approximation; the estimated maximum has Monte Carlo selection uncertainty. The inspected Stan implementation checks curvature and stores the approximation with favorable estimated ELBO. A good local approximation can provide initial states and scaling, but missing modes remain possible.

The paper's multipath importance step is best understood on an augmented space. If $J$ paths supply densities $q_j$, define

$$q(x, j) = J^{-1} q_j(x), \qquad \pi(x, j) = J^{-1} \pi(x). \tag{2.97}$$

The importance ratio is $\widetilde{\pi}(x)/q_j(x)$ for a draw from path $j$; the factors $J^{-1}$ cancel. This avoids evaluating every mixture component at every draw. It is different from the marginal-mixture ratio $\widetilde{\pi}(x)/[J^{-1}\sum_j q_j(x)]$, although both can define proper importance schemes with appropriate sampling. Pathfinder applies Pareto-smoothed importance resampling on the augmented construction. Finite resampling is an approximation; importance-weight concentration and tail diagnostics matter. It should be evaluated as an initializer before being credited with a reduction in total MCMC work.

## 2.8 Alternatives for residual stiffness and discontinuities

### 2.8.1 Delayed rejection with the reverse path included

If one global epsilon must be tiny in a small stiff region, delayed rejection can attempt a smaller step after a rejected proposal [Turok et al., 2026]. The principle is easiest to derive for ordinary proposal densities. First propose $y \sim q_1(x,\cdot)$ and accept with $\alpha_1(x,y)$. After rejection, propose $z \sim q_2(x,y,\cdot)$. The forward probability of this two-stage path, before the second decision, is $\pi(x)q_1(x,y)(1-\alpha_1(x,y))q_2(x,y,z)$. Its reverse counterpart starts from $z$, rejects $y$, and proposes $x$. Consequently the second-stage acceptance is

$$\alpha_2(x,y,z) = 1 \wedge \frac{\pi(z)q_1(z,y)[1-\alpha_1(z,y)]q_2(z,y,x)}{\pi(x)q_1(x,y)[1-\alpha_1(x,y)]q_2(x,y,z)}. \tag{2.98}$$

Multiplying either path density by its acceptance gives the minimum of the two path densities, proving equality of accepted flows. The factors for earlier rejections are essential. Halving epsilon after a rejection and applying only the ordinary endpoint MH ratio drops them.

Deterministic GHMC proposals require the corresponding reversible-map construction rather than pretending their transition laws have ordinary Lebesgue densities. The author implementation evaluates reverse *ghost* proposals to obtain the earlier rejection probabilities. Recursion can make work grow exponentially with retry depth. A bounded depth, evaluation counts including ghost trajectories, and forward/reverse tests are therefore part of a meaningful local comparison. Author experiments with reference-draw initialization do not establish cold-start performance on economic posteriors.

### 2.8.2 A true potential jump is not a steep smooth gradient

For a coordinate with Laplace kinetic energy $K(p) = |p|/m$, its speed away from $p = 0$ is $\dot{x} = \mathrm{sign}(p)/m$. At a boundary with potential jump $\Delta U = U_+ - U_-$, conservation of energy requires

$$|p_+| = |p_-| - m\Delta U. \tag{2.99}$$

If the right side is positive, the trajectory transmits with the same direction and reduced kinetic energy for an upward jump. Otherwise it reflects, reversing momentum while staying on the initial side. A downward jump increases kinetic energy. This describes the local physical rule behind Laplace-momentum discontinuous HMC; the complete numerical proposal also needs its reversible coordinate update and appropriate treatment of mixed smooth coordinates [Nishimura et al., 2020].

Changing NUTS to MAMS does not by itself implement those events. A DSGE regime boundary might instead create a discontinuous score of a continuous density, a discontinuity in the density, or an invalid region. These are different mathematical cases. Their target and boundary behavior must be determined before selecting a discontinuous method. This specialist option was examined at the paper level; no claim of an audited TensorFlow port follows.

## 2.9 State-space structure and likelihood cost

### 2.9.1 Associative filtering by unnormalized kernels

For fixed parameters, write a filtering factor as $f_i(x \mid z)g_i(z)$, where $f_i$ is a conditional state density and $g_i$ is the likelihood of the observations in that segment given its input state. Combine consecutive factors by

$$Z_{ij}(z) = \int g_j(y) f_i(y \mid z)\, dy, \tag{2.100}$$

$$g_{ij}(z) = g_i(z) Z_{ij}(z), \tag{2.101}$$

$$f_{ij}(x \mid z) = \frac{\int f_j(x \mid y) g_j(y) f_i(y \mid z)\, dy}{Z_{ij}(z)}. \tag{2.102}$$

Assume the integrals exist and the normalizing denominator is positive. Multiplication eliminates the normalization:

$$g_{ij}(z) f_{ij}(x \mid z) = \int [g_j(y) f_j(x \mid y)][g_i(z) f_i(y \mid z)]\, dy. \tag{2.103}$$

If $K_i(x, z) = g_i(z) f_i(x \mid z)$, the right side is kernel convolution. For three segments, both parenthesizations integrate $K_k(x, y_2) K_j(y_2, y_1) K_i(y_1, z)$ over $(y_1, y_2)$. Tonelli's theorem for nonnegative kernels, or Fubini under absolute integrability, permits the order to be exchanged. The unnormalized operation is therefore associative; the uniquely normalized $(f, g)$ pair inherits that property. This is the foundation of temporal parallel filtering [Särkkä and García-Fernández, 2020, Definition 2 and Theorem 3].

An associative scan computes every prefix. A balanced evaluation has $O(\log T)$ dependency depth and $O(T)$ total binary combinations. It needs enough processors and memory for the levels of the tree. Its arithmetic constants may be larger than those of a sequential Kalman recursion. Thus logarithmic depth is not logarithmic total work or a guarantee of speedup.

### 2.9.2 Gaussian closure and the likelihood constant

Gaussian factors make the convolution tractable. Let $r$ collect the retained variables at a join and let $y \in \mathbb{R}^m$ be the state being integrated out. Write the product of the two unnormalized factors in canonical form,

$$\exp\left\{a - \tfrac{1}{2} r^\top J_{rr} r - r^\top J_{ry} y - \tfrac{1}{2} y^\top J_{yy} y + h_r^\top r + h_y^\top y\right\}, \tag{2.104}$$

where $r \in \mathbb{R}^k$, $J_{rr} \in \mathbb{R}^{k\times k}$, $J_{ry} \in \mathbb{R}^{k\times m}$, $h_r \in \mathbb{R}^k$, and $h_y \in \mathbb{R}^m$. The block $J_{yy} \in \mathbb{R}^{m\times m}$ is symmetric positive definite and $J_{yr} = J_{ry}^\top$. Completing the square in $y$ gives

$$\bar{J} = J_{rr} - J_{ry}J_{yy}^{-1}J_{yr}, \tag{2.105}$$

$$\bar{h} = h_r - J_{ry}J_{yy}^{-1}h_y, \tag{2.106}$$

$$\bar{a} = a + \tfrac{m}{2}\log(2\pi) - \tfrac{1}{2}\log\det J_{yy} + \tfrac{1}{2}h_y^\top J_{yy}^{-1}h_y. \tag{2.107}$$

For verification, write $b_y = h_y - J_{yr}r$ and expand $\frac{1}{2}b_y^\top J_{yy}^{-1}b_y$ after integrating the centered Gaussian. The quadratic, linear and constant terms are exactly those above. Dropping $\bar{a}$ retains a proportional state density but loses the likelihood normalization required by parameter HMC. Deterministic state components may make a naive precision block singular; the formula then needs an appropriate constrained or square-root representation. Adding an undeclared ridge changes the target.

The serial Gaussian likelihood increment is

$$\ell_t = -\tfrac{1}{2}\{n_t\log(2\pi) + \log\det S_t + v_t^\top S_t^{-1}v_t\}, \tag{2.108}$$

where $v_t \in \mathbb{R}^{n_t}$ is the innovation and $S_t \in \mathbb{R}^{n_t\times n_t}$ is its symmetric positive-definite covariance. For a fixed observation pattern its differential is

$$d\ell_t = -\tfrac{1}{2}\operatorname{tr}(S_t^{-1}dS_t) - v_t^\top S_t^{-1}dv_t + \tfrac{1}{2}v_t^\top S_t^{-1}(dS_t)S_t^{-1}v_t. \tag{2.109}$$

This follows from $dS^{-1} = -S^{-1}(dS)S^{-1}$ and $d\log\det S = \operatorname{tr}(S^{-1}dS)$. The differentials include the recursive dependence of innovations and covariances on parameters. Agreement of filter values alone does not establish agreement of these total derivatives.

### 2.9.3 Correlated noises and missing components

Consider a transition and measurement

$$X' = AX + b + w, \qquad Y = HX' + d + v, \qquad \begin{pmatrix} w \\ v \end{pmatrix} \sim \mathcal{N}\left(0, \begin{pmatrix} Q & C \\ C^\top & R \end{pmatrix}\right). \tag{2.110}$$

Conditionally on $X = x$, the mean of $Y$ is $H(Ax+b)+d$, its covariance is

$$S = HQH^\top + HC + C^\top H^\top + R, \qquad \operatorname{Cov}(X', Y \mid X) = QH^\top + C. \tag{2.111}$$

The conditional Gaussian formula therefore yields

$$K = (QH^\top + C)S^{-1}, \tag{2.112}$$

$$X' \mid X = x, Y = y \sim \mathcal{N}\big((I-KH)(Ax+b) + K(y-d),\ Q - KSK^\top\big). \tag{2.113}$$

This supplies the $f$ factor; $g(x)$ is the Gaussian density of $y$ with the preceding mean and covariance $S$. Correlated noise does not destroy the abstract associative construction, but an implementation written only for independent noises must be extended through an equivalent formulation. If a matrix $O$ selects the observed components, replace $y, H, d, C, R$ by $Oy, OH, Od, CO^\top, ORO^\top$. An entirely missing observation instead has $g = 1$ and leaves the transition factor unchanged. Whole-time masks and component-specific masks are therefore different interfaces.

TensorFlow Probability exposes a parallel LGSSM option and a parallel-filter library. The inspected source returns likelihoods and supports time/batch masks. A direct replacement of MacroFinance's correlated-noise, component-masked recursion is not established by that source availability. Test the algebraic reformulation, likelihood, total score, dtype, compilation and memory behavior on the actual target. For nonlinear filters whose linearization or sigma points depend recursively on the current filtering law, an independently parallelized linear Gaussian scan is not automatically the same approximation.

### 2.9.4 Conditional particle methods for latent paths

If the unknown includes a long path, write its fixed-parameter smoothing density as

$$\pi_T(x_{1:T}) = Z_T^{-1}\gamma_T(x_{1:T}), \qquad \gamma_T = M_1(x_1)G_1(x_1)\prod_{t=2}^{T} M_t(x_t \mid x_{t-1})G_t(x_{t-1}, x_t). \tag{2.114}$$

Assume $0 < Z_T < \infty$ and proposal support covering the target. Ordinary particle filtering with $N$ particles uses proposal $r_1$ and $r_t$, samples ancestors with normalized weights $W_{t-1}$, and assigns incremental weights

$$w_1^i = \frac{M_1(x_1^i)G_1(x_1^i)}{r_1(x_1^i)}, \qquad w_t^i = \frac{M_t(x_t^i \mid x_{t-1}^{a_t^i})G_t(x_{t-1}^{a_t^i}, x_t^i)}{r_t(x_t^i \mid x_{t-1}^{a_t^i})}. \tag{2.115}$$

The particle-generation law is

$$\psi(\mathbf{x}, \mathbf{a}) = \prod_{i=1}^{N} r_1(x_1^i) \prod_{t=2}^{T}\prod_{i=1}^{N} W_{t-1}^{a_t^i} r_t(x_t^i \mid x_{t-1}^{a_t^i}). \tag{2.116}$$

For a selected final index $k$, define its ancestry by $b_T = k$ and $b_{t-1} = a_t^{b_t}$. The extended target

$$\Phi(\mathbf{x}, \mathbf{a}, k) = \frac{\pi_T(x_{1:T}^{b_{1:T}})}{N^T} \frac{\psi(\mathbf{x}, \mathbf{a})}{r_1(x_1^{b_1}) \prod_{t=2}^{T} W_{t-1}^{b_{t-1}} r_t(x_t^{b_t} \mid x_{t-1}^{b_{t-1}})} \tag{2.117}$$

has the desired selected-path marginal under the usual support conditions. To see the marginal, fix the selected path and its index sequence. The remaining quotient is the conditional particle simulation law for all off-path variables; integrating them gives one. Summing the $N^T$ possible index sequences cancels $N^{-T}$. Conversely, holding the particle system fixed, substitute (2.115). The ratio of path factors is

$$\frac{\gamma_T(x^b)}{r_1(x_1^{b_1}) \prod_{t=2}^{T} W_{t-1}^{b_{t-1}} r_t(x_t^{b_t} \mid x_{t-1}^{b_{t-1}})} = w_T^k \prod_{t=1}^{T-1} \sum_i w_t^i. \tag{2.118}$$

All intermediate selected weights cancel, so the conditional law of $k$ is $W_T^k$. Conditional particle simulation retaining the current path, followed by this terminal selection, is a Gibbs construction on $\Phi$ and leaves $\pi_T$ invariant [Andrieu et al., 2010, Sections 4.2–4.5, equation (31)]. Additional backward sampling or ancestor updates must preserve the same extended target. More particles affect mixing and cost, not the identity of that marginal.

### 2.9.5 Particle gradients and auxiliary marginalization

Particle-MALA and Particle-mGRAD adapt this construction to local, gradient-informed moves [Corenflos and Finke, 2024, Algorithms 3–7]. The key is to introduce auxiliary variables with known conditional law and correct the particle weights for the augmented target. In schematic form,

$$\bar{\pi}(x,u) = \pi_T(x)q(u \mid x). \tag{2.119}$$

First draw $u \mid x$, then use a conditional particle kernel invariant for $\bar{\pi}(x \mid u)$. Both conditional updates preserve $\bar{\pi}$; integrating $u$ returns $\pi_T$. A gradient shift in $q$ guides proposals, but its density must also appear in the conditional target and weights. Changing proposals while retaining bootstrap weights is wrong.

The Gaussian calculation underlying the GRAD family is explicit. Suppose the transition at a fixed parent is $X \sim \mathcal{N}(m,Q)$ and an auxiliary observation has density $u \mid X \sim \mathcal{N}(X, rI)$, $r > 0$, with $Q \in \mathbb{R}^{d\times d}$ symmetric positive definite. Completing the square gives

$$V = (Q^{-1} + r^{-1}I)^{-1} = rA, \qquad A = Q(Q+rI)^{-1}, \tag{2.120}$$

$$\mu = V(Q^{-1}m + r^{-1}u) = (I-A)m + Au. \tag{2.121}$$

If the reference state is $x$ and $u \sim \mathcal{N}(x + r\nabla \log G(x), rI)$, averaging the conditional proposal over this auxiliary draw gives

$$X' \sim \mathcal{N}\big((I-A)m + A[x + r\nabla \log G(x)], r(A + AA^\top)\big)\,. \tag{2.122}$$

The covariance follows from the law of total variance: $V$ from the conditional draw and $ArIA^\top$ from its random mean. For small prior covariance relative to $rI$, $A \approx Q/r$ and the proposal approaches the prior transition. For weak prior information, $A \approx I$ and it approaches $\mathcal{N}(x+r\nabla \log G(x), 2rI)$, a Langevin proposal. This explains the interpolation between prior-guided particle moves and MALA-like moves.

The conditional importance correction can also be derived explicitly. Denote the conditional Gaussian proposal with mean $\mu$ and covariance $V$ in (2.121) by $r_u(x)$, set $g(x) = \nabla \log G(x)$, and let the actual auxiliary density be $q(u \mid x) = \mathcal{N}(u; x + rg(x), rI)$. The Gaussian product identity is

$$\mathcal{N}(x; m, Q)\mathcal{N}(u; x, rI) = \mathcal{N}(u; m, Q + rI)\, r_u(x). \tag{2.123}$$

The target conditional on $u$ contains $\mathcal{N}(x; m, Q)G(x)q(u \mid x)$. Dividing by $r_u(x)$ and using the identity gives the unnormalized weight

$$w(x;u) = G(x)\mathcal{N}(u; m, Q+rI)\frac{\mathcal{N}(u; x + rg(x), rI)}{\mathcal{N}(u; x, rI)} \tag{2.124}$$

$$= G(x)\mathcal{N}(u; m, Q+rI)\exp\{g(x)^\top(u-x) - \tfrac{r}{2}\|g(x)\|^2\}. \tag{2.125}$$

The exponent follows by expanding $-\{\|u-x-rg(x)\|^2 - \|u-x\|^2\}/(2r)$. The normalizing factor depends on the parent through $m$ and must be retained when comparing ancestors. This calculation gives a complete local auxiliary correction, and explains why guidance cannot be added without adjusting the weights.

This one-particle marginal does not supply the complete Particle-mGRAD weights: particles can share auxiliary variables, so their joint law must also be marginalized. The paper derives those joint weight corrections. Its particular Particle-mGRAD construction

assumes the transition covariance $Q_t$ does not depend on the parent state; its more general auxiliary method allows parent-dependent covariance. The invariant distribution is the fixed-parameter smoothing law, not automatically the joint posterior in structural parameters and states. Theoretical interpolation results and high-dimensional stochastic-volatility examples support investigating it when latent paths are sampled. They do not establish an improvement over marginalized Kalman likelihood HMC. The inspected author implementation is JAX, and a TensorFlow implementation would be a separate project.

## 2.10 Warm-up, convergence and precision

Let $h(x)$ be a reported quantity: a structural parameter, an impulse response at a specified horizon, a predictive spread, or an indicator of an economically important event. For $\mu = \mathbb{E}_\pi h$, any estimator obeys

$$\mathbb{E}[(\widehat{\mu} - \mu)^2] = (\mathbb{E}\widehat{\mu} - \mu)^2 + \mathrm{Var}(\widehat{\mu}). \tag{2.126}$$

More independent chains can reduce the variance without removing a common initialization bias. An unadjusted integrator can introduce a stationary bias that survives an arbitrarily long run. A wrong likelihood changes $\mu$ itself. These errors cannot be diagnosed by a single acceptance statistic. Warm-up is intended to reduce initialization effects and learn a useful kernel; a Metropolis correction addresses the invariant distribution of a specified proposal; retained-sample diagnostics assess remaining disagreement and Monte Carlo precision. None implies the others.

### 2.10.1 Long-run variance and the Monte Carlo standard error

First suppose the retained chain is stationary. Write $Y_t = h(X_t) - \mu$ and $\gamma_k = \mathrm{Cov}(Y_0, Y_k)$. Directly expanding the variance of the sum gives

$$\mathrm{Var}\left(\frac{1}{n}\sum_{t=1}^{n} h(X_t)\right) = \frac{1}{n^2}\sum_{s=1}^{n}\sum_{t=1}^{n}\mathrm{Cov}(Y_s, Y_t) \tag{2.127}$$

$$= \frac{1}{n}\left[\gamma_0 + 2\sum_{k=1}^{n-1}\left(1 - \frac{k}{n}\right)\gamma_k\right]. \tag{2.128}$$

If the autocovariances are absolutely summable, the bracket converges to

$$\sigma_h^2 = \gamma_0 + 2\sum_{k=1}^{\infty}\gamma_k, \qquad \tau_h = \frac{\sigma_h^2}{\gamma_0}, \qquad n_{\mathrm{eff},h} = \frac{n}{\tau_h}. \tag{2.129}$$

For $\gamma_0 > 0$, $\tau_h$ is the integrated autocorrelation time. Negative correlations can make it less than one. A suitable Markov-chain central limit theorem supplies the stronger distributional conclusion $\sqrt{n}(\widehat{\mu} - \mu) \Rightarrow \mathcal{N}(0, \sigma_h^2)$; absolute summability alone is not a general proof of that theorem. The estimated Monte Carlo standard error is $\widehat{\mathrm{MCSE}}_h = \sqrt{\widehat{\sigma}_h^2/n}$. For a vector $h$, replace $\gamma_k$ by $\Gamma_k = \mathrm{Cov}(Y_0, Y_k)$ and set $\Sigma = \Gamma_0 + \sum_{k\geq 1}(\Gamma_k + \Gamma_k^\top)$. Each reported linear combination has its own variance $a^\top \Sigma a/n$. An ESS number divorced from its observable conceals this dependence.

Independent retained chains can be combined by variance addition. If chain $j$ contributes $n_j$ draws with long-run variance $\sigma_j^2$, the draw-weighted pooled mean has approximate variance

$$\mathrm{Var}(\widehat{\mu}_{\mathrm{pool}}) \simeq \frac{\sum_j n_j \sigma_j^2}{(\sum_j n_j)^2}. \tag{2.130}$$

This expression assumes the means estimate the same stationary expectation and cross-chain covariances vanish. Concatenating interacting ensemble chains and calling them one stationary chain does not establish those assumptions.

### 2.10.2 Batch means, lugsail correction and their limits

For a vector of observables $h : \mathbb{R}^d \to \mathbb{R}^p$, take $n = ab$, with $a > 1$ nonoverlapping batches of length $b$, and define $\bar{h}_k = b^{-1} \sum_{t=(k-1)b+1}^{kb} h(X_t)$. The batch-means covariance estimator is

$$\widehat{\Sigma}_b = \frac{b}{a-1} \sum_{k=1}^{a} (\bar{h}_k - \bar{h})(\bar{h}_k - \bar{h})^\top. \tag{2.131}$$

If a batch is long enough, its mean has covariance approximately $\Sigma/b$, which explains the factor $b$. Increasing the total run while keeping $b$ fixed does not generally remove the truncation bias. Consistency results require increasing batch length and number of batches, together with appropriate moment, mixing or strong-invariance conditions [Flegal and Jones, 2010, Section 3]; the multivariate lugsail extension is developed by Vats and Flegal [2021].

For integer $r > 1$ dividing $b$ and $0 \le c < 1$, the lugsail construction is

$$\widehat{\Sigma}_{\mathrm{lug}} = \frac{\widehat{\Sigma}_b - c\widehat{\Sigma}_{b/r}}{1-c}. \tag{2.132}$$

Its purpose is visible from the leading bias. Under the assumptions of Vats and Flegal [2021, Theorem 4], write $\mathbb{E}\widehat{\Sigma}_b - \Sigma = \mathcal{G}/b + o(b^{-1})$. Substituting the expansion at both batch sizes gives

$$\mathbb{E}\widehat{\Sigma}_{\mathrm{lug}} - \Sigma = \frac{1}{1-c}\left(\frac{\mathcal{G}}{b} - c\frac{r\mathcal{G}}{b}\right) + o(b^{-1}) \tag{2.133}$$

$$= \frac{1-rc}{1-c}\frac{\mathcal{G}}{b} + o(b^{-1}). \tag{2.134}$$

For a scalar process with predominantly positive correlations, $\mathcal{G} = -2\sum_{k\ge1} k\gamma_k$ is often negative. The choice $rc > 1$ reverses this leading bias; $rc = 1$ cancels it. For example, a stationary AR(1) correlation $\gamma_k = \gamma_0\phi^k$, $0 < \phi < 1$, gives

$$\sigma_h^2 = \gamma_0 \frac{1+\phi}{1-\phi}, \qquad \mathcal{G} = -\frac{2\gamma_0\phi}{(1-\phi)^2}. \tag{2.135}$$

The second equality follows by differentiating $\sum_{k\ge0}\phi^k = (1-\phi)^{-1}$ and multiplying by $\phi$. As $\phi$ approaches one, modest batches badly underrepresent the slow dependence. Lugsail changes the leading bias; it does not manufacture the missing long-time information.

The sign qualification matters for antithetic chains. For a stationary unit-variance AR(1) process, $\gamma_k = \rho^{|k|}$ with $|\rho| < 1$, define the variance of a normalized block sum by

$$V_b = \frac{1}{b}\operatorname{Var}\left(\sum_{t=1}^{b} X_t\right) = 1 + 2\sum_{k=1}^{b-1}\left(1 - \frac{k}{b}\right)\rho^k \tag{2.136}$$

$$= \frac{1+\rho}{1-\rho} - \frac{2\rho(1-\rho^b)}{b(1-\rho)^2}. \tag{2.137}$$

The first equality counts the $b-k$ pairs separated by lag $k$; the second uses the finite geometric sum and its derivative. For $a = \lfloor n/b \rfloor$ complete batches, the implemented estimator centers their means on the mean of those $ab$ observations. Expanding the centered sum of squares gives

$$\mathbb{E}\widehat{\sigma}_b^2 = \frac{b}{a-1}\left\{a\operatorname{Var}(\bar{X}_b) - a\operatorname{Var}(\bar{X}_{ab})\right\} = \frac{a}{a-1}\left(V_b - \frac{V_{ab}}{a}\right).$$

This is an exact finite-sample expectation under the stated stationary AR(1) law. With $n = 2048$, $b = 45$, $b/r = 15$, $c = 1/2$ and $\rho = -0.9$, the true long-run variance is 0.052632, whereas the expectations of ordinary and lugsail batch means are 0.064057 and 0.035102, respectively. Each batch size uses its own complete-batch count. For $\rho = 0.9$, the corresponding values are 19, 14.946816 and 20.492873. Thus lugsail is not universally conservative: reversing the leading bias can increase underestimation for negative dependence. These exact calculations motivate antithetic calibration; they do not establish interval coverage for HMC or explain a particular posterior reference alarm.

Subtracting covariance estimates can produce a negative scalar estimate or negative eigenvalues in finite samples. Such an estimate cannot be inserted into a square root or treated as zero Monte Carlo uncertainty. Numerical validity, adequate batch counts, sensitivity to batch length and additional sampling remain necessary. A declared positive-definiteness correction is a different estimator and needs its own justification. The lugsail paper discusses these finite-sample issues; its asymptotic results are not guarantees of finite-run coverage.

BayesFilter already implements the lugsail option in `bayesfilter/inference/hmc_precision.py`, and posterior assessment uses the precision machinery. The remaining task is to establish that each consumer supplies appropriate retained draws, observables and batch settings. A separate experimental use of lugsail for acceptance uncertainty did not pass its strong-dependence promotion study. That result does not remove the posterior MCSE estimator, and the existence of the estimator does not qualify the experimental acceptance policy.

### 2.10.3 Modern $\widehat{R}$ and many short chains

For $m$ chains of $n$ draws, define chain means $\bar{h}_j$, sample variances $s_j^2$, grand mean $\bar{h}$, and

$$W = \frac{1}{m}\sum_j s_j^2, \qquad B = \frac{n}{m-1}\sum_j(\bar{h}_j - \bar{h})^2, \qquad \widehat{V} = \frac{n-1}{n}W + \frac{B}{n}. \tag{2.138}$$

The classical statistic $\widehat{R} = \sqrt{\widehat{V}/W}$ compares between-chain disagreement with variation within chains. A common mean can hide differing scales or heavy tails, so modern practice

first splits each chain, then applies the statistic to rank-normalized draws. For $S$ pooled split-chain draws with midrank $r_i$, the normal score is

$$z_i = \Phi^{-1}\left(\frac{r_i - 3/8}{S + 1/4}\right). \tag{2.139}$$

Here $\Phi$ is the standard-normal CDF, so its inverse acts on probabilities strictly between zero and one. Apply the same calculation to the ranks of $|h_i - \text{median}(h)|$ and take the maximum of the rank and folded-rank results [Vehtari et al., 2021]. The folding checks scale disagreement that a location diagnostic can miss. Constant chains and undefined ratios require explicit treatment rather than automatic acceptance. Even agreement among dispersed chains cannot establish the absence of an unvisited mode.

When hundreds of chains each retain only a few draws, ordinary within-chain variance is poorly estimated. Nested $\widehat{R}$ uses $K$ superchains, each containing $M$ subchains sharing an initial state $X_{0k}$ but using independent randomness conditional on that state [Margossian et al., 2024, Definition 2.2 and Theorem 3.1]. Let $H_{km}$ be a subchain's mean after the chosen warm-up and $\bar{H}_k = M^{-1}\sum_m H_{km}$. If $\mu_n(x) = \mathbb{E}[H_{km} \mid X_{0k} = x]$ and $v_n(x) = \text{Var}(H_{km} \mid X_{0k} = x)$, the law of total variance yields

$$\text{Var}(\bar{H}_k) = \text{Var}\{\mathbb{E}[\bar{H}_k \mid X_{0k}]\} + \mathbb{E}\{\text{Var}(\bar{H}_k \mid X_{0k})\} \tag{2.140}$$

$$= \text{Var}\{\mu_n(X_{0k})\} + \frac{1}{M}\mathbb{E}\{v_n(X_{0k})\}. \tag{2.141}$$

Replication reduces the second term but leaves the first unchanged. This is why shared starts within a superchain are useful: independently dispersing every subchain would also average away the variation due to initialization.

For $N$ retained draws $h_{kmn}$ per subchain, the paper defines

$$\widehat{B}_\nu = \frac{1}{K-1}\sum_k (\bar{H}_k - \bar{H})^2, \tag{2.142}$$

$$\widetilde{B}_k = \frac{1}{M-1}\sum_m (H_{km} - \bar{H}_k)^2, \qquad \widetilde{W}_k = \frac{1}{M}\sum_m \frac{1}{N-1}\sum_n (h_{kmn} - H_{km})^2, \tag{2.143}$$

$$\widehat{W}_\nu = \frac{1}{K}\sum_k (\widetilde{B}_k + \widetilde{W}_k), \qquad \widehat{R}_\nu = \sqrt{1 + \widehat{B}_\nu/\widehat{W}_\nu}. \tag{2.144}$$

Set $\widetilde{B}_k = 0$ for $M = 1$ and $\widetilde{W}_k = 0$ for $N = 1$; at least one of $M, N$ must exceed one, and $K > 1$. The behavior and cutoff depend on the design, particularly $M$ and $N$. A familiar long-chain cutoff cannot simply be copied to an arbitrary ensemble. If subchains share adaptive statistics, conditional independence need not hold and the second line of (2.141) acquires cross-covariance terms. An independently learned, frozen kernel followed by a correctly designed diagnostic run is a simpler first application.

The first term in (2.141) measures variation of the conditional expectation across starts, not its distance from $\mu$. A constant but wrong $\mu_n(x)$ has zero such variation. Nested $\widehat{R}$ therefore remains a diagnostic whose usefulness depends on informative initial dispersion and the chain dynamics, not a proof of zero bias.

### 2.10.4 A stopping rule must name the economic quantity

For a scalar mean with a valid central limit approximation, one can require an estimated confidence half-width

$$z_{1-\alpha/2}\sqrt{\widehat{\sigma}_h^2/n} \le \delta_{\text{abs}} + \delta_{\text{rel}} s_h, \tag{2.145}$$

where the tolerances are chosen for the scientific use and $s_h$ is a specified scale, such as the posterior standard deviation. A relative tolerance based only on $|\widehat{\mu}|$ is unstable near a zero effect. Fixed-width stopping has an asymptotic theory under explicit assumptions; arbitrary repeated inspection of nominal fixed-sample intervals does not give a finite-sample sequential guarantee [Flegal and Gong, 2013]. Readiness checks, minimum information, a finite resource limit and an explicit inconclusive outcome are part of a usable procedure.

For a smooth nonlinear function $g(\mu)$, a delta-method variance is $\nabla g(\mu)^\top \Sigma \nabla g(\mu)/n$. This describes $g(\bar{h})$; the posterior expectation $\mathbb{E}_\pi g(h(X))$ is instead estimated by averaging $g(h(X_t))$ and needs that observable's long-run variance. For a quantile $q_p$ with continuous positive marginal density $f_h(q_p)$, linearizing the empirical CDF gives

$$\widehat{q}_p - q_p \simeq -\frac{\widehat{F}_n(q_p) - p}{f_h(q_p)}, \qquad \mathrm{Var}(\widehat{q}_p) \simeq \frac{\sigma^2_{\mathbf{1}\{h \le q_p\}}}{n f_h(q_p)^2}. \tag{2.146}$$

Thus a mean MCSE is not a quantile MCSE. For an event probability use the indicator's dependence, not an independent-binomial standard error unless independence is justified. Zero observed rare events produce an empirical variance of zero while leaving the probability uncertain. More sampling, an independently justified rare-event method, or an explicitly unresolved probability is needed. Heavy-tailed targets can lack the moments required for variance-based criteria; bounded observables and suitably justified quantile procedures may then be more appropriate.

These distinctions preserve the intended tuning interface. Acceptance compatibility, finite numerical behavior and movement determine which fixed-kernel candidates remain usable for further assessment. $\widehat{R}$ must not be reinstated as an acceptance-tuning requirement. Warm-up and posterior assessment can use rank diagnostics, precision and reference agreement in their own roles. Choosing an epsilon and a trajectory length is not a claim that the retained posterior has converged.

## 2.11 Practical choices for structural models

The least expensive useful change depends on the measured bottleneck. A representative MacroFinance MIDAS adapter differentiates a batched, masked Kalman likelihood with correlated noises and seven sampled parameters. There the observation horizon can dominate the work even though the parameter dimension is small. A stochastic-volatility benchmark with thousands of sampled latent states does not measure the same problem. The inspected BGS description in `dsge_hmc` instead declares an endpoint potential based on a particular UKF approximation and a deterministic position-only proposal field that is not asserted to be its exact score. That distinction affects the admissible methods before speed is considered.

A reversible shear integrator with an exact endpoint MH correction can remain valid with a position-only non-gradient force, as derived earlier. The equipartition identity and the stationary microcanonical dynamics use the actual score and cannot inherit that conclusion. BGS therefore needs either an eligible total score of its declared endpoint target or a separate derivation and validation of the proposed force-based method. Relabeling the field as a gradient would be wrong. The current capability registry describes that branch as conditional mechanics without exact-score retained-member authority. The operational reference and registry remain the entry points for an HMC consumer; this chapter does not add another tuner.

Table 2.2: Method selection depends on the sampled target and the measured cost. These are proposed investigations, not local speed rankings.

| Intervention | Most relevant circumstance | First discriminating evidence |
|---|---|---|
| Validated whitening and noncentering | Strong correlation, a funnel, or varying conditional scale | Exact transformed density and total score; inverse round trips; model-coordinate tails after freezing the map. |
| Randomized HMC and SNAPER | Resonance or costly trajectory selection on batched hardware | Same target, map and starts as corrected HMC; slow projections and squared observables; total time to declared precision. |
| Equivalent target/score optimization | Kalman, UKF or model-solution work dominates a gradient | Value and total-gradient parity, including masks, noise dependence and normalization; measured compilation and batch latency. |
| MAMS | Integration work remains costly after geometry repair | Sphere-measure correction, stable force flow, reversibility, reference agreement and equal-accuracy total gradient/time comparison. |
| LAPS | Large ensembles exhibit favorable measured batch scaling | Mean-shift and covariance counterexamples; independent warm-up assessment; total work and latency including its transient. |
| MALT or MEADS | Persistent dynamics may reduce residual resonance or random-walk behavior | Refreshment and momentum/slice-state correctness; cost of rejections and ensemble coordination. |
| Pathfinder | Initial exploration or geometry learning dominates | Initialization quality across dispersed starts, modes and boundaries; subsequent exact-kernel behavior and total setup cost. |
| Delayed rejection | Residual local stiffness survives a useful map | Reverse rejection factors and retry accounting; difficult-region movement versus extra gradients. |
| Particle-MALA/mGRAD | The latent trajectory itself must be sampled | Conditional smoothing reference, correct joint auxiliary weights, path mixing and scaling in horizon/state dimension. |

For the present TensorFlow setting, randomized-length HMC followed by SNAPER is the first sampler comparison worth making. The installed TFP source supplies trajectory adaptation. An experimental BayesFilter implementation now connects fixed-geometry ChEES-rate/SNAPER proposals, randomized corrected HMC and deterministic fully re-

freshed MAMS to the shared candidate-set tuners; The monograph's HMC tuning chapter specifies the supported interfaces. The implementation still needs scoped statistical, posterior and native GPU qualification before release or an efficiency claim. Whitening should continue alongside this work. Where a marginal Gaussian likelihood is expensive, investigate an equivalent parallel formulation or target/score optimization before attributing the cost to the sampler. MAMS is the next major kernel candidate. LAPS becomes attractive if those same model operations scale well with a large chain batch and its transient has independent diagnostic support.

The proposed order reflects implementation availability and mathematical applicability, not measured superiority. It can change after profiling. Custom QZ decompositions, equilibrium-solution derivatives, boundary handling and filter implementations make a JAX migration a separate engineering project. An author implementation in JAX is valuable for independent checks and algorithm details; it does not make a TensorFlow port either trivial or impossible.

## 2.12 Discriminating tests and model comparisons

Two comparisons are needed. A comparison with *matched geometry* fixes the posterior, data, numerical precision, coordinate map and model-coordinate starts, then allows each method its own valid tuning under a comparable resource budget. It measures the transition method. An *end-to-end* comparison also includes compilation, initialization, map fitting, tuning, fresh verification and retained sampling. It measures what a user pays. Both should include the current corrected broad-grid HMC and a simple randomized-length version before crediting a more elaborate method. BayesFilter excludes NUTS from implementation and local comparisons; its discussion here explains the literature and author benchmarks. Count every force evaluation, including integrator substages and rejected delayed-rejection proposals.

The primary question is time to a predeclared accuracy for stated posterior quantities. Numerical validity, reference disagreement, divergences and missing diagnostics can reject a candidate. Acceptance, ChEES/SNAPER objectives, equipartition and short-run ESS explain behavior but do not replace posterior checks. Paired repeated runs and uncertainty for the declared comparison are necessary for a supported ranking; passing a hard screen leaves a method viable, not superior.

Table 2.3: A model ladder that isolates mechanisms before expensive state-space applications.

| Model or construction | Misleading success | Discriminating observation |
|---|---|---|
| Gaussian at a full or half Hamiltonian period | Acceptance one; apparent movement from sign changes | Means, variances and squared-observable correlation; exact independent Gaussian reference and random-length control. |
| Rotated anisotropic Gaussian | Good average movement while one direction remains slow | Known covariance, eigenvector projections, worst declared estimand and diagonal versus full whitening. |

| Model or construction | Misleading success | Discriminating observation |
|---|---|---|
| Shifted Gaussian with correct covariance | Centered equipartition equals its target value | Known mean error, full diagnostic and separately identified implementation variant. |
| Wrong-correlation Gaussian or equal-variance mixture | Correct marginal variances hide dependence or distribution shape | Cross moments, fourth moments, event probabilities and full-matrix checks. |
| Exactly and partially whitened funnel | Good latent diagnostics hide map errors or model-coordinate tails | Analytic map and Jacobian; total score; model-coordinate moment and tail reference; residual curvature. |
| Heavy-tailed scale model | An apparently stable variance estimate assumes a nonexistent moment | Explicit moment conditions; bounded functions and quantiles; insufficient information as a valid outcome. |
| Separated mixture with unbalanced starts | All chains agree inside the same mode | Known mode weights, cross-mode transitions and independently dispersed initializations. |
| Constrained parameter near a boundary | Finite target values conceal a missing Jacobian or saturated score | Analytic or quadrature reference and directional derivatives near the boundary; appropriate discontinuity treatment. |
| LGSSM with masks and correlated noise | A faster filter evaluates a different likelihood | Sequential/parallel value and total-score agreement before posterior comparison; short/long horizons. |
| Nonlinear stochastic volatility or other nonlinear SSM | Samplers agree because they share the same biased filter | Small independent reference; separate likelihood-approximation error; conditional latent-path tests where applicable. |
| Representative MIDAS and BGS scopes | Synthetic performance does not transfer to model-solution work | Actual target/force identity, measured batch scaling, boundary and weak identification regimes, economic estimands. |

Transition-level tests should precede this ladder. For MAMS they include the excluded one-dimensional formula, zero force, velocities aligned and antialigned with the force, preservation of unit norm, forward/reverse round trips, analytic versus finite-difference sphere Jacobians, and large-argument stability. For MALT, compare the sum of local integration errors with the full forward/reverse path-density ratio. For delayed rejection, test a case in which the reverse first-stage rejection probability differs sharply from the

forward one. For Gaussian message passing, dropping one log-normalization term must fail a likelihood test even if the filtered mean still matches.

Posterior checks can use exact Gaussian sampling, numerical quadrature on small parameter targets and independent state-space reference calculations. Simulation-based calibration complements these checks by repeatedly drawing parameters and data from the model and assessing posterior ranks [Talts et al., 2020]. If the data-generating parameter and the posterior draws are jointly exchangeable conditional on the data, the parameter's rank is uniform with randomized tie-breaking; correlated MCMC draws require the corresponding dependence treatment before a rank test is interpreted. Calibration averaged over the prior can also conceal a failure in a rare but important regime, so boundary, weak-identification and long-horizon cases need explicit coverage. MCMC implementation tests based on invariance or reversible simulation provide another layer [Gandy and Scott, 2021]; no one finite suite proves correctness for every model.

This design makes failures informative. A failed sphere Jacobian is an implementation error. Correct transitions with poor model-coordinate tails after partial whitening indicate residual geometry or insufficient sampling. A correct sampler for the declared UKF posterior can still disagree with a more accurate likelihood because the target approximation differs. A method that passes all correctness checks but uses more total work is viable and possibly useful for latency, although a claimed work reduction would be unsupported. These distinctions determine the next repair.

## 2.13 Sources, versions and the scope of the conclusions

The literature review behind this chapter inspected the technical methods, relevant theory, experiments and appendices of the cited recent papers, as well as author implementations where available. The two motivating papers are the May 2025 revision of MAMS, arXiv:2503.01707v2, and the January 2026 LAPS preprint, arXiv:2601.16696v1. The unadjusted-bias discussion uses the June 2026 revision of arXiv:2412.08876; its original identifier year alone does not identify the examined argument. Bibliography entries record the inspected versions when a working paper is used.

Local copies, extracted text and source hashes are preserved under `.localresources/hmc-methods-review-20261007/`.

The source manifest records pinned revisions of BlackJAX and the authors' sampler benchmarks. It also records the TFP source, installed TFP 0.25.0 files, Stan Pathfinder, Particle-MALA, delayed-rejection GHMC and nested-$\widehat{R}$ sources. The implementation differences described in the LAPS and SNAPER sections refer to those snapshots. They must not be read as assertions that all versions implement the same equations.

The algebra in this chapter identifies preserved measures, required corrections and diagnostic limitations. Symbolic checks accompany the document review, but cannot establish geometric ergodicity, finite-time convergence, differentiability of an economic solution operator, or practical speed on a new consumer. The resulting recommendation is specific: first combine validated geometry with inexpensive randomized trajectory adaptation and an equivalent efficient likelihood; then evaluate MAMS, and evaluate LAPS when measured ensemble scaling warrants it. The state-space tests above are what would turn that recommendation into evidence for these repositories.

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